Polylogarithms and their Arithmetic Bridges

Volume 2: Signed Kernels, Zero Geometry and Experimental Discovery

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About this volume

Signed moment kernels, exact angular zero counts, all-depth compensation and certified computation develop the analytic geometry. The experimental chapter explains how computations generate proof obligations and how exact certificates settle them.

The four volumes form one collective manuscript. They share definitions, notation, source evidence and a bibliography. Chapter, theorem and equation numbers are retained from the unified manuscript so references in research reports remain stable. Cross-volume references link to the companion PDF; keep the four PDF files together when downloading them.

Volume 1	Polylogarithm Identities and Arithmetic Reductions
Volume 2	Signed Kernels, Zero Geometry and Experimental Discovery
Volume 3	Hurwitz Jets, Stieltjes Calculus and Gamma Correlations
Volume 4	Gamma Grids, CM Periods and Herglotz Arithmetic

A rigorous analytic proof, a finite exact certificate and an independently repeated numerical diagnostic carry different evidence. Computation drives conjecture formation and exposes missing structure; successful numerical recognition does not itself prove an identity or arithmetic independence. The source inventory and editorial ledger record the accepted corrections and the scope of material still undergoing analytic reconciliation.

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Chapter 5

Signed kernels and certified polylogarithm computation

The exact reductions suggest evaluation methods that expose more analytic structure. A signed density for $F_{a,b}(z) = \mathrm{Li}_{a,b}(z,1)$ proves a unique angular zero and makes Gaussian Euler errors one-sided. The general double integral supports Chebyshev and midpoint expansions, while Holder convolution provides uniform rational certificates for a wider alphabet. These methods connect experiments, geometry and finite arithmetic: their tail theorems and rounding contracts determine what a numerical comparison establishes. Write $D_{a,b}(x,y) = \mathrm{Li}_{a,b}(x,y)$. Throughout the signed-density discussion, $H_n^{(b)} = \sum_{j=1}^{n} j^{-b}$, $H_0^{(b)} = 0$, and $g_{a,b} = \operatorname{Im} F_{a,b}(\mathrm{i})$; real $b > 0$ is allowed where stated.

5.1 A signed-kernel representation

Definition 5.1. *For real $b > 0$ and $T > 0$, define*

$$\kappa_{1,b}(e^{-T}) = -\frac{T^b}{\Gamma(b+1)} + \frac{1}{\Gamma(b)} \int_T^\infty \frac{v^{b-1}}{e^v - 1}\, \mathrm{d}v. \tag{5.1}$$

For an integer $a \geq 2$, define successively

$$\kappa_{a,b}(u) = \int_u^1 \frac{\kappa_{a-1,b}(v)}{v}\, \mathrm{d}v, \qquad 0 < u < 1. \tag{5.2}$$

Theorem 5.2 (Moments and one sign change)**.** *For every integer $a \geq 1$ and real $b > 0$, the density $\kappa_{a,b}$ is integrable on $(0,1)$ and*

$$\int_0^1 u^{n-1} \kappa_{a,b}(u)\, \mathrm{d}u = \frac{H_{n-1}^{(b)}}{n^a}, \qquad n \geq 1. \tag{5.3}$$

In particular its total mass is zero. There is exactly one $r_{a,b} \in (0,1)$ such that

$$\kappa_{a,b}(u) < 0 \quad (0 < u < r_{a,b}), \qquad \kappa_{a,b}(u) > 0 \quad (r_{a,b} < u < 1). \tag{5.4}$$

The crossing is strict, and $r_{a,b} < r_{a-1,b}$ when $a \geq 2$.

Proof. First consider $a = 1$. The negative term in (5.1) has moment $-n^{-b-1}$. Substituting $u = e^{-T}$ and interchanging nonnegative integrands in the positive term gives

$$\frac{1}{\Gamma(b)} \int_0^\infty e^{-nT} \int_T^\infty \frac{v^{b-1}}{e^v - 1} \, \mathrm{d}v \, \mathrm{d}T$$
$$= \frac{1}{n\Gamma(b)} \int_0^\infty \frac{v^{b-1}(1 - e^{-nv})}{e^v - 1} \, \mathrm{d}v = \frac{1}{n} \sum_{k=1}^{n} k^{-b}.$$

The final equality uses the finite geometric identity $(1 - e^{-nv})/(e^v - 1) = \sum_{k=1}^{n} e^{-kv}$. Subtracting n^{-b-1} proves the base moment formula. At $n = 1$, both positive and negative terms have finite integral 1, so integrability and zero mass are established without subtracting divergent quantities.

As a function of T, the base density has derivative

$$\frac{\mathrm{d}}{\mathrm{d}T} \kappa_{1,b}(e^{-T}) = -\frac{T^{b-1}}{\Gamma(b)(1 - e^{-T})} < 0. \tag{5.5}$$

It is positive as $T \downarrow 0$: the limit is $\zeta(b) > 0$ if $b > 1$, and is $+\infty$ if $0 < b \leq 1$. It tends to $-\infty$ as $T \to \infty$. Thus it has exactly one crossing, with the sign pattern stated in the u coordinate.

For the induction, the integral operator in (5.2) is an L^1 contraction:

$$\int_0^1 \left| \int_u^1 \frac{f(v)}{v} \, \mathrm{d}v \right| \mathrm{d}u \leq \int_0^1 |f(v)| \, \mathrm{d}v.$$

Fubini's theorem gives the moment of the new density as $1/n$ times the previous moment; in particular it preserves zero mass. If the preceding crossing is r, the new density is positive on $[r, 1)$, while on $(0, r)$ its derivative is $-\kappa_{a-1,b}(u)/u > 0$. Zero mass forces it to be negative somewhere on that interval. Strict increase then gives exactly one crossing in $(0, r)$. This completes the induction. □

Corollary 5.3 (Integral continuation)**.** *For $z \in \mathbb{C} \setminus [1, \infty)$,*

$$F_{a,b}(z) = \int_0^1 \frac{z}{1 - zu} \kappa_{a,b}(u) \, \mathrm{d}u. \tag{5.6}$$

In particular, if $0 < \theta < \pi$,

$$\operatorname{Im} F_{a,b}(e^{\mathrm{i}\theta}) = \sin\theta \, J_{a,b}(\cos\theta), \qquad J_{a,b}(c) = \int_0^1 \frac{\kappa_{a,b}(u)}{1 - 2cu + u^2} \, \mathrm{d}u. \tag{5.7}$$

Proof. For $|z| < 1$, expand $z/(1 - zu)$ and use the moments. On compact subsets of the slit plane the denominator is bounded away from zero uniformly in $u \in [0, 1]$, so the integral is analytic there. This continues the disk identity. Taking the imaginary part of $e^{\mathrm{i}\theta}/(1 - ue^{\mathrm{i}\theta})$ gives (5.7). □

Lemma 5.4 (A sign test for zero-mass densities)**.** *Let κ have zero integral and the strict sign pattern (5.4), with crossing r. If q is strictly decreasing and the integral exists, then $\int q\kappa < 0$. If q is strictly increasing, the sign is positive.*

Proof. Subtract $q(r) \int \kappa = 0$. The product $(q(u) - q(r))\kappa(u)$ has the asserted strict sign on each side of r. □

5.2 The unique zero on the upper semicircle

Theorem 5.5 (Existence, uniqueness, and simplicity)**.** *Let $a \geq 1$ be an integer and let $b > 0$ be real. There is exactly one $\theta_{a,b} \in (0, \pi)$ such that $\operatorname{Im} F_{a,b}(e^{i\theta_{a,b}}) = 0$. It satisfies*

$$0 < \theta_{a,b} < \frac{\pi}{2}. \tag{5.8}$$

The imaginary part is positive for $0 < \theta < \theta_{a,b}$ and negative for $\theta_{a,b} < \theta < \pi$. At its zero the angular derivative is strictly negative. Consequently,

$$g_{a,b} < 0. \tag{5.9}$$

Proof. Suppress the indices in J and κ. The kernel $u \mapsto (1+u^2)^{-1}$ is strictly decreasing, so Lemma 5.4 gives $J(0) < 0$.

We next show that the limit of $J(c)$ as $c \uparrow 1$ is positive, possibly infinite. The negative density is supported in $(0, r)$, away from the singular endpoint, and its contribution has a finite limit. On $(r, 1)$ monotone convergence applies. Expanding $(1-u)^{-2}$, treating these two parts separately, and using (5.3) yields

$$\lim_{c\uparrow 1} J(c) = \int_0^1 \frac{\kappa(u)}{(1-u)^2}\,\mathrm{d}u = \sum_{n\geq 0} \frac{H_n^{(b)}}{(n+1)^{a-1}} > 0. \tag{5.10}$$

Every summand from $n = 1$ onward is positive. Thus continuity produces at least one zero $c_0 \in (0, 1)$.

For uniqueness, write $K_c(u) = (1 - 2cu + u^2)^{-1}$. If $c' > c$, then

$$\frac{\mathrm{d}}{\mathrm{d}u}\frac{K_{c'}(u)}{K_c(u)} = \frac{2(c'-c)(1-u^2)}{(1-2c'u+u^2)^2} > 0. \tag{5.11}$$

At a zero $J(c) = 0$, the signed density $K_c\kappa$ still has one sign change and zero mass. Applying Lemma 5.4 to the ratio in (5.11) gives $J(c') > 0$ for every $c' > c$. The same argument with a smaller parameter gives $J(c') < 0$ for $c' < c$. There can be only one zero in $(-1, 1)$, with the indicated signs.

Finally,

$$J'(c_0) = \int_0^1 K_{c_0}(u)\kappa(u)\frac{2u}{1-2c_0u+u^2}\,\mathrm{d}u > 0,$$

because the last factor has derivative $2(1-u^2)/(1-2c_0u+u^2)^2 > 0$. Equation (5.7) therefore has derivative $-\sin^2\theta_{a,b}\,J'(c_0) < 0$ at $\theta_{a,b} = \arccos c_0$. Since $c_0 > 0$, this zero is below $\pi/2$. The Gaussian inequality is its value at $\theta = \pi/2$. □

Remark 5.6 (The first zero is elementary). The identity $F_{1,1}(z) = \frac{1}{2}\log^2(1-z)$ gives

$$\operatorname{Im} F_{1,1}(e^{i\theta}) = \log\left(2\sin\frac{\theta}{2}\right)\frac{\theta-\pi}{2}.$$

Hence $\theta_{1,1} = \pi/3$ and $g_{1,1} = -\pi\log 2/8$. This example also checks the direction of the sign change in Theorem 5.5.

5.3 A four-term asymptotic for the zero

Put $\delta_{a,b} = \pi/2 - \theta_{a,b}$. The first surviving Fourier mode at $\pi/2$ is the $n = 3$ term, whereas the $n = 2$ term controls the derivative. Their ratio suggests $(2/3)^a$; the theorem below identifies both the next Fourier corrections and the first nonlinear correction.

Theorem 5.7 (Uniform zero asymptotic)**.** *For integer $a \to \infty$, uniformly for real $b \geq 1$, set $A = H_2^{(b)}$, $B = H_3^{(b)}$, and $C = H_4^{(b)}$. Then*

$$\delta_{a,b} = \frac{A}{2}\left(\frac{2}{3}\right)^a - \frac{C}{2}\left(\frac{2}{5}\right)^a + AB\left(\frac{1}{3}\right)^a - \frac{23A^3}{48}\left(\frac{8}{27}\right)^a + O\left(\left(\frac{2}{7}\right)^a\right). \tag{5.12}$$

The implied constant and the threshold for a can be chosen independently of $b \geq 1$.

Proof. For sufficiently large a, the boundary series and its derivative converge absolutely. Normalize the imaginary part by multiplying by 2^a, and write $\theta = \pi/2 - \delta$. The modes $n = 2, 3, 4, 5$ give

$$\begin{aligned} 0 = {} & \sin(2\delta) - Aq^a\cos(3\delta) - Bs^a\sin(4\delta) + Cr^a\cos(5\delta) \\ & + O\big(\delta\, 3^{-a} + (2/7)^a\big), \qquad q = \frac{2}{3}, \quad s = \frac{1}{2}, \quad r = \frac{2}{5}. \end{aligned} \tag{5.13}$$

The $n = 6$ mode is $O(\delta 3^{-a})$. For $n \geq 7$, the bound $H_{n-1}^{(b)} \leq 1 + \log n$ gives a uniform tail estimate: for $a \geq 3$,

$$\sum_{n\geq 7}(1+\log n)(2/n)^a \leq (2/7)^a \sum_{n\geq 7}(1+\log n)(7/n)^3.$$

The constant is finite and independent of b.

Since $1 \leq A \leq 3/2$ and B, C are bounded uniformly in b, the normalized expression is negative at $\delta = 0$ and positive at $\delta = q^a$ for all sufficiently large a. Theorem 5.5 identifies the root in that interval. Thus $\delta = O(q^a)$ uniformly.

Taylor expansion of (5.13) now gives

$$0 = 2\delta - Aq^a - 4Bs^a\delta + Cr^a + \frac{9}{2}Aq^a\delta^2 - \frac{4}{3}\delta^3 + O((2/7)^a). \tag{5.14}$$

For example, the omitted terms have exponential bases bounded by q^5, sq^3, rq^2, and $q/3$, all smaller than $2/7$. Insert first $\delta_0 = Aq^a/2$. The linear perturbation $-4Bs^a\delta_0$ requires the correction $AB(qs)^a$, while the $n = 5$ mode requires $-Cr^a/2$. The cubic contribution at δ_0 is

$$\frac{9}{2}Aq^a\delta_0^2 - \frac{4}{3}\delta_0^3 = \frac{23}{24}A^3q^{3a},$$

so its compensating correction is $-23A^3q^{3a}/48$. Substitution of the full displayed approximation in (5.14) leaves a residual $O((2/7)^a)$; all cross-products of the corrections have smaller exponential bases. The derivative of the normalized expression with respect to δ is $2 + o(1)$, uniformly on this interval. The mean-value theorem therefore converts the residual estimate into the error in (5.12). □

a	b	$\theta_{a,b}$, approximately	$\delta_{a,b}$, approximately
8	1	1.541868140889	0.02892818590626
8	2	1.546656477391	0.02413984940401
8	4	1.550268639150	0.02052768764464
12	1	1.565028587360	0.005767739435093
12	2	1.565988233290	0.004808093504888
12	4	1.566708834682	0.004087492112621
20	1	1.570570790993	0.0002255358018452
20	2	1.570608378737	0.0001879480579893
20	4	1.570636570306	0.0001597564888279

These entries are numerical illustrations from a 300-term Fourier sum, not certified root intervals. Existence, uniqueness, and the asymptotic estimate do not depend on the table. At $a = 20, 40, 80$ and $b = 1, 2, 8$, the computed next normalized remainder approaches $H_6^{(b)}/2$, as suggested by the $n = 7$ mode. A systematic expansion beyond the proved four terms is a separate research problem.

5.4 Certified Euler evaluation

5.4.1 A one-sided rational enclosure

Define the difference operator with the sign convention

$$(Df)(n) = f(n) - f(n+1).$$

This is the negative of the usual forward difference; it makes the Euler transformation particularly simple. Let

$$f_{a,b}(n) = \frac{H_{2n}^{(b)}}{(2n+1)^a}, \qquad d_k = D^k f_{a,b}(0), \qquad E_N(a,b) = \sum_{k=0}^{N-1} \frac{d_k}{2^{k+1}}. \tag{5.15}$$

Theorem 5.8 (Signed Euler certificate)**.** *For positive integers a, b,*

$$d_0 = 0, \qquad d_k < 0 \quad (k \geq 1), \qquad g_{a,b} = \sum_{k\geq 0} \frac{d_k}{2^{k+1}}. \tag{5.16}$$

For every $N \geq 1$,

$$E_N(a,b) - C_b 2^{-N} \leq g_{a,b} \leq E_N(a,b), \qquad C_b = \begin{cases} 1, & b = 1, \\ b/(b-1), & b \geq 2. \end{cases} \tag{5.17}$$

The approximants decrease strictly from $E_1 = 0$ to $g_{a,b}$.

Proof. The moment representation gives

$$d_k = \int_0^1 (1-u^2)^k \kappa_{a,b}(u)\, \mathrm{d}u. \tag{5.18}$$

For $k = 0$ this is zero mass. For $k \geq 1$ the weight is strictly decreasing, so Lemma 5.4 gives $d_k < 0$.

We bound either signed mass of the density. When $b = 1$,

$$\kappa_{1,1}(u) = \log \frac{u}{1-u}, \qquad \int_0^1 (\kappa_{1,1})_+ \,\mathrm{d}u = \log 2 < 1.$$

When $b \geq 2$, the positive term of (5.1) is at most $\zeta(b)$, so the positive mass is at most $\zeta(b) < 1 + 1/(b-1) = C_b$. Zero mass makes the negative mass equal to the positive mass. The L^1 contraction used in Theorem 5.2, together with preservation of zero mass, gives the same bound for every a. Since $0 \leq (1-u^2)^k \leq 1$, we obtain $|d_k| \leq C_b$.

The series in (5.16) is now absolutely convergent. Summing its geometric kernel gives

$$\sum_{k\geq 0} \frac{d_k}{2^{k+1}} = \int_0^1 \frac{\kappa_{a,b}(u)}{1+u^2}\,\mathrm{d}u = g_{a,b},$$

where the last equality is (5.6) at $z = \mathrm{i}$. All tail terms are negative, and their total absolute value is at most $C_b \sum_{k\geq N} 2^{-k-1} = C_b 2^{-N}$. This proves the certificate and strict monotonicity. □

The theorem is stronger than applying a general-purpose extrapolation rule and observing stability. Every endpoint in (5.17) is rational. For $N = 400$, the interval width is at most $2 \cdot 2^{-400} < 8 \cdot 10^{-121}$. The accompanying files round the enclosures outward to convenient decimal strings; that display rounding is separate from the internal rational width.

5.4.2 An exact linear-work finite formula

Computing a full triangular difference table uses quadratic work. A finite rearrangement avoids it.

Proposition 5.9 (Binomial-tail weights)**.** *For $N \geq 1$,*

$$E_N(a,b) = 2^{-N} \sum_{n=0}^{N-1} (-1)^n \frac{H_{2n}^{(b)}}{(2n+1)^a} \sum_{j=n+1}^{N} \binom{N}{j}. \tag{5.19}$$

At fixed a, b, the right side can be formed with $O(N)$ exact rational or integer updates after choosing N. All denominators divide

$$2^N \operatorname{lcm}(1, 2, \ldots, 2N)^{a+b}. \tag{5.20}$$

In particular, the number of bit operations is polynomial in the requested precision at fixed indices.

Proof. Expand $D^k f(0) = \sum_{n=0}^{k} (-1)^n \binom{k}{n} f(n)$ and interchange the two finite sums. The required weight identity is

$$\sum_{k=n}^{N-1} 2^{N-k-1} \binom{k}{n} = \sum_{j=n+1}^{N} \binom{N}{j}.$$

It follows by induction on N using Pascal's recurrence, or by comparing binomial-tail generating functions. This proves (5.19).

The binomial coefficients are generated successively by $\binom{N}{n} = \binom{N}{n-1}(N-n+1)/n$, and the cumulative tail is updated by subtracting one coefficient. Each harmonic number is updated by two rational summands. Thus only linearly many updates are needed. Each summand's denominator divides the indicated least-common-multiple power, and the additional factor is 2^N. Finally, the elementary bound $\operatorname{lcm}(1, \ldots, 2N) \leq (2N)!$ gives bit length $O(N + (a+b)N \log N)$ for a common denominator; the numerator has a comparable polynomial bound. Since $N = O(d)$ suffices for d decimal places by (5.17), even elementary integer arithmetic proves polynomial bit complexity. No optimal multiplication or sharp least-common-multiple estimate is required. □

5.4.3 Independent depth-one interval arithmetic

The right sides of the reductions are evaluated using rational enclosures separate from the Gaussian evaluator. For $s \geq 1$, the sequences $(2n+1)^{-s}$ and $(n+1)^{-s}$ are moments of positive densities, so all their D-differences are nonnegative. Their Euler sums give lower bounds for $\beta(s)$ and $\eta(s)$, with tails at most 2^{-N}. For integer $s \geq 2$, divide the latter interval by $1 - 2^{1-s}$ to enclose $\zeta(s)$.

For π, the implementation uses Machin's formula

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$$

and the alternating-series bounds for each rational arctangent series. The identity follows directly by tangent addition and the fact that the angle lies in the first quadrant. For $\log 2$, it uses

$$0 < \log 2 - 2 \sum_{n=0}^{N-1} \frac{1}{(2n+1)3^{2n+1}} \leq \frac{9}{4(2N+1)3^{2N+1}}. \tag{5.21}$$

The expansion is $2\operatorname{arctanh}(1/3)$; the bound replaces the remaining denominators by $2N+1$ and sums a geometric series.

Ordinary rational interval addition and multiplication then enclose every polynomial right side. An overlap check can detect an erroneous formula or implementation if the intervals separate. Overlap alone does not prove an identity; the proofs in Sections 4.6 and 4.4 do that.

5.4.4 Affine harmonic sequences as signed moments

The signed-measure formulation also certifies sequences whose harmonic and denominator steps differ. The Cayley continuation supplies the integer-order case; its Gamma-density argument extends without change to real outer order $a \geq 1$. It complements the sharper one-sided bounds above.

Theorem 5.10 (Affine signed-moment bound)**.** *Let r, d be positive integers, $a \geq 1$ real and $b > 0$ real. Set*

$$A_k = \frac{H_{rk}^{(b)}}{(dk+1)^a}, \qquad m_1 = 1, \qquad m_a = \frac{(a-1)^{a-1}e^{-(a-1)}}{\Gamma(a)} \quad (a > 1).$$

There is a finite signed measure ν on $[0,1]$, with no atom at one, such that $A_k = \int x^k \, d\nu(x)$ and

$$\|\nu\|_{\mathrm{TV}} \leq \frac{2r}{d} m_a b \zeta(b+1).$$

For $b > 1$ one also has $\|\nu\|_{\mathrm{TV}} \leq 2\zeta(b)$. Consequently the ordinary alternating series converges, and its N-term Euler transform E_N satisfies

$$\left| \sum_{k \geq 0} (-1)^k A_k - E_N \right| \leq C 2^{-N},$$

where C may be either applicable bound. For integer b, valid rational choices are $C = 10r/(3d)$ at $b = 1$ and $C = 10/3$ at $b \geq 2$.

Proof. Let Y have Gamma density $y^{a-1}e^{-y}/\Gamma(a)$, and let μ be the law of $X = e^{-dY}$. Then $\int x^k \, d\mu = (dk+1)^{-a}$. In the logarithmic coordinate $s = -\log x$, its density is

$$f(s) = \frac{s^{a-1}e^{-s/d}}{d^a \Gamma(a)} \mathbf{1}_{s>0}.$$

For every real $a \geq 1$ its maximum is m_a/d. Extended by zero to the negative axis, the total variation of its distributional derivative is $2m_a/d$: the rise and fall give this for $a > 1$, while at $a = 1$ the jump at zero and the decreasing exponential each contribute $1/d$. The L^1 distance from a bounded-variation density to its translate by $h \geq 0$ is at most h times this derivative variation. Indeed, integrate $f(s-h) - f(s) = -\int_{s-h}^{s} f'(v) \, dv$ and use Fubini; smooth convolution proves the same bound for the jump case. Thus, writing $D_\rho \mu$ for pushforward by $x \mapsto \rho x$,

$$\|\mu - D_{e^{-rt}}\mu\|_{\mathrm{TV}} \leq \min\left(2, \frac{2m_a r}{d} t\right).$$

The generalized harmonic integral gives the measure

$$\nu = \frac{1}{\Gamma(b)} \int_0^\infty \frac{t^{b-1}}{e^t - 1} (\mu - D_{e^{-rt}}\mu) \, dt.$$

This integral converges in total variation: the translation bound supplies an additional factor t at zero, and exponential decay controls infinity. Taking moments gives A_k, since the moments of the measure difference are $(dk+1)^{-a}(1-e^{-rkt})$. Its total-variation bounds follow from

$$\frac{1}{\Gamma(b)}\int_0^\infty \frac{t^b}{e^t-1}\,dt = b\zeta(b+1), \qquad \frac{1}{\Gamma(b)}\int_0^\infty \frac{t^{b-1}}{e^t-1}\,dt = \zeta(b) \quad (b>1).$$

Each constituent has no atom at one, and variation domination passes that property to ν. The finite alternating kernel is bounded by two and converges for $x<1$ to $(1+x)^{-1}$, so dominated convergence proves ordinary convergence. The exact remainder kernel in the Euler expansion is $2^{-N}(1-x)^N/(1+x)$, bounded by 2^{-N}; integrating against ν proves the stated error. Finally $m_a \le 1$, by bounding the maximum of the unit-scale Gamma density, and $\zeta(2) < 5/3$. For the displayed rational bound at $b=1$ one may also use only the integer a needed by the rational evaluator: then $m_a \le 1$ follows directly from $e^{a-1} \ge (a-1)^{a-1}/(a-1)!$. For the real-$a$ inequality, write the Gamma density as the convolution of an exponential density and a Gamma$(a-1,1)$ probability density when $a>1$; its supremum is at most one. For integer $b \ge 2$, $\zeta(b) \le \zeta(2)$ gives $10/3$. □

The real-order theorem does not make its centers rational at noninteger indices. The finite rational implementation uses integer a, b, r, d and binomial-tail Euler weights. No complete monotonicity of A_k is assumed: the representing measure is signed and has total mass zero, since $A_0 = 0$.

5.5 An exact kernel and the sharp Euler error

5.5.1 Separating the polynomial from the exponentially small part

For this section a, b are positive integers, $w = a+b$, and $T = -\log u$.

Theorem 5.11 (Explicit density)**.** *Define the polynomial*

$$P_{a,b}(T) = -\frac{T^{w-1}}{(w-1)!} + \sum_{\mu=b+1}^{w-1} (-1)^{\mu-b-1}\binom{\mu-1}{b-1}\zeta(\mu)\frac{T^{w-1-\mu}}{(w-1-\mu)!}. \tag{5.22}$$

Then, for $T>0$,

$$\kappa_{a,b}(e^{-T}) = P_{a,b}(T) + (-1)^{a-1}\sum_{j=0}^{b-1}\binom{w-j-2}{a-1}\frac{T^j}{j!}\operatorname{Li}_{w-1-j}(e^{-T}). \tag{5.23}$$

Consequently,

$$\kappa_{a,b}(u) = P_{a,b}(-\log u) + O\Big(u(1+|\log u|^{b-1})\Big) \quad (u \downarrow 0). \tag{5.24}$$

Proof. For $a=1$, expand $(e^v-1)^{-1} = \sum_{n\ge1} e^{-nv}$ in (5.1). The elementary incomplete-gamma integral for integer b gives

$$\frac{1}{(b-1)!}\int_T^\infty \frac{v^{b-1}}{e^v-1}\,\mathrm{d}v = \sum_{j=0}^{b-1}\frac{T^j}{j!}\operatorname{Li}_{b-j}(e^{-T}),$$

which is exactly the asserted base case.

Write $K_{a,b}(T) = \kappa_{a,b}(e^{-T})$. The recursion is $K_{a,b}(T) = \int_0^T K_{a-1,b}(s)\,\mathrm{d}s$. Differentiate the proposed right side of (5.23), using $\frac{\mathrm{d}}{\mathrm{d}T}\operatorname{Li}_s(e^{-T}) = -\operatorname{Li}_{s-1}(e^{-T})$ and Pascal's identity. The polylogarithmic coefficients reduce to those for $(a-1,b)$; the derivative of $P_{a,b}$ is $P_{a-1,b}$. When $a \geq 2$, the constant terms at $T = 0$ cancel because

$$\binom{w-2}{b-1} = \binom{w-2}{a-1}$$

and their signs are opposite. The asserted formula therefore has the correct derivative and initial value. This proves it by induction. Finally each $\operatorname{Li}_s(e^{-T})$ in the finite sum is $O(e^{-T})$ as $T \to \infty$, proving (5.24). □

5.5.2 The complete logarithmic error polynomial

The exact error is

$$E_N(a,b) - g_{a,b} = -2^{-N} \int_0^1 \kappa_{a,b}(u) \frac{(1-u^2)^N}{1+u^2}\,\mathrm{d}u. \tag{5.25}$$

This follows by summing only the first N terms of the geometric kernel in the proof of Theorem 5.8. Its localization near $u = 0$ explains why the polynomial in Theorem 5.11 determines the error.

Theorem 5.12 (Sharp error with all logarithmic terms)**.** *Define the polynomial of degree $w-1$*

$$\begin{aligned} Q_{a,b}(L) &= -\int_0^\infty e^{-t^2} P_{a,b}(L - \log t)\,\mathrm{d}t \\ &= -\sum_{j=0}^{w-1} \frac{(-1)^j P_{a,b}^{(j)}(L)}{j!\,2^{j+1}} \Gamma^{(j)}(1/2). \end{aligned} \tag{5.26}$$

For fixed a, b, as $N \to \infty$,

$$\begin{aligned} E_N(a,b) - g_{a,b} = 2^{-N}\Bigg\{ & N^{-1/2} Q_{a,b}\Big(\frac{1}{2}\log N\Big) \\ & + O\Big(N^{-1}(1+\log^{b-1} N) + N^{-3/2}(1+\log^{w-1} N)\Big)\Bigg\}. \end{aligned} \tag{5.27}$$

In particular,

$$E_N(a,b) - g_{a,b} \sim \frac{\sqrt{\pi}}{2^w (w-1)!} \frac{2^{-N}(\log N)^{w-1}}{\sqrt{N}}. \tag{5.28}$$

The leading coefficient depends only on the total weight, not on its split into a and b.

Proof. Choose a fixed $\varepsilon \in (0, 1/2)$. The part of (5.25) with $u \geq \varepsilon$ is bounded by $2^{-N}(1-\varepsilon^2)^N \|\kappa_{a,b}\|_1$ and is negligible at the displayed scales. In the remaining part substitute $u = t/\sqrt{N}$ and use (5.24). For $t \leq \varepsilon\sqrt{N}$,

$$\left| \frac{(1-t^2/N)^N}{1+t^2/N} - e^{-t^2} \right| \leq \frac{C}{N} e^{-t^2}(t^2 + t^4).$$

Indeed, $0 \leq -\log(1-x) - x \leq Cx^2$ for $0 \leq x \leq \varepsilon^2$, and the denominator changes the expression by at most t^2/N times the Gaussian bound.

The polynomial contribution, including the Jacobian $N^{-1/2}$, is therefore $N^{-1/2} Q_{a,b}(\frac{1}{2}\log N)$ up to $O(N^{-3/2}(1+\log^{w-1} N))$. Extending the t integral to infinity changes it only by an exponentially

small amount. The remainder in (5.24) contributes at most $O(N^{-1}(1+\log^{b-1} N))$; integrability of $te^{-t^2}|\log t|^j$ at both endpoints justifies this estimate.

To obtain the second line of (5.26), expand the polynomial and use

$$\int_0^\infty e^{-t^2}(\log t)^j \,\mathrm{d}t = 2^{-j-1}\Gamma^{(j)}(1/2),$$

which follows by differentiating $\frac{1}{2}\Gamma(s/2)$ at $s=1$. Finally the leading term of $-P_{a,b}$ is $T^{w-1}/(w-1)!$. Its Gaussian integral is $\sqrt{\pi}/2$, and $L=(\log N)/2$. This gives (5.28); both error terms in (5.27) are smaller than that leading term. □

For example,

$$Q_{1,1}(L) = \frac{\sqrt{\pi}}{2}\left(L + \frac{\gamma + 2\log 2}{2}\right), \tag{5.29}$$

where γ is Euler's constant. Keeping the lower logarithmic terms matters at practical truncation lengths. For $(a,b)=(1,1)$, the following ratios use the exact rational proxy $E_N - E_{2N}$; its distance from the true error is at most 2^{-2N}.

N	Ratio to the leading term	Ratio to the full $Q_{1,1}$ approximation
64	1.42496	0.967964
128	1.37730	0.980510
256	1.33791	0.988050
512	1.30501	0.992593

The asymptotic expansion is not substituted for the certified bound in the implementation. Its role is to explain the observed convergence and suggest more efficient future remainder enclosures.

Proposition 5.13 (Euler certificate for the odd-denominator harmonic family)**.** *For integer $p \geq 1$, let $f(n) = H_n/(2n+1)^p$, let $d_k = D^k f(0)$, and let $E_N^S = \sum_{k<N} d_k/2^{k+1}$. Then*

$$E_N^S - \frac{N+1}{3^p 2^N} \leq S_p \leq E_N^S, \qquad S_p = \sum_{n\geq 0} \frac{(-1)^n H_n}{(2n+1)^p}. \tag{5.30}$$

Proof. Use $H_n = \int_0^1 (1-t^n)/(1-t)\,\mathrm{d}t$ and the Mellin integral for $(2n+1)^{-p}$. Finite differencing gives

$$d_k = \frac{1}{\Gamma(p)}\int_0^1 (-\log u)^{p-1}\int_0^1 \frac{(1-u^2)^k - (1-u^2t)^k}{1-t}\,\mathrm{d}t\,\mathrm{d}u.$$

Thus $d_0 = 0$ and $d_k < 0$ for $k \geq 1$. The numerator has absolute value at most $ku^2(1-t)$, so $|d_k| \leq k/3^p$. Summing the Euler kernels yields

$$\sum_{k\geq 0}\frac{d_k}{2^{k+1}} = -\frac{1}{\Gamma(p)}\int_0^1 \frac{(-\log u)^{p-1}\log(1+u^2)}{1+u^2}\,\mathrm{d}u = S_p.$$

The final equality follows from the harmonic generating function, first with an Abel factor. The bound on d_k justifies summing and integrating, and $\sum_{k\geq N} k/2^{k+1} = (N+1)2^{-N}$ proves (5.30). □

At $N=400$, combine Proposition 5.13, Theorem 5.8, and the independent depth-one intervals. The exact rational computation encloses the difference between the two sides of (4.41) in

$$[-10^{-118},\, 10^{-118}]. \tag{5.31}$$

Both sides begin

$$-0.010493456297335043453356751006201751963510618877 32\ldots.$$

This is certified proximity, not proof of equality. The file `s4_interval_certificate.json` records the outward-rounded enclosures and the tail bound used.

5.5.3 The precise relation system being tested

We now specify a finite-dimensional linear question; it must not be confused with completeness of all polylogarithm relations. Let

$$X^{r,s}_{a,b} = \operatorname{Im} D_{a,b}(\mathrm{i}^r, \mathrm{i}^s), \qquad r,s \in \mathbb{Z}/4\mathbb{Z}, \qquad a+b=w.$$

Set $X^{r,s}=0$ when both r,s are even, and impose conjugation $X^{r,s}=-X^{-r,-s}$. Choose the lexicographically smaller of each conjugate pair. Remove the divergent imaginary coordinate $X^{0,1}_{1,w-1}$; its conjugate is already identified. This leaves $6w-7$ formal coordinates.

For every $p+q=w$ and colors r,s, take the imaginary parts of the finite product stuffle

$$D_{p,q}(x,y)+D_{q,p}(y,x)=\operatorname{Li}_p(x)\operatorname{Li}_q(y)-\operatorname{Li}_w(xy) \tag{5.32}$$

and product shuffle

$$\begin{aligned}&\sum_{j=0}^{p-1}\binom{q+j-1}{j}D_{q+j,p-j}(y,x/y)\\ &\quad+\sum_{j=0}^{q-1}\binom{p+j-1}{j}D_{p+j,q-j}(x,y/x)=\operatorname{Li}_p(x)\operatorname{Li}_q(y),\end{aligned} \tag{5.33}$$

where $x=\mathrm{i}^r$, $y=\mathrm{i}^s$. If exactly one factor is $\operatorname{Li}_1(1)$, retain the stuffle-minus-shuffle cancellation row: the common divergent double coordinate cancels, leaving the right side $-\operatorname{Im}\operatorname{Li}_w(xy)$. Such rows follow by subtracting before taking the regularized boundary limit. Zero rows are discarded; duplicate nonzero rows need not be discarded. This construction defines a rational coefficient matrix A_w and its depth-one right side. It is precisely the model implemented in `code/full_ds.py`.

These are only linear depth-two product rows in a fixed weight. They do not include every relation obtained from distribution, argument transformations, higher-depth elimination, or the full algebraic consequences of standard relations. Zhao's treatment [69] provides broader context for these distinctions.

5.5.4 An exact obstruction at weight five

Let $u_{4,1}=\operatorname{Im} D_{4,1}(\mathrm{i},-1)$. Splitting the inner harmonic sum into even and odd indices proves

$$S_4 = g_{4,1}+u_{4,1}. \tag{5.34}$$

Indeed, for an odd outer index $2n+1$, the sum of the two inner coefficients is $H_{2n}+\sum_{m=1}^{2n}(-1)^m/m = H_n$; even outer indices have zero imaginary part. Consequently, the left-hand coefficient vector needed for (4.31) is

$$T = 3g_{4,1}+3g_{3,2}+9g_{2,3}+7u_{4,1}. \tag{5.35}$$

Proposition 5.14 (A finite linear-module obstruction)**.** *The coefficient vector T in (5.35) is not in the rational row span of the explicitly defined weight-five matrix A_5. Therefore no rational linear combination of just those weight-five depth-two rows proves the S_4 identity.*

Proof. The matrix has 92 rows and 23 columns. In the canonical coordinate order above, define a vector v by the following nonzero entries, setting every other entry to zero:

(a,b,r,s)	v	(a,b,r,s)	v
$(1,4,1,0)$	$-1/12$	$(2,3,0,1)$	$1/24$
$(2,3,1,0)$	$1/8$	$(2,3,1,3)$	$-1/24$
$(3,2,0,1)$	$-1/8$	$(3,2,1,0)$	$-1/24$
$(3,2,1,3)$	$-1/24$	$(4,1,0,1)$	$1/12$

Substitution in the binomial rows (5.32)–(5.33) gives $A_5 v = 0$, while $T \cdot v = 1$. These are finite rational equalities; the complete integer matrix, coordinate order, target, and witness are included in `s4_linear_module_obstruction.json` and replayed by `verify.py`. Every row-span vector annihilates v, whereas T does not. This proves the assertion. □

The proposition concerns only the specified depth-two product rows. Theorem 4.8 adds a convergent octahedral duality and lifted relations through higher depths, outside this row vocabulary. The two statements are therefore compatible: the obstruction identifies the additional information an eventual successful proof had to supply.

5.5.5 Exact ranks from two polynomial reflections

Retain exactly the coefficient matrix A_w defined above, including the stuffle-minus-shuffle boundary rows and excluding the one divergent imaginary coordinate. No distribution or higher-depth relations are added. Let $A_w^{\neg g}$ denote the submatrix with the Gaussian columns deleted. The following theorem replaces the former rank conjecture.

Theorem 5.15 (All-weight ranks)**.** *For every odd $w \geq 3$,*

$$\operatorname{rank} A_w = 5w - 6, \qquad \operatorname{rank} A_w^{\neg g} = \frac{9w - 11}{2}. \tag{5.36}$$

For every even $w \geq 2$,

$$\operatorname{rank} A_w = 6w - 7, \qquad \operatorname{rank} A_w^{\neg g} = 5w - 6. \tag{5.37}$$

In particular, the Gaussian-supported row space has dimension $(w-1)/2$ in odd weight and $w-1$ in even weight. In odd weight it is exactly the span of the same-point shuffle rows.

Proof. Put $N = w-2$. Use six degree-N polynomials A, B, C, D, E, F to encode a homogeneous null vector. Their respective colors are $(0,1), (1,0), (1,1), (1,2), (1,3), (2,1)$. Coefficients of $X^{a-1}Y^{b-1}$ have $a + b = w$. Complete the missing coefficient of A by $v^{0,1}_{1,w-1} = -v^{1,0}_{w-1,1}$. This satisfies the missing stuffle row; the missing shuffle row follows from its retained difference with stuffle. The omitted coordinate occurs in no other row: outer index one in a shuffle sum forces one of its factors to be $\mathrm{Li}_1(1)$. Thus completion and restriction identify the matrix kernel with the full homogeneous polynomial solution space.

Writing $P^{\mathsf{T}}(X,Y) = P(Y,X)$, the stuffle equations are

$$A = -B^{\mathsf{T}}, \quad C = -C^{\mathsf{T}}, \quad D = -F^{\mathsf{T}}, \quad E = E^{\mathsf{T}}. \tag{5.38}$$

The shuffle equations are

$$\begin{aligned} E(X+Y,X)+A(X+Y,Y)&=0,\\ B(X+Y,X)+B(X+Y,Y)&=0,\\ C(X+Y,Y)-F(X+Y,X)&=0,\\ D(X+Y,Y)-D(X+Y,X)&=0. \end{aligned} \tag{5.39}$$

For general colors (r,s) the shuffle polynomial is $P_{s,r-s}(X+Y,X)+P_{r,s-r}(X+Y,Y)$; its coefficient of $X^{p-1}Y^{q-1}$ is the stated binomial row. The four equations above come from product colors $(0,1),(1,1),(1,2),(1,3)$. Swapping the factors, conjugating, and discarding even-even imaginary rows exhaust the other cases.

The first two equations give

$$E(X,Y)=B(X-Y,X), \qquad B(X,Y)+B(X,X-Y)=0.$$

With $\varepsilon=(-1)^N$, this implies $E(Y,X)=-\varepsilon E(X,Y)$. Since $E=E^{\mathsf{T}}$, the whole A,B,E block vanishes if N is even. If $N=2m-1$ is odd, its free polynomial B is precisely an anti-invariant under $Y\mapsto X-Y$, with basis

$$B_j=X^{N-2j-1}(2Y-X)^{2j+1}, \qquad 0\le j<m. \tag{5.40}$$

Indeed, $U=X, V=2Y-X$ changes this reflection into $V\mapsto -V$.

The remaining equations give

$$F(X,Y)=C(X,X-Y), \quad D(X,Y)=-C(Y,Y-X), \quad C=-C^{\mathsf{T}}.$$

They imply $D(X,X-Y)=-\varepsilon D(X,Y)$, while the last shuffle asks for equality. Hence the C,D,F block is zero when N is even. When $N=2m-1$, it has the basis determined by

$$C_j=X^jY^{N-j}-X^{N-j}Y^j, \qquad 0\le j<m. \tag{5.41}$$

Conversely, the displayed substitutions satisfy every homogeneous equation. The B_j and C_j are independent and are retained in full after deleting the missing coefficient of A. They therefore give an integral $\mathbb{Q}$-basis of the full nullspace.

For odd $w=2m+1$, the nullity is $2m=w-1$, proving $\operatorname{rank} A_w=(6w-7)-(w-1)=5w-6$. A null vector of $A_w^{\neg g}$ extends by zero on the Gaussian columns; this is exactly the C block, of dimension m. Its $5w-6$ columns thus have rank $5w-6-m=(9w-11)/2$. For even w all columns of A_w are independent, and so are all columns of its submatrix.

Projection of the row space onto the non-Gaussian columns gives the stated supported-row dimensions by taking the difference of the two ranks. In odd weight $w=2m+1$, the same-point shuffle coefficient on $g_{a,w-a}$ is $\binom{a-1}{p-1}+\binom{a-1}{w-p-1}$, $1\le p\le m$. Its first m columns form a triangular Pascal matrix with diagonal one. These m independent Gaussian-supported rows therefore exhaust that space. □

Exact membership and the harmonic-sum obstruction. Let K_w have the preceding integer null vectors as columns. A rational coefficient row t is in the specified row space exactly when $tK_w=0$. Otherwise any column with nonzero pairing is a separating certificate. The basis is constructible in $O(w^2)$ integer arithmetic operations and has $O(w)$-bit coefficients; coefficient recurrences and bit bounds are given in the accompanying article.

For odd $w = 2m+1$, choose $B = 0$, $C = Y^{2m-1} - X^{2m-1}$. It vanishes on every Gaussian column and has $D(1,0) = 1$. Since $S_{2m} = g_{2m,1} + \operatorname{Im} D_{2m,1}(\mathrm{i}, -1)$, it pairs to one with S_{2m}. Thus no rational linear combination of these fixed-weight product rows reduces S_{2m} to Gaussian doubles and the associated single-value right sides. In the formal quotient, adjoining it to the Gaussian span increases the dimension from m to $m+1$. For the weight-five target above, this witness gives $T \cdot v = 7$. This is a uniform proof-system obstruction, not a numerical independence theorem and compatible with the already proved S_4 identity, whose certificate uses a larger vocabulary.

Restoring the right sides: an explicit compiler

Set $\ell_k(r) = \operatorname{Li}_k(\mathrm{i}^r)$, with the formal regularization $\ell_1(0) = 0$, and define known degree-N polynomials

$$\begin{aligned} P_{r,s}(X,Y) &= \sum_{p+q=w} \operatorname{Im}(\ell_p(r)\ell_q(s)) X^{p-1} Y^{q-1}, \\ H_N(X,Y) &= \sum_{j=0}^{N} X^j Y^{N-j}, \\ S_{01} &= P_{0,1} - \beta(w) H_N, \qquad S_{12} = P_{1,2} + \beta(w) H_N. \end{aligned}$$

The same six letters now encode the actual imaginary double values, with the artificial missing coefficient $[X^0 Y^N]A = -g_{w-1,1} - \beta(w)$. Put

$$\begin{aligned} e(X,Y) &= P_{0,1}(Y, X-Y) - S_{01}(X, X-Y), \\ d(X,Y) &= S_{12}(X,Y) + P_{1,2}(X, Y-X), \\ b(X,Y) &= P_{1,1}(Y, X-Y), \\ h(X,Y) &= P_{1,3}(Y, Y-X) - e(Y, Y-X) + e(Y-X, Y). \end{aligned}$$

Theorem 5.16 (Even-weight polynomial solution)**.** *For even $w \geq 2$ the full solution is*

$$\begin{aligned} B(X,Y) &= \tfrac{1}{2}\big(b(X,Y) + h(X,Y)\big), \\ C(X,Y) &= \tfrac{1}{2}\big(P_{1,3}(X,-Y) + P_{1,1}(Y,X) - d(X-Y,-Y) + d(X-Y,X)\big), \\ A(X,Y) &= S_{01}(X,Y) - B(Y,X), \\ E(X,Y) &= B(X-Y,X) + e(X,Y), \\ F(X,Y) &= C(X, X-Y) - P_{1,2}(Y, X-Y), \\ D(X,Y) &= -C(Y, Y-X) + d(X,Y). \end{aligned}$$

Discarding the artificial coefficient gives explicit single-value reductions for every convergent imaginary double at fourth roots of unity.

Proof. The affine stuffle equations are $A + B^{\mathsf{T}} = S_{01}$, $C + C^{\mathsf{T}} = P_{1,1}$, $D + F^{\mathsf{T}} = S_{12}$, $E - E^{\mathsf{T}} = P_{1,3}$. The four shuffle right sides in (5.39) are now $P_{0,1}, P_{1,1}, P_{1,2}, P_{1,3}$, respectively. Elimination gives the last four displayed formulas and $B(X,Y) + B(X, X-Y) = b(X,Y)$. Even homogeneity and the E stuffle give the difference $B(X,Y) - B(X, X-Y) = h(X,Y)$, proving the formula for B. For C, the last shuffle is $D(u, u-v) - D(u,v) = P_{1,3}(v, u-v)$. Use $C(u-v, -v) = C(v-u, v) = P_{1,1}(v-u, v) - C(v, v-u)$ and then $(X,Y) = (v, v-u)$ to get the asserted formula. Actual values satisfy these equations, and the zero homogeneous kernel makes the solution unique. □

Two mixed leading-index families. Let $c_p = \operatorname{Im} D_{p,1}(\mathrm{i},\mathrm{i})$ and $e_p = \operatorname{Im} D_{p,1}(\mathrm{i},-\mathrm{i})$. For all $m \geq 1$,

$$c_{2m-1} = (m-1)\beta(2m) - \tfrac{1}{2}\log 2\,\beta(2m-1) - \sum_{k=1}^{m-1}(1-2^{1-2k})\zeta(2k)\beta(2m-2k), \tag{5.42}$$

$$e_{2m-1} = m\beta(2m) - \tfrac{1}{2}\log 2\,\beta(2m-1) - \sum_{k=1}^{m-1}\zeta(2k)\beta(2m-2k). \tag{5.43}$$

For a direct coefficient proof, evaluate $C(1,0)$ and $E(1,0)$ in the compiler. The resulting equations are

$$\begin{aligned} 2C(1,0) &= P_{1,3}(1,0) + P_{1,1}(0,1) - d(1,0) + d(1,1), \\ 2E(1,0) &= P_{1,1}(1,0) + P_{1,3}(1,0) + e(1,0) + e(0,1). \end{aligned}$$

In both lines the first two terms are $-\log 2\,\beta(w-1)$, and

$$\begin{aligned} d(1,1) - d(1,0) &= P_{1,2}(1,1) - P_{1,2}(1,-1) + N\beta(w), \\ e(1,0) + e(0,1) &= -P_{0,1}(1,1) + P_{0,1}(1,-1) + w\beta(w). \end{aligned}$$

Only even inner, respectively even outer, indices survive the differences. Using $\ell_{2k}(2) = -(1-2^{1-2k})\zeta(2k)$ and $\operatorname{Im}\ell_j(1) = \beta(j)$ proves both formulas.

Status and evidence. The rank theorem proves both individual ranks, rather than inferring them from their difference. The previous word "Equivalently" should therefore be read as "In particular." Independent matrix construction, exact integer kernel checks and modular lower-bound certificates cover every weight 2–41; 210 even-weight formulas through weight 12 satisfy all 816 product rows, and both leading-index formulas are checked through weight 16. These finite checks support the uniform proofs; they are not their substitute. The known general parity theorem already gives the relevant reduction phenomenon. The new contribution here is the explicit solution of this specified coefficient module. This row-system theorem asserts no numerical period independence. The S_4 identity is proved separately in Theorem 4.8. Historical conjectural receipts are retained as provenance rather than silently rewritten.

5.6 Real orders, the finite signed-measure threshold and its atoms

The recursive integer-order density is part of a continuous family. Regularized Hurwitz kernels give the exact finite signed-measure domain, including its two atomic edges. The construction below supplies the supercritical part of the later all-positive-order zero theorem. Positive Hausdorff generating functions supply a classical context [98]; here the signed and hypersingular measures are constructed explicitly with their separate integrability conditions. Write

$$A_n(a,b) = \frac{H_{n-1}^{(b)}}{n^a}, \qquad F_{a,b}(z) = \sum_{n\geq 1} A_n(a,b)z^n \quad (|z|<1).$$

We permit $a = 0$ to state the sharp endpoint. Throughout $b > 0$; write $w = a + b$ and $c_n = A_n(a,b)$ in these real-order extensions.

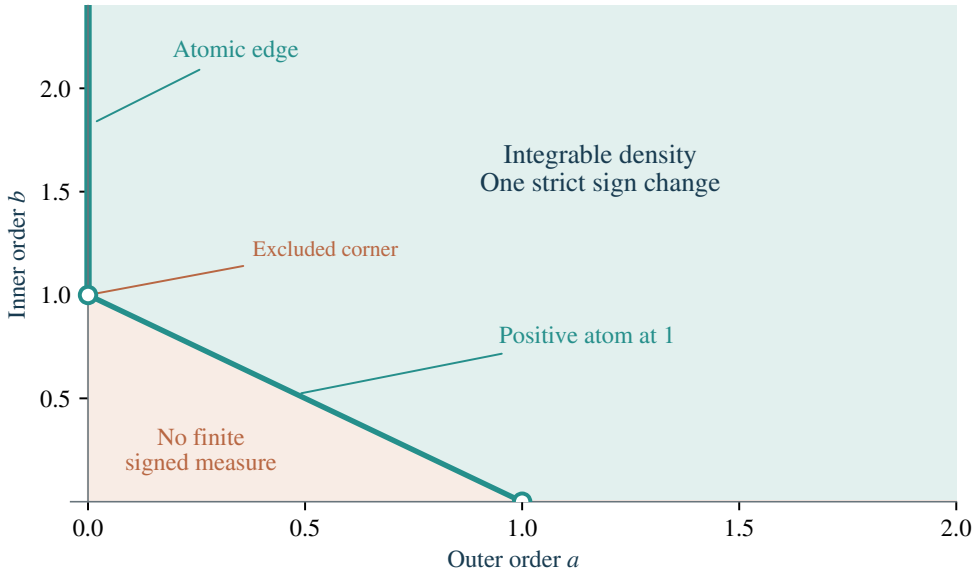


Figure 5.1: The exact finite signed-measure domain for $a \geq 0, b > 0$. The teal edges carry a positive atom at one. The point $(0, 1)$ is excluded, and the horizontal axis $b = 0$ is outside the stated domain. The entire positive quadrant has zero-free principal continuation by Theorem 5.26; finite signed-moment existence has the smaller domain shown here.

Theorem 5.17 (Exact finite-measure domain)**.** *For $a \geq 0$ and $b > 0$, a finite real signed measure $\nu_{a,b}$ on $[0, 1]$ satisfying*

$$\int_{[0,1]} u^{n-1}\, d\nu_{a,b}(u) = A_n(a, b) \qquad (n \geq 1) \tag{5.44}$$

exists if and only if

$$(a > 0 \text{ and } a + b \geq 1) \quad \text{or} \quad (a = 0 \text{ and } b > 1). \tag{5.45}$$

It is unique and has total mass zero. More precisely:

1. *If $a > 0$ and $a + b > 1$, it has an integrable density $\kappa_{a,b}(u)$ with exactly one strict sign change, from negative to positive as u increases, at a point $r_{a,b} \in (0, 1)$.*
2. *If $0 < b < 1$ and $a = 1 - b$, then $\nu_{a,b} = (1 - b)^{-1}\delta_1 + \kappa_{a,b}(u)\, du$, where $\kappa_{a,b}(u) < 0$ throughout $(0, 1)$.*
3. *If $a = 0$ and $b > 1$, then $\nu_{0,b} = \zeta(b)\delta_1 + \kappa_{0,b}(u)\, du$, where $\kappa_{0,b}(u) < 0$ throughout $(0, 1)$.*

In the two atomic cases all positive mass is at one and all negative mass is in $(0, 1)$. Outside (5.45), not even a finite signed measure with arbitrarily many sign changes can satisfy (5.44).

5.6.1 A regularized positive kernel

Set

$$q(t) = \frac{1}{1 - e^{-t}} - \frac{1}{t}, \qquad t > 0. \tag{5.46}$$

The elementary identities

$$q(t) = \frac{1}{2} + \frac{1}{2}\coth(t/2) - \frac{1}{t}, \qquad q'(t) = \frac{1}{t^2} - \frac{1}{4\sinh^2(t/2)} > 0$$

give $1/2 < q(t) < 1$, because $\sinh(t/2) > t/2$ and the endpoint limits are $1/2$ and 1. In particular q is positive, bounded and strictly increasing. Also $q(t) = 1/2 + O(t)$ as $t \downarrow 0$.

For later use put $k_v(T) = T^{v-1}/\Gamma(v)$ for $v > 0$. The positive convolution satisfies

$$C_{a,b}(T) = \frac{1}{\Gamma(a)\Gamma(b)} \int_0^T (T-t)^{a-1} t^{b-1} q(t)\, dt = k_{a+b}(T)\mathbb{E}[q(TV)], \quad V \sim \mathrm{Beta}(b, a), \tag{5.47}$$

$$\tfrac{1}{2} k_{a+b}(T) < C_{a,b}(T) < k_{a+b}(T), \tag{5.48}$$

$$\int_0^\infty e^{-nT} C_{a,b}(T)\, dT = n^{-a}\left(\zeta(b, n) - \frac{n^{1-b}}{b-1}\right) \quad (b \neq 1). \tag{5.49}$$

At $b = 1$ the last expression is $n^{-a}(\log n - \psi(n))$. These identities follow by absolutely convergent convolution integrals.

The classical Hurwitz integral and one elementary subtraction give

$$\zeta(b, x) = \frac{x^{1-b}}{b-1} + \frac{1}{\Gamma(b)} \int_0^\infty e^{-xt} t^{b-1} q(t)\, dt, \qquad x > 0,\ b > 0,\ b \neq 1. \tag{5.50}$$

For $b > 1$ this is obtained by splitting $1/(1 - e^{-t}) = 1/t + q(t)$ in the usual Gamma integral. The integral involving q is holomorphic for $\Re b > 0$, so the identity theorem extends the same formula to $0 < b < 1$. This is analytic continuation of an absolutely convergent regularized integral, not subtraction of two divergent real integrals. The standard starting integral and recurrence are DLMF 25.11.25 and 25.11.4 [93]. For $b = 1$ the corresponding formula is

$$\psi(x) = \log x - \int_0^\infty e^{-xt} q(t)\, dt, \tag{5.51}$$

which is DLMF 5.9.13 [95] and also follows by taking the finite part of (5.50) at $b = 1$.

For $0 < b < 1$, we shall use $\zeta(b) < 0$. Indeed the alternating Dirichlet series $\eta(b)$ is positive, while $\eta(b) = (1 - 2^{1-b})\zeta(b)$ and $1 - 2^{1-b} < 0$.

5.6.2 Explicit densities and their Laplace transforms

Put $T = -\log u$ and $K_{a,b}(T) = \kappa_{a,b}(e^{-T})$. For $a > 0$, $a + b > 1$ and $b \neq 1$, define

$$K_{a,b}(T) = \frac{\zeta(b) T^{a-1}}{\Gamma(a)} - \frac{T^{a+b-2}}{(b-1)\Gamma(a+b-1)} - \frac{1}{\Gamma(a)\Gamma(b)} \int_0^T (T-t)^{a-1} t^{b-1} q(t)\, dt. \tag{5.52}$$

When $b > 1$, an equivalent form is

$$K_{a,b}(T) = \frac{\zeta(b) T^{a-1}}{\Gamma(a)} - \frac{1}{\Gamma(a)\Gamma(b)} \int_0^T \frac{(T-t)^{a-1} t^{b-1}}{1 - e^{-t}}\, dt. \tag{5.53}$$

At $b = 1$, define

$$K_{a,1}(T) = \frac{T^{a-1}}{\Gamma(a)}\big(\gamma + \psi(a) - \log T\big) - \frac{1}{\Gamma(a)} \int_0^T (T-t)^{a-1} q(t)\, dt. \tag{5.54}$$

The Gamma factors and the term $\gamma + \psi(a)$ in this last formula are essential; in particular it reduces at $a = 1$ to $K_{1,1}(T) = -\log(e^T - 1)$, agreeing with the original signed kernel.

These functions are integrable against $e^{-T}\, dT$. At zero, (5.52) has only the integrable powers T^{a-1}, T^{a+b-2} and $O(T^{a+b-1})$; the logarithmic factor in (5.54) is likewise integrable for $a > 0$. At infinity all terms have at most polynomial times logarithmic growth. For $b > 1$, the integrability of the convolution in (5.53) at its lower endpoint uses $b > 1$ explicitly.

By the Gamma integral and the convolution rule, the Laplace transform of (5.52) is

$$\begin{aligned}\int_0^\infty e^{-nT} K_{a,b}(T)\, dT &= n^{-a}\left[\zeta(b) - \frac{n^{1-b}}{b-1} - \frac{1}{\Gamma(b)} \int_0^\infty e^{-nt} t^{b-1} q(t)\, dt\right] \\ &= n^{-a}\big(\zeta(b) - \zeta(b,n)\big) = \frac{H_{n-1}^{(b)}}{n^a}.\end{aligned}$$

Each convolution interchange is justified absolutely; q is bounded and the remaining factors have Gamma integrals. Formula (5.54) has Laplace transform

$$n^{-a}\left(\gamma + \log n - \int_0^\infty e^{-nt} q(t)\, dt\right) = n^{-a}(\gamma + \psi(n)) = \frac{H_{n-1}}{n^a}.$$

Here the Laplace transform of $T^{a-1}\log T$ is obtained by differentiating the Gamma integral, justified by absolute integrability. The substitution $u = e^{-T}$ changes $u^{n-1}\, du$ to $e^{-nT}\, dT$, proving the desired moment normalization. In particular $n = 1$ gives total mass zero.

5.6.3 The one-sign-change proof

For $b > 1$, divide (5.53) by T^{a-1} and rescale $t = Tv$:

$$T^{1-a} K_{a,b}(T) = \frac{\zeta(b)}{\Gamma(a)} - \frac{1}{\Gamma(a)\Gamma(b)} \int_0^1 (1-v)^{a-1} v^{b-1} \frac{T^b}{1 - e^{-Tv}}\, dv.$$

For fixed $v > 0$, the logarithmic derivative of $T^b/(1 - e^{-Tv})$ is

$$\frac{b}{T} - \frac{v}{e^{Tv} - 1} > \frac{b-1}{T} > 0.$$

Thus the rescaled density is strictly decreasing. Its limit at zero is $\zeta(b)/\Gamma(a) > 0$, and its limit at infinity is $-\infty$. This proves one strict positive–negative crossing in the T coordinate.

For $b = 1$, the rescaled expression is

$$\Gamma(a) T^{1-a} K_{a,1}(T) = \gamma + \psi(a) - \log T - T \int_0^1 (1-v)^{a-1} q(Tv)\, dv.$$

It is strictly decreasing from $+\infty$ to $-\infty$, since q is positive and increasing. For $0 < b < 1$, instead divide by T^{a+b-2} to obtain

$$T^{2-a-b}K_{a,b}(T) = \frac{1}{(1-b)\Gamma(a+b-1)} + \frac{\zeta(b)}{\Gamma(a)}T^{1-b} - \frac{T}{\Gamma(a)\Gamma(b)}\int_0^1 (1-v)^{a-1}v^{b-1}q(Tv)\,dv. \tag{5.55}$$

Now the first term is positive, the second is strictly decreasing because $\zeta(b) < 0$, and the final term is strictly decreasing because q is positive and increasing. The endpoint limits are a positive constant and $-\infty$. Thus there is again exactly one strict crossing. The derivatives of these rescaled expressions are strictly negative; differentiation is dominated by the same integrable beta weights. At a zero of K, multiplication by the positive rescaling factor does not affect the sign of its derivative. Reversing $T = -\log u$ gives the asserted negative–positive sign change and a strictly positive u derivative at its unique crossing.

5.6.4 The critical atom and the exact boundary

Let $0 < b < 1$ and $a = 1 - b$. Replace the middle term of (5.52) by an atom of mass $(1-b)^{-1}$ at $u = 1$, and retain the density

$$K_{1-b,b}(T) = \frac{\zeta(b)T^{-b}}{\Gamma(1-b)} - \frac{1}{\Gamma(1-b)\Gamma(b)}\int_0^T (T-t)^{-b}t^{b-1}q(t)\,dt. \tag{5.56}$$

Both terms are strictly negative and integrable against $e^{-T}\,dT$. Its moments plus the atom are

$$\frac{1}{1-b} + n^{b-1}\left[\zeta(b) - \frac{1}{\Gamma(b)}\int_0^\infty e^{-nt}t^{b-1}q(t)\,dt\right] = \frac{H_{n-1}^{(b)}}{n^{1-b}},$$

by (5.50). Thus the measure has total mass zero, and its negative mass is exactly $(1-b)^{-1}$.

For $a = 0$, $b > 1$, the formula is especially simple:

$$\nu_{0,b} = \zeta(b)\delta_1 + K_{0,b}(-\log u)\,du, \qquad K_{0,b}(T) = -\frac{T^{b-1}}{\Gamma(b)(1-e^{-T})}. \tag{5.57}$$

The ordinary Hurwitz integral gives its moments immediately. Its negative mass is exactly $\zeta(b)$.

Every finite signed measure has bounded moments, since $|\int u^{n-1}\,d\nu| \le \|\nu\|_{\mathrm{TV}}$. If $0 < b < 1$, elementary integral comparison gives $H_{n-1}^{(b)} \sim n^{1-b}/(1-b)$, so the moments diverge to infinity when $a + b < 1$. If $a = 0, b = 1$, they diverge logarithmically. These are exactly the excluded cases for $a \ge 0, b > 0$. Finally, two finite signed measures with the same moments integrate every polynomial equally; polynomial density in $C([0,1])$ makes the measures identical. This proves Theorem 5.17.

5.7 Consequences for complex zeros and angular zeros

Theorem 5.18 (Nonvanishing throughout the finite-measure domain)**.** *For all parameters in* (5.45)*, $F_{a,b}$ has an analytic continuation to $\mathbb{C} \setminus [1,\infty)$ and has no zero there except for its double zero at the origin. Every upper open semicircle $z = \rho e^{i\theta}$, $0 < \rho \le 1$, $0 < \theta < \pi$, has exactly one simple zero of $\operatorname{Im} F_{a,b}(z)$, and it satisfies*

$$\arccos\rho < \theta_{a,b}(\rho) < \frac{\pi}{2}.$$

The imaginary part is positive below this angle and negative above it; its angular derivative at the zero is strictly negative. In the atomic cases, unit-circle values mean analytic continuation or Abel boundary values: the original Fourier summands do not tend to zero, so ordinary Fourier convergence is not asserted.

Proof. The signed transform

$$F_{a,b}(z) = z \int_{[0,1]} \frac{d\nu_{a,b}(u)}{1 - zu}$$

follows by geometric expansion in the disk and extends analytically to the slit plane by uniform domination on compact sets. Let r be the crossing from Theorem 5.17, or let $r = 1$ in the atomic cases. Zero mass gives

$$F_{a,b}(z) = \frac{z^2}{1 - rz} P_{a,b}(z), \qquad P_{a,b}(z) = \int_{[0,1]} \frac{u - r}{1 - zu}\, d\nu_{a,b}(u). \tag{5.58}$$

The measure $(u - r)\, d\nu_{a,b}(u)$ is positive and nonzero, including when $r = 1$: its atom disappears but its density is strictly positive on $(0, 1)$. Consequently $\operatorname{Im} P_{a,b}(z)$ has the strict sign of $\operatorname{Im} z$, while $P_{a,b}(x) > 0$ for $x < 1$. Apart from the explicit z^2, the factor $(1 - rz)^{-1}$ in (5.58) is analytic and nonzero on the slit plane. Since $P_{a,b}(0) = A_2 = 2^{-a}$, the zero at zero has multiplicity exactly two.

For angular zeros put

$$J_\rho(c) = \int_{[0,1]} \frac{d\nu(u)}{1 - 2\rho c u + \rho^2 u^2}, \qquad \operatorname{Im} F(\rho e^{i\theta}) = \rho \sin\theta\, J_\rho(\cos\theta).$$

The zero-mass one-crossing sign test holds for our atomic measures as well: subtract the value of the test function at r, and use the ordered supports of the two signs. It gives $J_\rho(0) < 0$. When $\rho < 1$, the kernel at $c = \rho$ is strictly increasing in u, so $J_\rho(\rho) > 0$. When $\rho = 1$ and the crossing is interior, separate the two signs before expanding $(1 - u)^{-2}$ to obtain

$$\lim_{c \uparrow 1} J_1(c) = \int_{[0,1]} \frac{d\nu(u)}{(1 - u)^2} = \sum_{n \geq 1} n A_n > 0,$$

allowing $+\infty$. The negative part is supported away from one, so this is the same justified monotone-limit argument as in the original signed-kernel theorem. In the atomic cases the limiting signed integral at $c = 1$ need not be defined, and we do not use that display. Instead the limit is $+\infty$ directly: the atom contributes $M/[2(1 - c)]$, and the negative density contributes $o((1 - c)^{-1})$. For the latter assertion, multiply its integral by $1 - c$ and use dominated convergence: the multiplier $(1 - c)/(1 - 2cu + u^2)$ is bounded by 1 for $0 \leq c < 1$ and tends to zero for every $u < 1$. The negative density is integrable and has no atom at one.

For $c' > c$, the quotient of the two kernels has derivative

$$\frac{d}{du} \frac{1 - 2\rho c u + \rho^2 u^2}{1 - 2\rho c' u + \rho^2 u^2} = \frac{2\rho(c' - c)(1 - \rho^2 u^2)}{(1 - 2\rho c' u + \rho^2 u^2)^2} > 0 \quad (0 < u < 1).$$

At any zero, apply the sign test to the signed measure obtained by weighting ν with the positive kernel. It follows that J_ρ is positive at every larger c and negative at every smaller c; this proves uniqueness and the signs. Finally its derivative at its zero is positive, because the additional factor $2\rho u/(1 - 2\rho c u + \rho^2 u^2)$ is strictly increasing in u. The angular derivative is therefore $-\rho \sin^2\theta\, J'_\rho(\cos\theta) < 0$. □

Corollary 5.19 (Ordinary and analytic boundary values)**.** *For $a, b > 0$ with $a + b > 1$, the defining series converges ordinarily at every $|z| = 1, z \neq 1$, and equals its analytic continuation. At $a + b \leq 1$, or on the axis $a = 0, b > 0$, its terms do not tend to zero at those boundary points. There the stated values are analytic or Abel values.*

Proof. In the absolutely continuous finite-measure case, expand the finite geometric sum and use domination of $z(1 - (zu)^N)/(1 - zu)$ against the total variation; there is no atom at one. Harmonic asymptotics give a positive nonzero limit or unbounded coefficients in the remaining cases, so the term test fails. The axis formula and the subsequent Stieltjes factorization still give well-defined continuation and radial Abel limits away from the cut. □

5.8 A positive two-variable representation at all positive orders

The threshold in Theorem 5.17 concerns a *one-variable finite signed measure*. It is not a threshold for analytic continuation or every positivity property.

Theorem 5.20 (Positive double kernel)**.** *For every $a, b > 0$ and $z \in \mathbb{C} \setminus [1, \infty)$,*

$$F_{a,b}(z) = z^2 \int_{0<y<x<1} \frac{W_{a,b}(x, y)}{(1 - xz)(1 - yz)} \, \mathrm{d}y \, \mathrm{d}x, \tag{5.59}$$

where

$$W_{a,b}(x, y) = \frac{(-\log x)^{a-1} (\log(x/y))^{b-1}}{\Gamma(a)\Gamma(b)}. \tag{5.60}$$

The weight is strictly positive and has total mass 2^{-a}.

Proof. For $|z| < 1$, expand the two denominators. Substitution $y = xv$ gives, for nonnegative integers j, k,

$$\int_{0<y<x<1} W_{a,b}(x, y) x^j y^k \, \mathrm{d}y \, \mathrm{d}x = \frac{1}{(j + k + 2)^a (k + 1)^b}.$$

Summing over $j + k = n - 2$ yields c_n. Absolute convergence justifies the rearrangement, and taking $j = k = 0$ gives the total mass. On each compact subset of $\mathbb{C} \setminus [1, \infty)$, both denominators are bounded away from zero uniformly in $x, y \in [0, 1]$. The integral is consequently holomorphic there and supplies the stated continuation. □

Corollary 5.21 (Disk nonvanishing below and above the threshold)**.** *For all $a, b > 0$, $F_{a,b}$ has no nonzero zeros in $\{z : |z| \leq 1,\ z \neq 1\}$. More precisely, when $\operatorname{Im} z > 0$ and $|z| \leq 1$,*

$$\operatorname{Im} \frac{F_{a,b}(z)}{z} > 0. \tag{5.61}$$

At any upper-half-disk zero of $\operatorname{Im} F_{a,b}(z)$, the value $F_{a,b}(z)$ itself is strictly negative real.

Proof. A direct calculation gives

$$\operatorname{Im} \frac{z}{(1 - xz)(1 - yz)} = \frac{(\operatorname{Im} z)(1 - xy|z|^2)}{|1 - xz|^2 |1 - yz|^2} > 0$$

for $0 < y < x < 1$, proving (5.61). The lower half follows by conjugation. On the real interval $z < 1$, the integral $F(z)/z^2$ is strictly positive. Finally, if $F(z) = R \in \mathbb{R}$ with $\operatorname{Im} z > 0$, then $\operatorname{Im}(R/z) = -R \operatorname{Im} z/|z|^2 > 0$, so $R < 0$. □

Corollary 5.22 (Gaussian negativity and strict log-convexity)**.** *For every $a, b > 0$,*

$$-g_{a,b} = \int_{0<y<x<1} W_{a,b}(x,y) \frac{x+y}{(1+x^2)(1+y^2)} \, \mathrm{d}y \, \mathrm{d}x > 0, \tag{5.62}$$

and

$$\frac{H_2^{(b)}}{4\,3^a} < -g_{a,b} < \frac{H_2^{(b)}}{3^a}. \tag{5.63}$$

The function $(a,b) \mapsto \Gamma(a)\Gamma(b)(-g_{a,b})$ is strictly log-convex on the positive quadrant.

Proof. Set $z = \mathrm{i}$ in (5.59) and rationalize the denominator. Since $1 < (1+x^2)(1+y^2) < 4$ and $\int W(x+y) = c_3 = H_2^{(b)}/3^a$, the bounds follow. For log-convexity apply Hölder's inequality to the positive integral with the gamma factors removed. Its parameter dependence is the exponential of $(a-1)\log(-\log x) + (b-1)\log\log(x/y)$. Equality in Hölder would require a nontrivial real linear combination of these two logarithms to be constant almost everywhere on the open triangle, which is impossible. This proves strictness for every nontrivial line segment of parameter pairs. □

5.8.1 All-order integral identities

Theorem 5.23 (Hurwitz–polylogarithm transport)**.** *For $a, b > 0$, $b \neq 1$, and $z \in \mathbb{C} \setminus [1, \infty)$,*

$$F_{a,b}(z) = \zeta(b)\,\mathrm{Li}_a(z) - \frac{\mathrm{Li}_{a+b-1}(z)}{b-1} - \frac{1}{\Gamma(b)} \int_0^\infty t^{b-1} q(t)\,\mathrm{Li}_a(ze^{-t})\,\mathrm{d}t. \tag{5.64}$$

At $b = 1$ the pole-free identity is

$$F_{a,1}(z) = \gamma\,\mathrm{Li}_a(z) - \partial_a\,\mathrm{Li}_a(z) - \int_0^\infty q(t)\,\mathrm{Li}_a(ze^{-t})\,\mathrm{d}t. \tag{5.65}$$

In particular, the index $a+b-1$ in (5.64) is allowed to be zero or negative.

Proof. Inside the disk, multiply (5.50), with $x = n$, by $n^{-a}z^n$ and sum. Exponential decay at infinity, boundedness of q, and $b > 0$ justify termwise integration. The Hurwitz difference is $H_{n-1}^{(b)}$, so coefficient comparison gives (5.64). At $b = 1$, use $H_{n-1} = \gamma + \log n - \int_0^\infty e^{-nt} q(t)\,\mathrm{d}t$; the sum of $(\log n) n^{-a} z^n$ is $-\partial_a\,\mathrm{Li}_a(z)$. The integral expressions on the right are locally holomorphic on the slit plane. Near $t = 0$, $\mathrm{Li}_a(ze^{-t})$ stays bounded on compact slit sets, and at infinity it is $O(e^{-t})$. Analytic continuation therefore proves the identities throughout the claimed domain. Standard conventions for Li_s are those of DLMF 25.12 [96]. □

These identities are useful even when no signed Hausdorff measure exists. They are transformations to a depth-one polylogarithm integral, not assertions that a depth-two constant has been reduced to a finite basis of depth-one periods.

5.9 Below the measure threshold: positive differences settle the zero question

The failure of finite signed-moment representability below $a + b = 1$ does not imply a failure of positivity after a discrete derivative. Consecutive differences supply a finite positive measure and

resolve the slit-plane nonvanishing and angular-uniqueness questions left open by the real-order reports. They also extend the right-endpoint radial mechanism from the critical line to the entire subcritical region.

Write $c_n = H_{n-1}^{(b)}/n^a$ and $F_{a,b}(z) = \sum_{n\ge1} c_n z^n$. For $a > 0$, $b > 0$, $w = a + b < 1$, put $s = w - 1 \in (-1, 0)$ and

$$k_v(T) = T^{v-1}/\Gamma(v), \tag{5.66}$$

$$\mathfrak{h}(t) = (1 - e^{-t})^{-1} - t^{-1}, \tag{5.67}$$

$$\mathcal{C}_{a,b}(T) = \frac{1}{\Gamma(a)\Gamma(b)} \int_0^T (T - t)^{a-1} t^{b-1} \mathfrak{h}(t)\, dt, \tag{5.68}$$

$$R_{a,b}(T) = -\zeta(b)k_a(T) - \frac{k_s(T)}{1 - b} + \mathcal{C}_{a,b}(T). \tag{5.69}$$

Here k_s is a signed function, not a positive Gamma density: $\Gamma(s) = \Gamma(s+1)/s < 0$. Since $0 < b < 1$, $\zeta(b) < 0$, and $1/2 < \mathfrak{h}(t) < 1$, all three terms of $R_{a,b}$ are strictly positive. The regularized Hurwitz identity used here is

$$\zeta(b, n) = \frac{n^{1-b}}{b - 1} + \frac{1}{\Gamma(b)} \int_0^\infty e^{-nt} t^{b-1} \mathfrak{h}(t)\, dt;$$

it follows from the convergent Hurwitz integral by subtraction and analytic continuation in b, as in the preceding construction [93].

Theorem 5.24 (Positive Hausdorff differences)**.** *For $a, b > 0$ with $a + b < 1$, define*

$$d\omega_{a,b}(u) = (1 - u)R_{a,b}(-\log u)\, du, \qquad 0 < u < 1.$$

This is a finite positive measure, with strictly positive density and mass 2^{-a}. For every $n \ge 1$,

$$c_n = \int_0^1 (1 - u^{n-1})R_{a,b}(-\log u)\, du, \tag{5.70}$$

$$c_{n+1} - c_n = \int_0^1 u^{n-1}\, d\omega_{a,b}(u). \tag{5.71}$$

The first measure has infinite mass, but every displayed integral is absolutely convergent. Thus the shifted sequence c_{n+1} is a discrete Bernstein sequence, and its differences are strictly completely monotone. Here $\Delta f_n = f_{n+1} - f_n$. In particular, for all $j \ge 0$ and $n \ge 1$,

$$(-1)^j \Delta^j (c_{n+1} - c_n) = \int_0^1 u^{n-1}(1 - u)^j\, d\omega_{a,b}(u) > 0.$$

The positive difference measure is uniquely determined by these moments.

Proof. At $T = 0$, the leading term of R is a positive constant times T^{w-2}; the other terms have orders T^{a-1} and T^{w-1}. Multiplication by $1 - e^{-T}$ therefore makes it integrable, since $w > 0$. At infinity all terms grow at most polynomially and the change of variables $u = e^{-T}$ supplies e^{-T}. This proves finiteness of ω, convergence of (5.70), and divergence of the unweighted mass at $u = 1$.

For $s \in (-1, 0)$, integration by parts gives the convergent identity

$$\int_0^\infty (e^{-T} - e^{-nT})k_s(T)\, dT = 1 - n^{-s}.$$

Both boundary terms vanish: near zero their order is T^{s+1}, and infinity has exponential decay. For k_a the same identity is the ordinary Gamma integral. The convolution has Laplace transform

$$\int_0^\infty e^{-nT}\mathcal{C}_{a,b}(T)\,dT = n^{-a}\left(\zeta(b,n)+\frac{n^{1-b}}{1-b}\right).$$

Every convolution interchange is absolute. Insert these three identities into (5.69), with the factor $e^{-T}-e^{-nT}$ retained. The constant and power terms cancel, leaving $n^{-a}(\zeta(b)-\zeta(b,n))=c_n$. No divergent Laplace transforms are subtracted. Subtract the two convergent formulas at $n+1$ and n to obtain (5.71). At $n=1$ this gives mass $c_2=2^{-a}$. The difference formula proves every strict sign, and polynomial density on $[0,1]$ proves uniqueness of the finite measure. □

On the critical line $a=1-b>0$, use instead $R_{a,b}=-\zeta(b)k_a+\mathcal{C}_{a,b}$. Its finite mass is $1/a$, and its Laplace transform is $1/a-c_n$, so the same formulas for $\omega=(1-u)R(-\log u)\,du$ hold. This is precisely the negative density of the critical signed measure; its positive atom at one disappears after multiplication by $1-u$. On the whole axis $a=0, b>0$, the difference measure is simply

$$d\omega_{0,b}(u)=\frac{(-\log u)^{b-1}}{\Gamma(b)}\,du, \qquad c_{n+1}-c_n=n^{-b},$$

and $F_{0,b}(z)=z\,\mathrm{Li}_b(z)/(1-z)$. These measures also have strictly positive densities, with mass 2^{-a} and no endpoint atoms.

Corollary 5.25 (A sharp complementary positivity classification)**.** *For $a\ge 0, b>0$, the consecutive differences $c_{n+1}-c_n$ are moments of a finite positive measure on $[0,1]$ if and only if $a=0$ or $a+b\le 1$. In the same parameter region the function*

$$\Phi_{a,b}(t)=(t+1)^{-a}\{\zeta(b)-\zeta(b,t+1)\}, \qquad t\ge 0,$$

with $\gamma+\psi(t+1)$ replacing the braces at $b=1$, is a Bernstein function: $\Phi(0)=0$, $\Phi(t)>0$ for $t>0$, and $(-1)^j\Phi^{(j+1)}(t)>0$ for all $j\ge 0$.

Proof. The preceding proof works for every real $n\ge 1$, since the Gamma and regularized Hurwitz identities used there do not require integer n. Thus $\Phi(t)=\int(1-u^t)\,d\rho(u)$ with $d\rho=d\omega/(1-u)$. Differentiation gives

$$(-1)^j\Phi^{(j+1)}(t)=\int_0^1(-\log u)^{j+1}u^t\,d\rho(u)>0.$$

All these integrals converge even at $t=0$: the endpoint at one is controlled by $\int(1-u)\,d\rho<\infty$, and the logarithmic-coordinate density at zero has exponential decay times polynomial growth. The same representation follows on the axis from the ordinary Hurwitz integral, or its convergent difference at $b\le 1$.

Conversely, for $a>0, a+b>1$, elementary harmonic asymptotics give $c_n\to 0$ while $c_2=2^{-a}>0$. The sequence cannot be nondecreasing, so its differences cannot all be positive-measure moments and its continuous interpolation cannot be a Bernstein function. This proves both sharp parameter statements. □

Theorem 5.26 (Global nonvanishing at all positive inner orders)**.** *For every $a\ge 0$, $b>0$, the principal continuation of $F_{a,b}$ has no zero on $\mathbb{C}\setminus[1,\infty)$ except its double zero at zero. For $a>0$, $a+b\le 1$, and also for $a=0, b>0$, the stronger factorization is*

$$F_{a,b}(z)=\frac{z^2}{1-z}P_{a,b}(z), \qquad P_{a,b}(z)=\int_0^1\frac{d\omega_{a,b}(u)}{1-zu}. \tag{5.72}$$

In particular $P(0)=2^{-a}$, $\operatorname{Im}P(z)$ has the strict sign of $\operatorname{Im}z$, and $P(x)>0$ for every real $x<1$.

Proof. The coefficient of z^m in $(1-z)F(z)/z^2$ is $c_{m+2}-c_{m+1}$, including $m=0$ because $c_1=0$. The difference moments give (5.72) in the disk. The finite positive measure gives holomorphic continuation to the slit plane, where its imaginary-part and real-axis signs prove nonvanishing. The mass proves multiplicity exactly two. For $a>0, a+b>1$, the one-crossing finite signed measure already gives $F=z^2P_r/(1-rz)$ with a nonzero positive Stieltjes transform and $0<r<1$. The same argument applies. This proves the complete parameter statement while retaining the sharp boundary for finite signed moments. □

5.9.1 Angular uniqueness and a full subcritical radial theorem

Theorem 5.27 (Right-endpoint Stieltjes geometry)**.** *For $a,b>0$ with $a+b\le 1$, and for $a=0, b>0$, every open upper semicircle of radius $0<\rho\le 1$ has exactly one simple zero of $\operatorname{Im} F_{a,b}$. Its angle satisfies*

$$\arccos\rho<\theta_{a,b}(\rho)<\arccos(\rho/2), \qquad \tfrac{1}{2}<\eta_{a,b}(\rho)=\cos\theta_{a,b}(\rho)/\rho<1.$$

The imaginary part is positive before that angle and negative after it. The normalized quantity $\eta_{a,b}$ is strictly increasing in ρ. Together with the supercritical angular theorem, this proves angular uniqueness for every $a\ge 0, b>0$, including the formerly conjectural subcritical region. Boundary values are analytic or Abel values where the ordinary Fourier series does not converge.

Proof. The following argument works for any finite positive measure ω with a strictly positive density on $(0,1)$. Set $x=\rho^2$, $m=2\cos\theta/\rho-1$, and

$$D(u)=1+x\{u^2-(1+m)u\}, \qquad G(m,x)=\int_0^1 \frac{m-u}{D(u)}\,d\omega(u).$$

From (5.72), direct rationalization gives

$$\operatorname{Im} F(\rho e^{i\theta})=\frac{\rho^3\sin\theta}{|1-\rho e^{i\theta}|^2}G(m,x).$$

For $0<m<1$ all denominators are positive, and

$$G_m=\int_0^1 \frac{1-xu}{D(u)^2}\,d\omega(u)>0.$$

We have $G(0,x)<0$. For $x<1$, $G(1,x)>0$. At $x=1$, its negative part for $u>m\ge 1/2$ is bounded in magnitude by $2\omega((m,1))$, which tends to zero. Its positive part tends pointwise to $d\omega(u)/(1-u)$ and has a strictly positive limit inferior by Fatou's lemma, with infinity allowed. Hence $G(m,1)>0$ for some $m<1$. This proves a unique root in $(0,1)$ even when $\omega/(1-u)$ has infinite mass. Outside that interval the numerator has a fixed sign, proving uniqueness on the full upper arc. At the zero, the angular derivative is $-2\rho^2\sin^2\theta\,G_m/|1-\rho e^{i\theta}|^2<0$.

At a root, the probability measure proportional to $d\omega/D$ has mean m. For $d(u)=u^2-(1+m)u$, the identities $d(m)=-m$ and $d(u)-d(m)=(u-m)(u-1)$ give

$$G_x=-\frac{1}{1-xm}\int_0^1 \frac{(u-m)^2(1-u)}{D(u)^2}\,d\omega(u)<0.$$

Consequently

$$\frac{dm}{dx}=\frac{\displaystyle\int (u-m)^2(1-u)D(u)^{-2}\,d\omega(u)}{\displaystyle(1-xm)\int (1-xu)D(u)^{-2}\,d\omega(u)}>0, \qquad \eta'(\rho)=\rho m'(\rho^2)>0.$$

All differentiations are dominated locally at the interior root, including at $x = 1$ since $m < 1$. The implicit-function theorem proves smooth dependence. This argument uses only the finite measure ω; it does not apply a finite-measure limit theorem to its infinite unweighted predecessor. □

5.9.2 One-sided Euler certificates even for divergent boundary series

Theorem 5.28 (Subcritical Euler continuation and a sharp fixed-pair constant)**.** *For the parameters of Theorem 5.27, put* $A_k = c_{2k+1}$, $g_{a,b} = \operatorname{Im} F_{a,b}(i)$ *and*

$$E_N = \sum_{j=0}^{N-1} 2^{-j-1} \sum_{k=0}^{j} (-1)^k \binom{j}{k} A_k.$$

Then E_N *decreases strictly to the analytic Gaussian value, and*

$$-g_{a,b} = \frac{1}{2} \int_0^1 \frac{1+u}{1+u^2}\, d\omega(u), \tag{5.73}$$

$$E_N - g_{a,b} = 2^{-N} \int_0^1 \frac{(1+u)(1-u^2)^{N-1}}{1+u^2}\, d\omega(u) > 0. \tag{5.74}$$

The least constant in $E_N - g \le C2^{-N}$ *for all* $N \ge 1$ *is* $C = -2g$, *and*

$$2^{-a} < -2g_{a,b} < \frac{1+\sqrt{2}}{2}\, 2^{-a} < \frac{5}{4}.$$

Thus $E_N - (5/4)2^{-N} < g_{a,b} < E_N$ *is a rational enclosure throughout this domain. It concerns analytic continuation when the ordinary alternating series diverges.*

Proof. For $j \ge 1$, the finite binomial difference of A_k is

$$-\int_0^1 (1+u)(1-u^2)^{j-1}\, d\omega(u),$$

while the order-zero difference is $A_0 = 0$. Sum the geometric tail against the finite measure to get (5.74). Evaluating (5.72) at i gives the same limiting value. The scaled errors decrease strictly to zero, and their largest value is $2(E_1 - g) = -2g$. Finally $(1+u)/(1+u^2)$ has minimum one at the endpoints and maximum $(1+\sqrt{2})/2$ at $u = \sqrt{2} - 1$. The positive density makes both bounds strict, and the measure has mass 2^{-a}. Since $\sqrt{2} < 3/2$, the rational bound follows. □

The global integer-outer-order normalized-radius conjecture remains open. The new radial theorem covers the full subcritical and critical region and the zero outer-order axis; it does not assert the same radial direction in the supercritical region, where fractional turning points are possible. Finite signed-moment existence, analytic nonvanishing, angular uniqueness and radial motion now have explicitly different scopes.

Corollary 5.29 (An exact Gaussian identity on the outer-order axis)**.** *For every* $b > 0$, *with analytic or Abel interpretation when necessary,*

$$g_{0,b} = -\frac{1}{2}\left(\beta(b) + 2^{-b}\eta(b)\right), \qquad \eta(b) = \sum_{n\ge1} \frac{(-1)^{n-1}}{n^b}.$$

In particular $g_{0,1} = -\pi/8 - \log 2/4$ *and* $g_{0,2} = -G/2 - \pi^2/96$.

Proof. The defining residue classes give $\mathrm{Li}_b(i) = -2^{-b}\eta(b) + i\beta(b)$. Since $F_{0,b}(i) = i\,\mathrm{Li}_b(i)/(1-i) = (-1+i)\,\mathrm{Li}_b(i)/2$, its imaginary part is the claimed expression. Both Dirichlet series converge for $b > 0$; the depth-two series retains its stated boundary interpretation. These formulas also identify the sharp fixed-pair Euler constant as $\beta(b) + 2^{-b}\eta(b)$ on this axis. □

Theorem 5.30 (Strict Gaussian decrease in the real outer order)**.** *For every fixed $b > 0$, the positive magnitude $-g_{a,b}$ decreases strictly as a runs through $[0,\infty)$. In particular, for $a > 0$,*

$$0 < -2g_{a,b} < \beta(b) + 2^{-b}\eta(b) < \frac{1+\sqrt{2}}{2}.$$

The supremum over nonnegative outer orders is the explicit axis value. This theorem is a magnitude comparison. The separate contraction proof in Section 5.20 establishes the Euler remainder interpretation of the same axis expression throughout the parameter domain.

Proof. The outer Gamma integral, first by coefficient comparison in the disk and then by locally dominated continuation to the slit plane, is

$$F_{a,b}(z) = \frac{1}{\Gamma(a)} \int_0^\infty t^{a-1} F_{0,b}(ze^{-t})\, dt \qquad (a > 0).$$

Near zero its boundary value at $z = i$ is bounded; at infinity the integrand without t^{a-1} is $O(e^{-2t})$. Let $h_b(t) = e^t[-\operatorname{Im} F_{0,b}(ie^{-t})]$. The positive axis measure gives, with $\rho = e^{-t}$,

$$h_b(t) = \int_0^1 \frac{(1+u)\rho^2}{(1+\rho^2)(1+u^2\rho^2)}\, d\omega_{0,b}(u).$$

The derivative in ρ of the displayed kernel is

$$\frac{2\rho(1+u)(1-u^2\rho^4)}{(1+\rho^2)^2(1+u^2\rho^2)^2} > 0 \qquad (0 < \rho \le 1,\ 0 < u < 1).$$

Thus h_b is positive, bounded and strictly decreasing in t. The Gamma integral is equivalently $-g_{a,b} = \mathbb{E}[h_b(T_a)]$, where $T_a \sim \mathrm{Gamma}(a,1)$. Independent Gamma variables satisfy $T_{a+d} \overset{d}{=} T_a + T_d$; strict decrease of h_b proves strict decrease of the expectation. At $a = 0$ take $T_0 = 0$, and bounded continuity gives the same strict comparison and the limit as $a \downarrow 0$. For $a > 0$, differentiation also gives the exact negative covariance $\partial_a(-g_{a,b}) = \mathrm{Cov}(\log T_a, h_b(T_a)) < 0$. The axis identity and its positive-measure bound complete the proof. □

Theorem 5.31 (Gaussian monotonicity at every fixed total order)**.** *For every $w > 0$, $a \mapsto -2g_{a,w-a}$ decreases strictly on $(0,w)$. Its endpoint limits give the exact strict inequalities*

$$2\{\beta(w) - \beta(w-1)\} < -2g_{a,w-a} < \beta(w) + 2^{-w}\eta(w).$$

Here $\beta(w-1)$ uses analytic continuation when necessary. In particular, this proves the previously conjectural strict decrease on the entire critical line $w = 1$.

Proof. In the positive double kernel put $x = e^{-s}$, $y = e^{-(s+t)}$, then $T = s + t$ and $R = s/T$. Its Gaussian value becomes

$$-g_{a,b} = \mathbb{E}[K(R,T)], \qquad K(R,T) = \frac{X(X+Y)}{(1+X^2)(1+Y^2)}, \quad X = e^{-RT}, \quad Y = e^{-T},$$

where $T \sim \mathrm{Gamma}(w,1)$ and $R \sim \mathrm{Beta}(a, w-a)$ are independent. The Jacobian gives $e^{-2s-t} = e^{-T}X$; the remaining power is T^{w-1}, so every normalization is explicit. For fixed $T > 0$ the derivative with respect to X is

$$\frac{2X + Y - YX^2}{(1+X^2)^2(1+Y^2)} > 0.$$

Thus K decreases strictly in R. The beta score along fixed w is an increasing function of R, namely $\log(R/(1-R))$ minus its mean. Differentiation therefore gives a strictly negative covariance. All logarithmic beta moments are finite for $0 < a < w$. At $a \downarrow 0$, $R \to 0$ in probability; at $a \uparrow w$, $R \to 1$. Bounded convergence gives the two endpoint values. The first is the axis formula. The second also follows from $F_{w,0}(z) = \mathrm{Li}_{w-1}(z) - \mathrm{Li}_w(z)$ and its analytic value at i. □

5.9.3 A unique global Gaussian maximum and the sharp critical Euler constant

Put $C(b) = \beta(b) + 2^{-b}\eta(b)$ and $\phi(T) = (1+e^{-T})/(1+e^{-2T})$. The axis measure gives $C(b) = \mathbb{E}[\phi(T_b)]$ for $T_b \sim \mathrm{Gamma}(b,1)$. The function ϕ increases and then decreases, with its unique maximum at $T = \log(1+\sqrt{2})$.

Theorem 5.32 (A unique, nondegenerate axis maximum)**.** *The function C has exactly one stationary point b_* on $(0,\infty)$. It is a nondegenerate global maximum; $C' > 0$ below b_* and $C' < 0$ above it. Moreover,*

$$1 < b_* < 2, \qquad \frac{227}{200} < C(b_*) < \frac{1+\sqrt{2}}{2}.$$

The supremum of $-2g_{a,b}$ over $a \geq 0, b > 0$ is this maximum, attained only at $(a,b) = (0, b_*)$. Over $a, b > 0$ it is a supremum approached at the same axis point.

Proof. The gamma law tends to zero in probability as $b \downarrow 0$ and to infinity in probability as $b \to \infty$. Hence $C(b) \to 1$ at both ends. For every finite $b > 0$, $1 < C(b) < \max \phi$, since the density is positive throughout $(0,\infty)$. Thus an interior maximum exists.

We prove a strict two-intersection property. For $1 < \lambda < \max \phi$, the function $f(T) = \phi(T) - \lambda$ has sign pattern negative, positive, negative, with exactly two sign changes $T_1 < T_2$. If $C(b_i) = \lambda$ at three distinct orders $b_1 < b_2 < b_3$, each of the three integrals $\int f(T) T^{b_i - 1} e^{-T}\, dT$ vanishes. Write $v = T^{b_2 - b_1}$ and $p = (b_3 - b_1)/(b_2 - b_1) > 1$. Subtract from v^p its secant line through the values at T_1, T_2, and multiply by $-T^{b_1 - 1}$. Strict convexity gives a linear combination of the three powers with the same sign pattern as f. Its integral against $f(T)e^{-T}$ is strictly positive, contradicting the three vanishing integrals. Thus every horizontal level has at most two intersections.

At a stationary point b_0, take $\lambda = C(b_0)$. If also $C''(b_0) = 0$, the Mellin integral and its first two derivatives vanish there. Their linear combination with kernel $-T^{b_0 - 1}(\log T - \log T_1)(\log T - \log T_2)$ again has strictly positive integral, a contradiction. Therefore every stationary point is nondegenerate. It cannot be a local minimum: the end limits one, the minimum itself, and the two adjacent larger values would give three distinct intersections with its own horizontal level. Every stationary point is consequently a strict maximum, and there can be only one. This proves both derivative signs and nondegeneracy.

The exact axis certificate at $b = 3/2$ gives $C(3/2) > 227/200$. For comparison, $\pi < 22/7$ and $\log 2 < 25/36$ give $C(1) < 571/504 < 227/200$. The latter logarithmic bound follows by summing the first two terms of its artanh series and bounding the remaining positive geometric tail. Also the first five terms of the alternating Catalan series give $G < 37/40$, hence $C(2) = G + \pi^2/48 < 37/40 + 121/588 < 227/200$. Strict unimodality now forces $1 < b_* < 2$. The magnitude theorem proves the remaining supremum assertions. □

The numerical axis experiment finds $b_* \approx 1.30221658710124120959$ and $C(b_*) \approx 1.13656110333950956095$. This experiment supplies decimal diagnostics; the preceding proof gives the broad bracket $1 < b_* < 2$ and exact value bounds independently. The sharper rational enclosures in Section 5.10 subsequently certify the displayed location and height. Neither uniqueness proof relies on a numerical scan.

Corollary 5.33 (The sharp uniform constant on and below the critical line)**.** *For every $a, b > 0$ with $a + b \le 1$ and every integer $N \ge 1$,*

$$0 < E_N - g_{a,b} < C_{\text{critical}} 2^{-N}, \qquad C_{\text{critical}} = \frac{\pi}{4} + \frac{\log 2}{2}.$$

This is the least constant uniform over the critical line and is also the least constant uniform over the whole positive subcritical triangle. Thus the critical constant conjecture and its stronger strict-monotonicity claim are both settled. The rational choice $57/50$ is valid on this triangle.

Proof. On the critical line, Theorem 5.31 bounds $-2g$ strictly by its left endpoint $C(1) = C_{\text{critical}}$. For $a + b < 1$, the outer-order magnitude theorem gives $-2g_{a,b} < C(b)$, and $b < 1 < b_*$ gives $C(b) < C(1)$. Apply the fixed-pair Euler constant theorem. Sharpness follows from the critical endpoint limit $a \downarrow 0, b \uparrow 1$ and the error at $N = 1$, where $E_1 = 0$. Finally $C(1) < 571/504 < 57/50$. The conclusion does not extend the same bound to arbitrary supercritical orders or to the entire outer-order axis. □

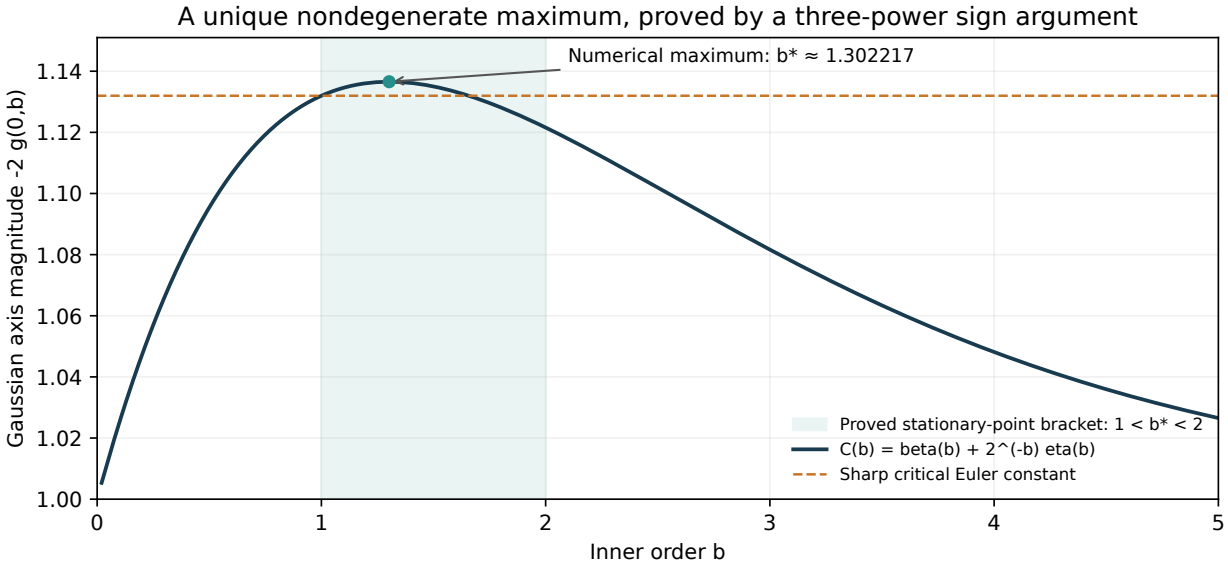


Figure 5.2: Numerical values of the exact Gaussian axis expression and its stationary-point marker. The shaded interval is the proved bracket for the unique maximum; the horizontal line is the proved sharp critical Euler constant. The marker was plotted numerically; its parameter and height are subsequently certified in Section 5.10. The universal Euler interpretation is established separately by the tail-kernel contraction in Theorem 5.64.

5.10 Strict Gaussian envelope shape and a certified maximum

The incoming extremal continuation [99] independently develops the axis maximum and strengthens its description. Its strict normalized Mellin comparison gives a second proof of unimodality, Turan inequalities and a rigorous enclosure of the maximum. We retain the earlier three-power proof as a distinct route and reconcile the common axis normalization before using the stronger certificate.

Lemma 5.34 (A strict normalized Mellin comparison)**.** *Suppose $f : (0,\infty) \to (0,\infty)$ is strictly log-concave, and that its Mellin integrals and their first two parameter derivatives are finite and admit differentiation under the integral. Then*

$$M(p) = \frac{1}{\Gamma(p)} \int_0^\infty T^{p-1} f(T)\, dT \quad (p > 0) \quad \textit{satisfies} \quad (\log M)''(p) < 0.$$

Proof. Normalize $T^{p-1}f(T)$ to a probability density P. Choose the gamma density Q of shape p and rate $\lambda > 0$ so that $\psi(p) - \log\lambda = \mathbb{E}_P \log T$. The two densities have matching mass and matching logarithmic mean. Their log likelihood ratio is $\log f(T) + \lambda T$ plus a constant, hence strictly concave. Its positive set is an interval, and matching mass forces both signs. A single crossing at T_0 is impossible: $\log(T/T_0)(P-Q)$ would have one strict sign while its integral is zero by the two matching moments. There must therefore be two crossings $T_1 < T_2$ and the sign pattern of $P - Q$ is negative, positive, negative.

It follows that

$$\int (\log T - \log T_1)(\log T - \log T_2)(P-Q)(T)\, dT < 0.$$

The constant and linear terms vanish by the matching moments. Thus $\mathrm{Var}_P(\log T) < \mathrm{Var}_Q(\log T) = \psi_1(p)$. Differentiation of $\log M$ gives exactly this variance difference. □

Theorem 5.35 (Strict envelope log-concavity)**.** *Put $D(b) = C(b) - 1$, with C defined in Theorem 5.32. For every $b > 0$,*

$$D(b) > 0, \qquad (\log D)''(b) < 0.$$

Consequently $D(b)^2 > D(b-h)D(b+h)$ for $0 < h < b$. Moreover $2^b D(b)$ increases strictly from zero to one.

Proof. Subtract the ordinary gamma integral for one from the axis integral:

$$D(b) = \frac{1}{\Gamma(b)} \int_0^\infty T^{b-1} f(T)\, dT, \qquad f(T) = \frac{e^{-T}(1-e^{-T})}{2\cosh T}.$$

Here $f(T) = T/2 + O(T^2)$ near zero, $f(T) < e^{-2T}$ at infinity, and

$$(\log f)''(T) = -\frac{e^T}{(e^T-1)^2} - \operatorname{sech}^2 T < 0.$$

All differentiated integrals are locally dominated on $b > 0$, so the lemma applies. Strict midpoint concavity gives the Turan inequality. The endpoint limits $D(b) \to 0$ at zero and infinity also give a second proof of a unique nondegenerate maximum: at its stationary point $D'' = D(\log D)'' < 0$.

For the final assertion let T_b have gamma shape b and rate two. Then

$$2^b D(b) = \mathbb{E}\frac{1 - e^{-T_b}}{1 + e^{-2T_b}}.$$

The displayed function increases strictly from zero to one. Gamma addition gives strict increase with b. The gamma laws tend to zero and infinity in probability at the respective parameter endpoints; bounded convergence proves the limits. □

For example, the Turan inequality at $b = 2, h = 1$ gives the explicit constant comparison

$$\left(G + \frac{\pi^2}{48} - 1\right)^2 > \left(\frac{\pi}{4} + \frac{\log 2}{2} - 1\right)\left(\frac{\pi^3}{32} + \frac{3\zeta(3)}{32} - 1\right).$$

5.10.1 Outward rational enclosure of the axis maximum

Theorem 5.36 (Certified location and height)**.** *The unique maximum satisfies the rational inequalities*

$$1.302216587101241209592371\mathbf{70} < b_* < 1.302216587101241209592371\mathbf{71},$$
$$1.136561103339509560952586375779942387681723\mathbf{540} < C(b_*)$$
$$C(b_*) < 1.136561103339509560952586375779942387681723\mathbf{541}.$$

In particular $C(b_*) < 57/50$.

Proof. The incoming standard-library verifier `code/certify_gaussian_envelope.py` was replayed on an isolated copy. We give the analytic budgets used in its finite certificate, so its transcendental enclosures have explicit proof premises. For $d = 1, 2$, put

$$A_d(b) = \sum_{j\ge 0} \frac{(-1)^j}{(dj+1)^b}, \qquad T_{d,N}(b) = 2^{-N} \sum_{j=0}^{N-1} \frac{(-1)^j}{(dj+1)^b} \sum_{h=j+1}^{N} \binom{N}{h}.$$

These give $A_1 = \eta$ and $A_2 = \beta$. For T with gamma shape b and rate one, finite geometric summation gives

$$A_d - T_{d,N} = 2^{-N}\mathbb{E}\frac{(1-e^{-dT})^N}{1+e^{-dT}} \in (0, 2^{-N}).$$

Differentiating the gamma density introduces the centered score $\log T - \psi(b)$. Cauchy–Schwarz and $\sqrt{\psi_1(b)} \le \pi/\sqrt{6} < 2$ for $b \ge 1$ yield $|A'_d - T'_{d,N}| < 2^{1-N}$. For $C_N = T_{2,N} + 2^{-b}T_{1,N}$ this proves

$$0 < C - C_N \le (1 + 2^{-b})2^{-N}, \qquad |C' - C'_N| < 4\,2^{-N} \quad (1 \le b \le 2).$$

The finite terms are enclosed on an outward-rounded 10^{90} grid. After reducing an integer logarithm to $1 \le r \le 2$, its expansion

$$\log r = 2\sum_{j=0}^{M-1} \frac{v^{2j+1}}{2j+1} + R_M, \quad v = \frac{r-1}{r+1}, \quad 0 \le R_M \le \frac{2v^{2M+1}}{(2M+1)(1-v^2)}$$

uses $M = 160$. Exponentials are scaled to $[0, 1]$, bounded by their positive Taylor polynomial through degree 180 and twice the first omitted term, then recovered by four outward-rounded squarings and positive reciprocals. Multiplication by the logarithm gives the derivative enclosure with the same rational arithmetic.

With $N = 160$, the resulting certificate proves $C'(\ell) > 0$ and $C'(u) < 0$ at the displayed location endpoints. Its separate checks give $C'(1) > 0.03260$ and $C'(2) < -0.03561$. Strict log-concavity of D provides the tangent upper bound

$$D(b_*) \le D(\ell)\exp\left(\frac{C'(\ell)}{D(\ell)}(u - \ell)\right).$$

The outward upper endpoint of this expression and the lower bound $C(\ell)$ give the displayed value interval. The fresh exact receipt is `verification/fifth-replay/extremal/results/gaussian_envelope_certificate.json`. The final comparison with $57/50$ is rational. No numerical optimizer or floating-point library evaluation is a premise. □

The old plotted stationary-point marker remains a diagnostic. The enclosures above now certify its location and the axis height to the displayed widths; they do not certify every plotted value. The further inverse generalized-power expansion in the incoming report is retained as a separate analytic integration task.

5.11 Cartesian motion of the complete real-order zero arc

The normalized radial motion can have different signs above the critical line, but its Cartesian abscissa has one direction throughout the entire nonnegative-outer-order domain.

Theorem 5.37 (Strict outward abscissa motion)**.** *For every $a \geq 0, b > 0$, put $s = \rho^2$ and $X_{a,b}(s) = \rho\cos\theta_{a,b}(\rho)$. The unique upper-arc zero has a real-analytic abscissa on $(0,1]$, extending real analytically through zero, with*

$$0 < X_{a,b}(s) < s, \qquad X'_{a,b}(s) > 0 \quad (0 < s \leq 1).$$

This is an abscissa theorem; it does not assert one direction for $\cos\theta/\rho$ in every supercritical parameter pair.

Proof. For $a > 0, a + b > 1$, let ν be the finite signed measure of Theorem 5.17. Set

$$G(X,s) = \int_0^1 \frac{d\nu(u)}{D(u)}, \qquad D(u) = 1 - 2Xu + su^2.$$

The angular root satisfies $G = 0$ and $0 < X < s \leq 1$. At that root the measure $d\tau = d\nu/D$ has zero total mass and the same ordered signs. The sign test gives

$$G_X = 2\int \frac{u}{D}\,d\tau > 0, \qquad \left(\frac{u}{D}\right)' = \frac{1-su^2}{D^2} > 0,$$
$$G_s = -\int \frac{u^2}{D}\,d\tau < 0, \qquad \left(\frac{u^2}{D}\right)' = \frac{2u(1-Xu)}{D^2} > 0.$$

Thus $X' = -G_s/G_X > 0$. At $(X,s) = (0,0)$, $G = 0$ and $G_X = 2c_2 > 0$, proving analytic extension by the implicit-function theorem.

For $a + b \leq 1$ and on the full axis, the right-endpoint factorization gives $X = s\eta(\sqrt{s})$, with $\eta > 0$ and $\partial_s\eta > 0$ by Theorem 5.27. Its positive-measure equation has derivative $G_m = \omega([0,1]) > 0$ at $s = 0$, so m and $\eta = (1+m)/2$ are analytic there. This proves the same derivative conclusion. In every case the root is away from the cut at $s = 1$, and all denominators are locally uniformly nonzero; the analytic implicit argument also applies there. □

5.11.1 The opposite endpoint mechanism in the large-inner-order limit

For fixed $a > 0$, the disk coefficients tend as $b \to \infty$ to $c_1 = 0$, $c_n = n^{-a}$ for $n \geq 2$. The limit function is $F_{a,\infty}(z) = \mathrm{Li}_a(z) - z$. Its signed measure is the positive Gamma moment law minus a unit atom at zero. Thus its normalized-radius motion has the opposite direction from the right-endpoint case.

Theorem 5.38 (Strict decrease for the left-endpoint limit)**.** *For every $a > 0$, the unique upper-arc zero of $\operatorname{Im} F_{a,\infty}$ satisfies*

$$\arccos(\rho/2) < \theta_{a,\infty}(\rho) < \pi/2, \qquad 0 < \eta_{a,\infty}(\rho) < \tfrac{1}{2}, \qquad \eta'_{a,\infty}(\rho) < 0$$

for $0 < \rho \leq 1$.

Proof. Let $U = e^{-Y}$, $Y \sim \mathrm{Gamma}(a, 1)$, and let $d\omega(u) = u\, d\Pr(U \le u)$. The measure has strictly positive density on $(0, 1)$ and mass 2^{-a}. Moment expansion gives

$$F_{a,\infty}(z) = z^2 \int_0^1 \frac{d\omega(u)}{1 - zu}.$$

Set $x = \rho^2$, $m = 2\cos\theta/\rho$, and

$$D(u) = 1 + x(u^2 - mu), \qquad G(m, x) = \int_0^1 \frac{m - u}{D(u)}\, d\omega(u).$$

Direct rationalization shows that the imaginary part has the sign of G. We have $G(0, x) < 0 < G(1, x)$, and $G_m = \int D^{-2}\, d\omega > 0$. Outside $0 < m < 1$ the numerator has a fixed sign. Thus the zero is unique and simple on the full upper arc. At the root, differentiation gives

$$G_x = \int \frac{u(u - m)^2}{D(u)^2}\, d\omega(u) > 0, \qquad \frac{dm}{dx} = -\frac{\int u(u - m)^2 D(u)^{-2}\, d\omega(u)}{\int D(u)^{-2}\, d\omega(u)} < 0.$$

All denominators are uniformly positive at $0 < m < 1$, $0 \le x \le 1$. Hence $\eta'(\rho) = \rho m'(\rho^2) < 0$, proving the claims. The theorem concerns the exact limit family, not an unproved transfer of its radial sign to every finite inner order. □

5.12 Critical defects and the emergence of the atom

Theorem 5.39 (Completely monotone critical defects)**.** *Let $a = 1 - b \in (0, 1)$ and*

$$D_n = \frac{1}{a} - \frac{H^{(b)}_{n-1}}{n^a}, \qquad n \ge 1.$$

Then $D_n = \int_0^1 u^{n-1}\, \mathrm{d}\sigma_{a,b}(u)$ where $d\sigma_{a,b}(u) = -K_{1-b,b}(-\log u)\, du$ is the strictly positive critical negative part. If $\Delta D_n = D_{n+1} - D_n$, then $(-1)^k \Delta^k D_n > 0$ for every $k \ge 0, n \ge 1$. Moreover,

$$-\zeta(b) n^{-a} + \frac{1}{2n} < D_n < -\zeta(b) n^{-a} + \frac{1}{n}. \tag{5.75}$$

Every finite Hankel matrix $(D_{j+k+1})_{j,k=0}^{r}$ is positive definite, and $D_n D_{n+2} > D_{n+1}^2$.

Proof. Subtract the critical signed-measure moments from its atomic mass. Then

$$(-1)^k \Delta^k D_n = \int_0^1 u^{n-1}(1 - u)^k\, \mathrm{d}\sigma(u) > 0.$$

On $w = 1$, $k_w(T) = 1$, so (5.48) reads $1/2 < C_{a,b}(T) < 1$. Integrating these inequalities against $e^{-nT}\, \mathrm{d}T$ gives (5.75). For a nonzero real vector $(v_0, \dots, v_r)$,

$$\sum_{j,k=0}^{r} v_j v_k D_{j+k+1} = \int_0^1 \left(\sum_{j=0}^{r} v_j u^j \right)^2 \mathrm{d}\sigma(u) > 0,$$

since a nonzero polynomial cannot vanish on an interval and the density is positive everywhere inside $(0, 1)$. Strict log-convexity follows either from its 2×2 minors or from strict Cauchy–Schwarz with the weight $u^{n-1}\, \mathrm{d}\sigma$. □

The theorem also gives a continuous version. For $q > 0$,

$$D(q) = \frac{1}{a} - q^{-a}\big(\zeta(b) - \zeta(b,q)\big) = \int_0^\infty e^{-qT}\big[-\zeta(b)k_a(T) + C_{a,b}(T)\big]\,\mathrm{d}T. \tag{5.76}$$

Thus $(-1)^j D^{(j)}(q) > 0$ for all integers $j \geq 0$. All derivatives are justified by exponential domination. These are inequalities for specific harmonic/Hurwitz expressions, not claims about independence of their special values.

5.13 Elementary critical-kernel identities and Hurwitz inequalities

In this section $0 < b < 1$, $a = 1 - b$, and $C_b(T) = C_{1-b,b}(T) = \mathbb{E}[q(TV)]$ with $V \sim \mathrm{Beta}(b, 1-b)$. In particular $1/2 < C_b(T) < 1$ and C_b is strictly increasing.

Theorem 5.40 (An elementary convergent series)**.** *For every $T > 0$,*

$$C_b(T) = \frac{1}{2} + \sum_{k=1}^{\infty} \frac{\sin\big(b\arctan(T/(2\pi k))\big)}{\pi k\big(1 + (T/(2\pi k))^2\big)^{b/2}}. \tag{5.77}$$

Every summand is positive. The tail after $K \geq 1$ terms is bounded by

$$0 < C_b(T) - \frac{1}{2} - \sum_{k=1}^{K}(\mathit{summand}) \leq \frac{bT}{2\pi^2 K}. \tag{5.78}$$

For $|T| < 2\pi$ it also has the convergent expansion

$$C_b(T) = \frac{1}{2} + \sum_{j=1}^{\infty} \frac{B_{2j}(b)_{2j-1}}{(2j)!(2j-1)!} T^{2j-1}, \tag{5.79}$$

where $(b)_m$ is the rising factorial and B_{2j} is a Bernoulli number, not a Bernoulli polynomial evaluated at b.

Proof. The classical partial fraction for coth gives

$$q(t) = \frac{1}{2} + 2t\sum_{k=1}^{\infty} \frac{1}{t^2 + (2\pi k)^2}. \tag{5.80}$$

This follows from DLMF 4.36.3 after rescaling [97]. The summands are positive for $t > 0$, so Tonelli permits averaging over V. For $|z| < 1$, the beta moments yield

$$\mathbb{E}\frac{1}{1 - zV} = \sum_{j \geq 0} z^j \frac{(b)_j}{j!} = (1-z)^{-b}.$$

Both sides are analytic for $z \in \mathbb{C} \setminus [1, \infty)$, so the same identity holds there. At $z = \mathrm{i}q$, its imaginary part is

$$\mathbb{E}\frac{qV}{1 + q^2V^2} = \frac{\sin(b\arctan q)}{(1+q^2)^{b/2}}.$$

Taking $q = T/(2\pi k)$ proves (5.77). Since $\sin(b\arctan q) \leq bq$ and the denominator factor is at least one, its tail is at most $bT(2\pi^2)^{-1}\sum_{k>K} k^{-2} \leq bT/(2\pi^2 K)$.

The usual Bernoulli expansion of (5.46) has radius 2π: $q(t) = 1/2 + \sum_{j\geq 1} B_{2j}t^{2j-1}/(2j)!$. For $|T| < 2\pi$ it is uniformly absolutely convergent for $0 \leq V \leq 1$. Average termwise and use $\mathbb{E}V^{2j-1} = (b)_{2j-1}/(2j-1)!$ to obtain (5.79). □

For instance,

$$C_{1/2}(T) = \frac{1}{2} + \frac{T}{24} - \frac{T^3}{2304} + \frac{T^5}{122880} + O(T^7). \tag{5.81}$$

The companion program regenerates exact coefficients through degree 13 at $b = 1/10, 1/2, 9/10$ by two algebraically equivalent constructions. The numerical tests of (5.77) use independent beta quadrature, with the analytic tail bound displayed separately.

Theorem 5.41 (A completely monotone normalized Hurwitz expression)**.** *For $0 < b < 1$ and $q > 0$, define*

$$L_b(q) = q^{b-1}\zeta(b, q) + \frac{1}{1-b}. \tag{5.82}$$

Then

$$L_b(q) = \int_0^\infty e^{-qT} C_b(T)\, \mathrm{d}T, \qquad \frac{1}{2q} < L_b(q) < \frac{1}{q}. \tag{5.83}$$

In particular, $(-1)^j L_b^{(j)}(q) > 0$ for all $j \geq 0$. The function

$$qL_b(q) = q^b\zeta(b, q) + \frac{q}{1-b} \tag{5.84}$$

decreases strictly from 1 to 1/2 as q increases from zero to infinity.

Proof. Formula (5.83) is (5.49) with $a = 1 - b$. Bounds and strict complete monotonicity follow from $1/2 < C_b < 1$ and differentiating the positive Laplace integral. Substitution $T = s/q$ gives

$$qL_b(q) = \int_0^\infty e^{-s} C_b(s/q)\, \mathrm{d}s.$$

Strict increase of C_b proves strict decrease with q. The endpoint limits follow by dominated convergence and the limits $C_b(0+) = 1/2$, $C_b(\infty) = 1$. □

The beta average is the common source of the elementary series and the Hurwitz inequalities. The inequalities, finite-difference signs, and Hankel determinants are analytic consequences of a positive integral, not numerical pattern extrapolations.

5.14 A proved local part of the normalized-radius conjecture

The earlier *Angular Zero Continuation* report [94] asks whether

$$\eta_{a,b}(\rho) = \frac{\cos\theta_{a,b}(\rho)}{\rho}$$

decreases on the whole interval $0 < \rho \leq 1$ for every integer $a \geq 2$, $b > 0$. At $a = 1$ it predicts decreasing behavior for $b > 1$, increasing behavior for $0 < b < 1$, and the constant $1/2$ for $b = 1$. That report identifies the sign of the first nonconstant Taylor coefficient as a smaller explicit open problem. We solve that problem for the full stated parameter range and describe the intervening fractional-order threshold.

Throughout this section put

$$U = 1 + 2^{-b}, \qquad V = 1 + 2^{-b} + 3^{-b}, \qquad W = 1 + 2^{-b} + 3^{-b} + 4^{-b}.$$

The parameter ρ in this section is a disk radius. It is unrelated to the Lerch deformation parameter used later.

Lemma 5.42 (The local radial coefficient)**.** *For fixed $a \geq 1$, $b > 0$, $\eta_{a,b}$ extends as a real analytic even function near $\rho = 0$, and*

$$\eta_{a,b}(\rho) = \frac{U}{2}\left(\frac{2}{3}\right)^a + K(a,b)\rho^2 + O_{a,b}(\rho^4), \tag{5.85}$$

where

$$K(a,b) = UV\,3^{-a} - \frac{W}{2}\left(\frac{2}{5}\right)^a - \frac{U^3}{2}\left(\frac{8}{27}\right)^a. \tag{5.86}$$

Proof. Set $\delta = \pi/2 - \theta$ and divide $\operatorname{Im} F_{a,b}(\rho e^{i\theta})$ by $2^{-a}\rho^2$. The resulting equation is analytic near $(\rho, \delta) = (0,0)$ and starts with $\sin(2\delta)$; its δ derivative there is 2. The implicit-function theorem gives a unique analytic solution, odd under $(\rho, \delta) \mapsto (-\rho, -\delta)$. Writing $\delta = \alpha\rho + \beta\rho^3 + O(\rho^5)$, the modes with outer indices $2, 3, 4, 5$ give

$$\alpha = \frac{U}{2}(2/3)^a, \qquad \beta = UV\,3^{-a} - \frac{W}{2}(2/5)^a - \frac{23U^3}{48}(8/27)^a.$$

All higher modes enter only at order ρ^5 or later in the normalized equation. Since $\eta = \sin\delta/\rho$, its quadratic coefficient is $\beta - \alpha^3/6$, which is (5.86). This also rederives the earlier report's local coefficient without assuming its sign. □

Theorem 5.43 (Complete sign in the conjectured local range)**.** *The coefficient* (5.86) *satisfies*

1. *$K(a,b) < 0$ for every real $a \geq 2$ and every $b > 0$;*
2. *$K(1,b) < 0$ if $b > 1$, $K(1,b) > 0$ if $0 < b < 1$, and $K(1,1) = 0$;*
3. *for $b \geq 1$, $K(a,b) < 0$ whenever $a > 1$;*
4. *for each $0 < b < 1$, there is exactly one $a_*(b) \in (1,2)$ with $K(a_*(b), b) = 0$. The coefficient is positive on $1 \leq a < a_*(b)$ and negative for $a > a_*(b)$.*

The function $a_*(b)$ is real analytic on $(0,1)$. Every strict coefficient sign gives the corresponding strict monotonicity of $\eta_{a,b}$ on some interval $0 < \rho < \rho_0(a,b)$. No all-radius monotonicity claim is made here.

5.14.1 Monotonicity in the outer order

Define

$$G_a = W + U^3\left(\frac{20}{27}\right)^a - 2UV\left(\frac{5}{6}\right)^a.$$

Then $K(a,b) = -\frac{1}{2}(2/5)^a G_a$, so it suffices to determine the sign of G_a. We first show

$$\frac{\partial G_a}{\partial a} > 0 \qquad (a \geq 1, b > 0). \tag{5.87}$$

Put $x = 2^{-b}$ and $y = 3^{-b}$. Since $y \geq x^2$ and $0 < x < 1$,

$$\frac{U^2}{V} \leq \frac{(1+x)^2}{1+x+x^2} \leq \frac{4}{3}.$$

The last inequality is equivalent to $(1-x)^2 \geq 0$. Elementary alternating-series bounds for $\log(1+t)$ give

$$\log(27/20) < \frac{7273}{24000}, \qquad \log(6/5) > \frac{9}{50}.$$

Consequently

$$\frac{U^2}{2V}\frac{\log(27/20)}{\log(6/5)} < \frac{7273}{6480} < \frac{9}{8} \leq \left(\frac{9}{8}\right)^a.$$

This is precisely the inequality obtained by differentiating G_a and moving its negative term to the other side. It proves (5.87).

5.14.2 A finite positive certificate at $a = 2$

Because $\log_2 3 > 3/2$, $y \leq x^{3/2}$. The coefficient of y in G_2 is $-(7+25x)/18 < 0$. With $x = t^2$, $0 < t < 1$, we obtain

$$G_2 \geq P_6(t) := \frac{800t^6 - 2025t^5 + 1833t^4 - 567t^3 - 192t^2 + 233}{1458}. \tag{5.88}$$

This polynomial has the degree-six Bernstein representation

$$P_6(t) = \sum_{j=0}^{6} b_j \binom{6}{j} t^j (1-t)^{6-j},$$

with the exact coefficient vector

$$(b_0, \ldots, b_6) = \left(\frac{233}{1458}, \frac{233}{1458}, \frac{367}{2430}, \frac{665}{5832}, \frac{55}{486}, \frac{95}{1458}, \frac{41}{729}\right). \tag{5.89}$$

Every coefficient is positive, and the Bernstein basis is nonnegative and sums to one on $[0,1]$. Hence $G_2 > 0$. Together with (5.87), this proves the first assertion of the theorem. The accompanying verifier expands (5.89) over $\mathbb{Q}$ and checks equality with (5.88); no floating-point polynomial sign test is used.

5.14.3 The sign change at $a = 1$, $b = 1$

A direct simplification gives

$$27G_1 = 20x^3 + 42x^2 - 3x + 2 - (45x + 18)y. \tag{5.90}$$

If $b > 1$, let $t = \sqrt{2x} \in (0,1)$. Since $y = (2x)^{\log_2 3}/3 \leq t^3/3$, substitution in (5.90) yields

$$G_1 \geq (1-t)P_5(t), \qquad P_5(t) = \frac{-5t^5 + 10t^4 - 11t^3 + t^2 + 4t + 4}{54}.$$

The degree-five Bernstein coefficients of P_5 are

$$\left(\frac{2}{27}, \frac{4}{45}, \frac{19}{180}, \frac{14}{135}, \frac{1}{10}, \frac{1}{18}\right). \tag{5.91}$$

They are all positive, so $G_1 > 0$ for $b > 1$.

Suppose instead $0 < b < 1$, so $1/2 < x < 1$. Put $p = \log_2 3$. The exact integer inequality $2^{11} < 3^7$ gives $p > 11/7$, while $p < 2$. For $y(x) = x^p$ this implies

$$y(1/2) = \frac{1}{3}, \quad y'(1/2) > \frac{22}{21}, \quad y''(x) = p(p-1)x^{p-2} > \frac{44}{49} \quad (1/2 < x < 1).$$

Taylor's formula with an integral remainder therefore gives

$$y \geq \frac{1}{3} + \frac{22}{21}(x - 1/2) + \frac{22}{49}(x - 1/2)^2.$$

Using the negative coefficient of y in (5.90), we find

$$G_1 \leq -\frac{(2x-1)(10x^2 - 337x + 334)}{2646} < 0.$$

Indeed the quadratic is decreasing on $[1/2, 1]$ and its value at one is $7 > 0$. At $b = 1$, direct substitution $x = 1/2$, $y = 1/3$ gives $G_1 = 0$. The exact identity $F_{1,1}(z) = \frac{1}{2}\log^2(1 - z)$ gives $\eta_{1,1}(\rho) = 1/2$ for every $0 < \rho \leq 1$.

Finally, G_a is strictly increasing in $a \geq 1$. The signs of G_1 and G_2 give all the remaining assertions and the unique threshold. Its analyticity follows from the analytic implicit-function theorem and (5.87). Differentiating (5.85) shows that a nonzero K controls the sign of η' for sufficiently small positive ρ.

Remark 5.44 (What remains of the conjecture). This proves the local sign statement explicitly requested in the earlier report, including all real $a \geq 2$. It does not show that a derivative cannot change sign farther from the origin. At the fractional threshold $a = a_*(b)$, the quadratic coefficient vanishes; its quartic successor is needed even for local monotonicity. The fractional-turning development below proves opposite quartic signs at two inner orders, producing both maxima and minima. The next section proves the complete $a = 1$ full-radius branch. The higher-integer global problem and the general quartic-sign classification require separate arguments; the low-order local coefficient does not exclude later changes of radial direction.

5.15 Global normalized-radius motion at leading index one

The local $a = 1$ sign extends to the full disk. Here the zero curve is defined by the imaginary part; unrestricted slit-plane nonvanishing was already proved above. The continuation [103] supplies a broader Green-kernel theorem, an outer-radius cutoff, moment inequalities, and exact sums over inner orders. The proof needed for the present assertion follows here.

Theorem 5.45 (The complete leading-index-one branch)**.** *Let $b > 0$, and let $\theta_{1,b}(\rho)$ be the unique upper-arc zero of $\operatorname{Im} F_{1,b}(\rho e^{i\theta})$ for $0 < \rho \leq 1$. Then $\eta_b(\rho) = \cos\theta_{1,b}(\rho)/\rho$ is strictly decreasing for $b > 1$, strictly increasing for $0 < b < 1$, and identically $1/2$ for $b = 1$. Its analytic even extension at zero has value $\eta_b(0) = (1 + 2^{-b})/3$.*

Define

$$w_b(u) = \frac{(-\log u)^{b-1}}{\Gamma(b)},$$
$$Q_b(u) = (1-u)\int_0^u \frac{w_b(t)}{1-t}\,dt + u\int_u^1 \frac{w_b(t)}{t}\,dt. \tag{5.92}$$

The ratio $\min(u,t)(1-\max(u,t))/(t(1-t))$ lies in $[0,1]$. Dominated convergence therefore extends Q_b continuously to zero at both endpoints. Its first derivative is integrable, since each positive term in

$$Q_b'(u) = \int_u^1 \frac{w_b(t)}{t}\,dt - \int_0^u \frac{w_b(t)}{1-t}\,dt$$

has integral 1. Thus Q_b is absolutely continuous, positive inside, and

$$Q_b''(u) = -\frac{w_b(u)}{u(1-u)} < 0. \tag{5.93}$$

It is strictly log-concave as well. Direct integration of the Green kernel gives

$$\int_0^1 u^j Q_b(u)\,du = \frac{H_{j+1}^{(b)}}{(j+1)(j+2)}.$$

Hence coefficient comparison and integration by parts establish

$$F_{1,b}(z) = z^2 \int_0^1 \frac{Q_b(u)}{(1-zu)^2}\,du = z\int_0^1 \frac{-Q_b'(u)}{1-zu}\,du \tag{5.94}$$

on the principal slit plane. The mean of the probability density $2Q_b$ is $\mu_b = (1+2^{-b})/3$.

Lemma 5.46 (A unique nondegenerate Cauchy mode)**.** *For every $y>0$, the function $A_y(x) = \int_0^1 Q_b(u)((x-u)^2+y^2)^{-1}\,du$ has one stationary point $m_b(y)\in(0,1)$, and $A_y''(m_b(y))<0$.*

Proof. With $K(v) = (v^2+y^2)^{-1}$, A_y' is positive on $x\le 0$ and negative on $x\ge 1$. At any stationary point x, the signed integrand $Q_b(x-v)K'(v)$ has zero integral, positive sign for $v<0$, and negative sign for $v>0$. The score $S(v) = Q_b'(x-v)/Q_b(x-v)$ is strictly increasing by strict log-concavity. Integration by parts, then subtraction of $S(0)$, gives

$$A_y''(x) = \int_{x-1}^{x} Q_b(x-v)K'(v)(S(v)-S(0))\,dv < 0.$$

The integrability of Q_b' and the zero endpoint values justify the calculation even if endpoint scores diverge. Every stationary point is consequently a strict downward crossing of A_y', so there is only one. The implicit-function theorem makes m_b real analytic. □

Lemma 5.47 (Reflected curvature controls the height derivative)**.** *Put*

$$I(x,t) = \int_0^1 \frac{(u-x)Q_b(u)}{((u-x)^2+t)^2}\,du.$$

At its zero $x = m_b(\sqrt{t})$, $I_t>0$ for $b>1$ and $I_t<0$ for $0<b<1$. For $b=1$ the zero is $x=1/2$ for all t.

Proof. Suppose $b>1$, so w_b is strictly decreasing. Extend Q_b by zero outside $[0,1]$, and let $\Delta_x(v) = Q_b(x+v) - Q_b(x-v)$ for $v\ge 0$. At $x=1/2$, it is strictly convex between two zero endpoint values, because

$$\Delta_{1/2}''(v) = \frac{w_b(1/2-v) - w_b(1/2+v)}{1/4-v^2} > 0.$$

Thus $I(1/2,t)<0$, and the unique mode lies below $1/2$. For $0<x<1/2$ and $0<v<x$,

$$\Delta_x''(v)=\frac{w_b(x-v)}{(x-v)(1-x+v)}-\frac{w_b(x+v)}{(x+v)(1-x-v)}>0.$$

The first numerator is larger and its denominator smaller. Therefore $\Delta_x(v)/v$ is strictly increasing on $(0,x)$. Also $\Delta_x(x)=Q_b(2x)>0$, and $\Delta_x(v)>0$ on $x<v<1-x$. At the mode, the paired equation

$$I(x,t)=\int_0^\infty \frac{v\Delta_x(v)}{(v^2+t)^2}\,dv=0$$

forces a negative part, so Δ_x has exactly one negative-to-positive crossing $v_0\in(0,x)$. Consequently

$$I_t=\int_0^\infty \frac{v\Delta_x(v)}{(v^2+t)^2}\left(-\frac{2}{v^2+t}+\frac{2}{v_0^2+t}\right)dv>0.$$

For $0<b<1$, reflect the increasing weight w_b under $u\mapsto 1-u$ and apply the same argument to the resulting decreasing weight; the sign reverses. At $b=1$ the density is symmetric, so the mode is $1/2$. □

Proof of Theorem 5.45. For $z=(x-\mathrm{i}y)^{-1}$, (5.94) gives

$$\operatorname{Im} F_{1,b}(z)=-2yI(x,y^2).$$

Thus the upper-half-plane imaginary zero has $x=m_b(y)$. To convert this to disk radius, set $s=\rho^2$ and $\mathcal{I}(x,s)=I(x,s^{-1}-x^2)$. At the angular zero $x=\eta_b(\sqrt{s})$.

The signed density $\kappa=-Q_b'$ has zero mass and one strict negative-to-positive crossing. Put

$$J_\rho(c)=\int_0^1 \frac{\kappa(u)}{1-2\rho cu+\rho^2u^2}\,du.$$

At its unique zero, $J_\rho'(c)>0$: the signed measure obtained by multiplying κ by the positive denominator reciprocal still has zero mass and one crossing, and the derivative score $2\rho u/(1-2\rho cu+\rho^2u^2)$ is strictly increasing in u for $0<\rho\le 1$. Subtraction of its value at the crossing proves the sign. The root lies in $0<c<\rho$, since $J_\rho(0)<0<J_\rho(\rho)$; the upper endpoint at $\rho=1$ is interpreted by monotone convergence of its positive part.

Since $\operatorname{Im} z=sy$, comparison of the two integral formulas gives

$$J_{\sqrt{s}}(\sqrt{s}\,x)=-\frac{2}{s}\mathcal{I}(x,s),\qquad \mathcal{I}_x<0 \text{ at the root.}$$

At fixed x, $\mathcal{I}_s=-s^{-2}I_t$. Hence

$$\frac{d}{ds}\eta_b(\sqrt{s})=\frac{s^{-2}I_t}{\mathcal{I}_x}. \tag{5.95}$$

Lemma 5.47 gives all strict signs. Symmetry gives $\eta_1=1/2$. The local analytic extension and its value at zero follow either from the existing local coefficient calculation or, after substituting $c=\rho\eta$, from $J/\rho^2=\eta-\mu_b+O(\rho^2)$. □

Corollary 5.48 (The local coefficient is a central moment)**.** *Let U_b have density $2Q_b$ and mean μ_b. Then*

$$K(1,b)=-2\mathbb{E}(U_b-\mu_b)^3.$$

More generally, every odd central moment of order $2m+1$, $m\ge 1$, has the sign of $b-1$.

Proof. At $x = \mu_b$, the mean identity gives $\int_0^\infty v\Delta_x(v)\,dv = 0$. For $b > 1$, the reflected-curvature argument gives exactly one negative-to-positive crossing v_0. Multiplication by the increasing factor $v^{2m} - v_0^{2m}$ shows $\int v^{2m+1}\Delta_x(v)\,dv > 0$. Reflection handles $b < 1$ and symmetry handles $b = 1$. The raw moments from (5.92) yield

$$\mathbb{E}(U_b - \mu_b)^3 = \frac{H_4^{(b)}}{10} - \frac{H_2^{(b)}H_3^{(b)}}{6} + \frac{2(H_2^{(b)})^3}{27},$$

which agrees with $-K(1, b)/2$. □

Remark 5.49 (Scope). The theorem resolves the full-radius branch $a = 1$, not the remaining higher-outer-order problem. Monotonicity should also not be replaced by a uniform convexity claim: the coefficient of ρ^4 in η_2 is $-263/777600$, whereas in η_4 it is $504047/3149280000$. The separate article proves these values and provides exact verification artifacts.

The isolated exact replay certifies 1,938 finite coefficient and moment checks and eighteen rational angular brackets, including fractional inner orders 1/4 and 1/2; the separate symbolic replay checks five identities. These catch finite algebra and transcription errors. The continuum monotonicity theorem rests on the Green-density and single-crossing proof above, not on the historical quadrature grid. Fresh receipts are in `verification/seventh-replay/leading-one/`.

5.16 A proved failure of fractional global monotonicity

The local sign threshold does not extend directly to a classification of radial monotonicity on the whole unit disk. There are fractional orders for which $\eta_{a,b}(\rho) = \cos\theta_{a,b}(\rho)/\rho$ first increases and then decreases. This is an analytic theorem, supported by a finite exact certificate for its hypotheses.

5.16.1 The next local coefficient

Put $p_n = H_{n-1}^{(b)}/n^a$ and

$$\mu = \frac{p_3}{2p_2}, \qquad K = -\frac{4p_3\mu^2 - 4p_4\mu + p_5}{2p_2}.$$

Thus $\mu = \eta_{a,b}(0)$ and K is the previously identified quadratic coefficient. The next coefficient is

$$Q = -\frac{(8p_3\mu - 4p_4)K + 8p_4\mu^3 - 12p_5\mu^2 + 6p_6\mu - p_7}{2p_2}, \tag{5.96}$$

and the convergent local expansion is

$$\eta_{a,b}(\rho) = \mu + K\rho^2 + Q\rho^4 + O(\rho^6). \tag{5.97}$$

The coefficients and remainder are real analytic in a, b locally within the parameter range under consideration.

To check (5.96), write $c = \rho\eta$ and use $\sin(n\theta) = \sin\theta\, U_{n-1}(c)$. Division of the imaginary-part equation by $\rho^3 \sin\theta$ gives

$$\begin{aligned} 0 = {} & 2p_2\eta - p_3 + \rho^2(4p_3\eta^2 - 4p_4\eta + p_5) \\ & + \rho^4(8p_4\eta^3 - 12p_5\eta^2 + 6p_6\eta - p_7) + O(\rho^6). \end{aligned}$$

Every omitted mode has order at least ρ^6. This equation is analytic in $a, b, t = \rho^2, \eta$ near $t = 0$; it follows either from the convergent disk series or from the parity of each Chebyshev term. Its derivative in η at zero is $2p_2 > 0$. The analytic implicit-function theorem therefore gives (5.97), and substitution proves the formula for Q.

Theorem 5.50 (Emergence of a strict radial maximum)**.** *Fix $b = 1/2$. There is a unique zero $a_\star$ of $K(a, 1/2)$ in the rational interval*

$$1.31298577512189 < a_\star < 1.31298577512190. \tag{5.98}$$

There exist $\delta > 0$ and $\rho_0 > 0$ such that, for every $a \in (a_\star - \delta, a_\star)$, the function $\eta_{a,1/2}(\rho)$ has exactly one critical point in $(0, \rho_0)$. It is a strict local maximum: the function increases before it and decreases after it within that interval. Writing this point as $\rho_{\max}(a)$, one has

$$\rho_{\max}(a) = \left(\frac{\partial_a K(a_\star, 1/2)}{2Q(a_\star, 1/2)}\right)^{1/2} (a_\star - a)^{1/2} + O((a_\star - a)^{3/2}). \tag{5.99}$$

The positive prefactor is rigorously enclosed by

$$44.0912731287 < \left(\frac{\partial_a K(a_\star, 1/2)}{2Q(a_\star, 1/2)}\right)^{1/2} < 44.0912738633.$$

In particular these fractional normalized-radius curves are not monotone on $(0, 1]$, despite having $K > 0$.

Proof. The accompanying standard-library verifier `certify_fractional_turning.py` uses only integer and rational arithmetic. Let I be the closed rational interval in (5.98). Its output proves

$$K(\inf I, 1/2) > 8.16 \cdot 10^{-17}, \qquad K(\sup I, 1/2) < -8.72 \cdot 10^{-17},$$

and, throughout I,

$$-0.016895831331 < \partial_a K(a, 1/2) < -0.016895831329,$$
$$-4.345546 \cdot 10^{-6} < Q(a, 1/2) < -4.345545 \cdot 10^{-6}.$$

These signs alone give existence and uniqueness of $a_\star$ in I, independently of the more general local-threshold theorem. That theorem also identifies it as the unique threshold in $(1, 2)$.

Here is the complete arithmetic basis of the certificate. For $2 \le n \le 7$, it encloses $\log n$ by the first 120 terms of

$$\log n = 2 \sum_{j \ge 0} \frac{z^{2j+1}}{2j+1}, \qquad z = \frac{n-1}{n+1},$$

with positive tail bounded by $2z^{241}/(241(1 - z^2))$. For a nonnegative rational x, the first 60 terms after the constant term of e^x have tail at most

$$\frac{x^{61}}{61!} \frac{1}{1 - x/62};$$

all arguments used here are less than three. Monotonicity encloses $e^{a \log n}$ at both ends of the interval arguments, and reciprocation encloses n^{-a}. The factors $j^{-1/2}$ in the harmonic numbers are bounded by exact integer-square-root operations. All arithmetic operations round outwards to multiples of 2^{-128}. Substitution in the formulas for K, Q and their elementary derivatives proves the displayed signs. Exact rational endpoints, including the tighter prefactor enclosure, are recorded in `fractional_turning_certificate.json`. The written tail bounds and integer inequalities make this a finite exact proof certificate; no floating-point sign decision is used.

Write $\eta_{a,1/2}(\rho) = \Phi(a,t)$, $t = \rho^2$, using the analytic function constructed above. Set $U(a,t) = \partial_t \Phi(a,t)$. Then

$$U(a_\star, 0) = 0, \qquad \partial_t U(a_\star, 0) = 2Q(a_\star, 1/2) < 0, \qquad \partial_a U(a_\star, 0) = \partial_a K(a_\star, 1/2) < 0.$$

The analytic implicit-function theorem gives a unique local solution $t = t(a)$ of $U(a,t) = 0$, with

$$t(a) = \frac{\partial_a K(a_\star, 1/2)}{2Q(a_\star, 1/2)}(a_\star - a) + O((a_\star - a)^2).$$

It is positive for $a < a_\star$ sufficiently close. On a sufficiently small fixed rectangle, $\partial_t U < 0$; at its upper t boundary, $U < 0$, while $U(a,0) = K(a,1/2) > 0$ for $a < a_\star$. Thus there is exactly one such positive critical point and the derivative $\partial_\rho \eta = 2\rho U(a,\rho^2)$ changes from positive to negative there. At the critical point its second derivative is $4t\partial_t U < 0$. Taking the positive square root of the expansion for $t(a)$ proves (5.99). □

5.16.2 Exploration beyond the existence theorem

A separate 70-digit diagnostic evaluates the defining disk series and its derivative at $b = 1/4, 1/2, 3/4$, at the local threshold and at offsets $-0.02, -10^{-4}, +0.02$, for radii $0.3, 0.7, 0.9, 0.98$. For each b, the offset -10^{-4} gives a positive derivative at 0.3 and a negative derivative at 0.98. Thus the same phenomenon appears numerically at all three sampled inner orders; the theorem above proves it at $b = 1/2$.

For the exact rational pair $a = 21/16$, $b = 1/2$, the diagnostic finds

$$\eta'(0.9) = 1.2937181281564777 \cdot 10^{-6}, \qquad \eta'(0.98) = -2.6349934635137804 \cdot 10^{-6}.$$

These two particular signs are numerical observations, not an extra certified instance of the theorem's unspecified interval in a. The disk-series truncation errors for F and zF' are bounded analytically by 10^{-50}, but rounding is not interval-enclosed. The existence theorem relies instead on the exact finite coefficient certificate above. Extending this argument to other b requires determining the sign of $Q(a_\star(b), b)$. The complementary minimum theorem below proves that this sign is not always negative.

5.16.3 A complementary minimum and a change of quartic sign

The sign of the quartic coefficient along the local threshold also changes. In particular, the negative sign at $b = 1/2$ cannot extend to every $0 < b < 1$. The following exact certificate exhibits the opposite local bifurcation at a rational inner order close to one.

Theorem 5.51 (Emergence of a strict radial minimum)**.** *Set $b_0 = 999/1000$. The unique local threshold $a_0 = a_\star(b_0)$ satisfies the rational enclosure*

$$1.00057049936559087278 < a_0 < 1.00057049936559087279. \tag{5.100}$$

Moreover,

$$Q(a_0, b_0) > 0, \qquad \partial_a K(a_0, b_0) < 0.$$

There exist $\delta > 0$ and $\rho_0 > 0$ such that, for every $a \in (a_0, a_0 + \delta)$, the function $\eta_{a,b_0}(\rho)$ has exactly one critical point in $(0, \rho_0)$. It is a strict local minimum: the function decreases before it and increases after it within this interval. Its location satisfies

$$\rho_{\min}(a) = \left(\frac{-\partial_a K(a_0, b_0)}{2Q(a_0, b_0)}\right)^{1/2} (a - a_0)^{1/2} + O((a - a_0)^{3/2}), \tag{5.101}$$

where the prefactor has the certified enclosure

$$17076.2396129683 < \left(\frac{-\partial_a K(a_0, b_0)}{2Q(a_0, b_0)}\right)^{1/2} < 17076.2396548521.$$

Thus fractional normalized-radius curves can also fail global monotonicity while their initial quadratic coefficient is negative. The theorem asserts a sufficiently small neighborhood; it gives no numerical lower bound on δ.

Proof. Let I be the closed rational interval in (5.100). The exact interval verifier `certify_fractional_minimum.py` proves

$$K(\inf I, b_0) > 4.08027 \cdot 10^{-23}, \qquad K(\sup I, b_0) < -1.29944 \cdot 10^{-22}.$$

Throughout I, it also proves

$$2.92779195 \cdot 10^{-11} < Q(a, b_0) < 2.92779198 \cdot 10^{-11},$$
$$-0.017074763285092 < \partial_a K(a, b_0) < -0.017074763285091.$$

The arithmetic uses the same outward dyadic intervals and rational logarithm/exponential remainder bounds as the maximum certificate. For this rational inner order, each j^{-b_0} is enclosed as the reciprocal of $e^{b_0 \log j}$, for $1 \le j \le 6$; each n^{-a} is enclosed in the same way, for $2 \le n \le 7$. All exponential arguments are less than three. Substitution into the finite formulas for K, Q and $\partial_a K$ establishes the signs. Exact rational endpoints and the sharper prefactor interval are in `fractional_minimum_certificate.json`. Thus both the small positive quartic coefficient and the much smaller endpoint values of K are certified without floating-point arithmetic.

Write $\eta_{a,b_0}(\rho) = \Phi(a, t)$ with $t = \rho^2$, and let $U = \partial_t \Phi$. At $(a_0, 0)$ one has

$$U = 0, \qquad U_t = 2Q(a_0, b_0) > 0, \qquad U_a = \partial_a K(a_0, b_0) < 0.$$

The analytic implicit-function theorem gives a unique local zero $t = t(a)$ with

$$t(a) = \frac{-\partial_a K(a_0, b_0)}{2Q(a_0, b_0)}(a - a_0) + O((a - a_0)^2).$$

It is positive for $a > a_0$ sufficiently close. On a sufficiently small fixed rectangle, $U_t > 0$; hence U has exactly one zero in the indicated positive t interval and changes from negative to positive there. Since $\eta_\rho = 2\rho U$, this is the asserted minimum, with $\eta_{\rho\rho} = 4tU_t > 0$ at the critical point. Taking its positive square root proves (5.101). □

Corollary 5.52 (A zero of the quartic coefficient along the threshold)**.** *There exists $b_\dagger \in (1/2, 999/1000)$ for which*

$$Q(a_\star(b_\dagger), b_\dagger) = 0.$$

Proof. The threshold equation has nonzero a derivative at every $a_\star(b)$, so $a_\star(b)$ is real analytic on $(0, 1)$. The finite formula for Q then makes $q(b) = Q(a_\star(b), b)$ real analytic there. The exact maximum and minimum certificates give $q(1/2) < 0 < q(999/1000)$, so the intermediate value theorem applies. □

The finite-coefficient diagnostic `locate_quartic_sign_transition.py`, run at 100 decimal digits, locates one apparent sign change at

$$b_\dagger \approx 0.99749378987342042107460559 31, \qquad a_\star(b_\dagger) \approx 1.00143018096149513812935172 94.$$

These displayed decimal locations are diagnostic, not certified enclosures. No uniqueness of $b_\dagger$ is claimed. The 11-point finite-coefficient sweep has negative $q(b)$ at every requested sample from 0.001 through 0.99, and positive $q(0.999)$; the latter sign is proved by the independent exact certificate. Determining the number of zeros of q and the higher-order radial behavior at its zeros remains a concrete research problem.

5.17 The two atomic edges and the logarithmic corner

Theorem 5.53 (Boundary-layer scale and weak convergence)**.** *Fix $0 < b < 1$, let $a = 1 - b + \varepsilon$ with $\varepsilon > 0$, and let $T_\varepsilon = -\log r_{a,b}$. Define*

$$\tau_\varepsilon = \left(\frac{\Gamma(1-b+\varepsilon)}{(1-b)(-\zeta(b))\Gamma(\varepsilon)} \right)^{1/(1-b)}.$$

Then, as $\varepsilon \downarrow 0$,

$$T_\varepsilon = \tau_\varepsilon \left(1 + O_b\left(\frac{\tau_\varepsilon}{\varepsilon} \right) \right), \qquad 0 < T_\varepsilon < \tau_\varepsilon, \tag{5.102}$$

and in particular

$$1 - r_{1-b+\varepsilon,b} \sim \left(\frac{\Gamma(1-b)}{(1-b)(-\zeta(b))} \varepsilon \right)^{1/(1-b)}. \tag{5.103}$$

The measures converge weakly to the critical measure in Theorem 5.17, and their positive masses converge to $(1-b)^{-1}$ while their positive supports shrink to $u = 1$.

Proof. In (5.55), put $A_\varepsilon = ((1-b)\Gamma(\varepsilon))^{-1}$ and $B_\varepsilon = (-\zeta(b))/\Gamma(1-b+\varepsilon)$. The zero equation is

$$A_\varepsilon - B_\varepsilon T^{1-b} - T C_\varepsilon(T) = 0, \quad C_\varepsilon(T) = \frac{1}{\Gamma(a)\Gamma(b)} \int_0^1 (1-v)^{a-1} v^{b-1} q(Tv)\, dv.$$

Since $1/2 < q < 1$ and $a + b = 1 + \varepsilon$,

$$\frac{1}{2\Gamma(1+\varepsilon)} < C_\varepsilon(T) < \frac{1}{\Gamma(1+\varepsilon)}.$$

By definition $B_\varepsilon \tau_\varepsilon^{1-b} = A_\varepsilon$, so the equation at $T = \tau_\varepsilon$ has strictly negative left side and $T_\varepsilon < \tau_\varepsilon$. Furthermore

$$0 \le 1 - (T_\varepsilon/\tau_\varepsilon)^{1-b} = \frac{T_\varepsilon C_\varepsilon(T_\varepsilon)}{A_\varepsilon} \le \frac{(1-b)\tau_\varepsilon}{\varepsilon}.$$

The final equality in the upper bound uses $\Gamma(1+\varepsilon) = \varepsilon\Gamma(\varepsilon)$. Now $\tau_\varepsilon = O_b(\varepsilon^{1/(1-b)})$, hence $\tau_\varepsilon/\varepsilon \to 0$. The last inequality proves (5.102). Expanding the Gamma ratio and using $1 - e^{-T} \sim T$ proves (5.103).

For weak convergence, the positive term in (5.52), regarded as a measure in the T coordinate, is

$$\frac{1}{1-b} \frac{e^{-T} T^{\varepsilon-1}}{\Gamma(\varepsilon)}\, dT.$$

It is $(1-b)^{-1}$ times a Gamma$(\varepsilon,1)$ law. Its mean is ε, so it converges weakly to mass $(1-b)^{-1}$ at $T=0$, equivalently at $u=1$. The two negative components in (5.52) converge in $L^1(e^{-T}\,dT)$ to those in (5.56). For the power term, dominated convergence holds with an exponent bounded above -1. For the convolution, either use the rescaled beta integral and the same domination, or use L^1 continuity of the Gamma densities and the integrable convolution factor $e^{-t}t^{b-1}q(t)$.

The unique positive support is $0<T<T_\varepsilon$, which shrinks to zero. The Gamma$(\varepsilon,1)$ mass in this interval tends to one: since T_ε is a fixed power of ε to leading order,

$$\frac{1}{\Gamma(\varepsilon)}\int_0^{T_\varepsilon} e^{-T}T^{\varepsilon-1}\,dT=\frac{T_\varepsilon^{\varepsilon}}{\Gamma(1+\varepsilon)}(1+o(1))\longrightarrow 1.$$

The negative components have vanishing mass on the shrinking interval by their L^1 convergence to an integrable limit. Thus the actual positive mass of the signed density tends to $(1-b)^{-1}$, completing the proof. □

5.17.1 The other edge and the logarithmic corner

The second atomic edge has a complementary power law. The corner $b=1$ has an exponential scale and no finite limiting measure.

Proposition 5.54 (Two further endpoint scales)**.** *For fixed $b>1$, as $a\downarrow 0$,*

$$1-r_{a,b}\sim\big(\Gamma(b)\zeta(b)a\big)^{1/(b-1)}. \tag{5.104}$$

The measures converge weakly to $\nu_{0,b}$, and their positive masses converge to $\zeta(b)$. For $b=1$, set $\sigma_a=\exp(\gamma+\psi(a))$. Then

$$-\log r_{a,1}=\sigma_a\big(1+O(\sigma_a/a)\big),\qquad 0<-\log r_{a,1}<\sigma_a, \tag{5.105}$$

where

$$\sigma_a=\exp\big(-1/a+\zeta(2)a-\zeta(3)a^2+O(a^3)\big).$$

If $M_{a,1}$ denotes either signed mass of $\nu_{a,1}$, then

$$M_{a,1}\sim\frac{1}{ea}. \tag{5.106}$$

Thus the finite signed measures have unbounded total variation at this corner, consistently with the impossibility of a finite measure at $a=0,b=1$.

Proof. For $b>1$, the equation obtained from (5.52) after division by T^{a-1} is

$$0=\frac{\zeta(b)}{\Gamma(a)}-\frac{T^{b-1}}{(b-1)\Gamma(a+b-1)}-\frac{T^b}{\Gamma(a)\Gamma(b)}\int_0^1(1-v)^{a-1}v^{b-1}q(Tv)\,dv.$$

The last integral, after its Gamma normalization, lies between $1/(2\Gamma(a+b))$ and $1/\Gamma(a+b)$. The ratio of this last term to the preceding negative term is therefore $O_b(T)$, uniformly for small positive a. The balancing point of the first two terms is

$$\left(\frac{(b-1)\Gamma(a+b-1)\zeta(b)}{\Gamma(a)}\right)^{1/(b-1)}\sim\big(\Gamma(b)\zeta(b)a\big)^{1/(b-1)}.$$

The strict monotonicity of the rescaled density, or the same two-sided comparison used in Theorem 5.53, makes the actual zero asymptotic to this point. The positive term is $\zeta(b)$ times a Gamma$(a,1)$ law in the T coordinate. The negative term in (5.53), after multiplication by e^{-T}, is the convolution of that Gamma probability density with $e^{-t}t^{b-1}/(\Gamma(b)(1-e^{-t}))$, an integrable function of mass $\zeta(b)$. As $a \downarrow 0$, this convolution converges in L^1 to the latter function; Gamma$(a,1)$ is an approximate identity since its mean tends to zero. This proves weak convergence. As before, the Gamma mass below the crossing tends to one, since $a \log T \to 0$, and the limiting negative density is integrable. Hence the positive mass tends to $\zeta(b)$.

For $b = 1$, put $J_a(T) = \int_0^1 (1-v)^{a-1} q(Tv)\, dv$, so that $1/(2a) < J_a(T) < 1/a$. At the crossing T_a, formula (5.54) says

$$\log(\sigma_a/T_a) = T_a J_a(T_a), \qquad 0 < \log(\sigma_a/T_a) < \sigma_a/a.$$

This gives (5.105). The digamma expansion $\psi(a) = -1/a - \gamma + \zeta(2)a - \zeta(3)a^2 + O(a^3)$ gives the stated scale.

Finally the positive mass is the integral of $e^{-T}T^{a-1}[\log(\sigma_a/T) - TJ_a(T)]/\Gamma(a)$ over $0 < T < T_a$. Omitting the factor e^{-T}, the TJ_a term, and the vanishing interval between T_a and σ_a changes the result by a relative $o(1)$. To see this explicitly, the first omission has relative error at most σ_a; the TJ_a integral is at most $\sigma_a^{a+1}/(a(a+1)\Gamma(a))$, whereas the main integral is $\sigma_a^a/(a^2\Gamma(a))$. For the removed interval, write $T = \sigma_a v$ and use $1 - T_a/\sigma_a = O(\sigma_a/a)$; its relative contribution is $O(\sigma_a^2)$. Therefore

$$M_{a,1} \sim \frac{\sigma_a^a}{a^2\Gamma(a)} \sim \frac{1}{ea},$$

as asserted. □

5.18 Power-scale corrections and exponential mass concentration

The atom in Theorem 5.17 raises a quantitative question: how does an absolutely continuous positive part collapse into that atom? Two distinct scales occur. Its outermost point approaches the endpoint by a power of the distance to the critical line. Most of its mass lies much closer to the endpoint, on an exponential scale described by a limiting exponential distribution.

Fix $b \in (0,1)$, and write

$$p = 1 - b, \qquad a = p + \varepsilon, \qquad K_\varepsilon(T) = K_{p+\varepsilon,b}(T), \qquad 0 < \varepsilon \le \varepsilon_0 < b. \tag{5.107}$$

Let T_ε be the unique logarithmic crossing of K_ε, so $K_\varepsilon > 0$ on $(0, T_\varepsilon)$. All constants below may depend on the fixed b; the local expansion constants may also depend on its chosen radius T_0.

Lemma 5.55 (A uniform local kernel expansion)**.** *For any fixed $T_0 < 2\pi$, uniformly for $0 \le \varepsilon \le \varepsilon_0$ and $0 < T \le T_0$,*

$$\begin{aligned} K_\varepsilon(T) = T^{\varepsilon-1}\Bigg\{ &\frac{\varepsilon}{p\Gamma(1+\varepsilon)} + \frac{\zeta(b)}{\Gamma(p+\varepsilon)}T^p - \frac{T}{2\Gamma(1+\varepsilon)} \\ &- \frac{bT^2}{12\Gamma(2+\varepsilon)} + O_{b,T_0}(T^4)\Bigg\}. \end{aligned} \tag{5.108}$$

More generally the bracket has the convergent expansion

$$\begin{aligned} &\frac{\varepsilon}{p\Gamma(1+\varepsilon)} + \frac{\zeta(b)}{\Gamma(p+\varepsilon)}T^p - \frac{T}{2\Gamma(1+\varepsilon)} \\ &\quad - \sum_{\ell \ge 1} \frac{B_{2\ell}\Gamma(b+2\ell-1)}{(2\ell)!\,\Gamma(b)\Gamma(\varepsilon+2\ell)}T^{2\ell}, \qquad 0 < T < 2\pi. \end{aligned} \tag{5.109}$$

Here $B_{2\ell}$ are the Bernoulli numbers, with $B_2 = 1/6$.

Proof. In (5.52), put $a = p + \varepsilon$ and $a + b = 1 + \varepsilon$. The common regularizer has the convergent expansion

$$q(t) = \frac{1}{2} + \sum_{\ell \geq 1} \frac{B_{2\ell}}{(2\ell)!} t^{2\ell - 1}, \qquad |t| < 2\pi.$$

Its positive convolution therefore has expansion

$$C_{a,b}(T) = \frac{T^{\varepsilon}}{2\Gamma(1+\varepsilon)} + \sum_{\ell \geq 1} \frac{B_{2\ell}\Gamma(b + 2\ell - 1)}{(2\ell)!\Gamma(b)\Gamma(\varepsilon + 2\ell)} T^{\varepsilon + 2\ell - 1}.$$

The beta integral evaluates each coefficient. Substitute this into (5.52) and use $\Gamma(1+\varepsilon) = \varepsilon\Gamma(\varepsilon)$ to obtain (5.109). For uniform remainders choose $T_0 < T_1 < 2\pi$; Cauchy estimates bound the regularizer's Taylor remainder on $|t| \leq T_0$, while $v^{b-1}(1-v)^{p-1}$ dominates all beta weights. The Gamma factors are bounded on the fixed compact parameter interval. This proves the uniform truncation and the full locally convergent series. □

Theorem 5.56 (Crossing scale and its first correction)**.** *Define*

$$\tau_{\varepsilon} = \left\{ \frac{\Gamma(p+\varepsilon)\varepsilon}{p[-\zeta(b)]\Gamma(1+\varepsilon)} \right\}^{1/p}. \tag{5.110}$$

As $\varepsilon \downarrow 0$,

$$T_{\varepsilon} = \tau_{\varepsilon} - \frac{\tau_{\varepsilon}^2}{2\varepsilon} + O_b\left(\frac{\tau_{\varepsilon}^3}{\varepsilon^2} \right). \tag{5.111}$$

In particular, with

$$K_b = \left\{ \frac{\Gamma(1-b)}{(1-b)[-\zeta(b)]} \right\}^{1/(1-b)}, \tag{5.112}$$

one has

$$T_{\varepsilon} \sim K_b \varepsilon^{1/(1-b)}, \qquad 1 - r_{p+\varepsilon,b} \sim K_b \varepsilon^{1/(1-b)}. \tag{5.113}$$

Proof. For every fixed $T > 0$, the kernel depends continuously on ε and its limit $K_{p,b}(T)$ is strictly negative by (5.56). The unique crossing therefore satisfies $T_{\varepsilon} \to 0$. At that crossing, multiply the bracket in (5.108) by $\Gamma(1+\varepsilon)$ and write

$$C_{\varepsilon} = \frac{[-\zeta(b)]\Gamma(1+\varepsilon)}{\Gamma(p+\varepsilon)} > 0.$$

This gives

$$\frac{\varepsilon}{p} = C_{\varepsilon} T_{\varepsilon}^{p} + \frac{T_{\varepsilon}}{2} + \frac{bT_{\varepsilon}^2}{12(1+\varepsilon)} + O_b(T_{\varepsilon}^4). \tag{5.114}$$

Since $p < 1$, the last three terms are lower order than T_{ε}^{p}. Hence $T_{\varepsilon}/\tau_{\varepsilon} \to 1$.

Put $y = T_{\varepsilon}/\tau_{\varepsilon}$ and $q = \tau_{\varepsilon}/\varepsilon$. The leading equivalence implies $q = O_b(\varepsilon^{b/p}) \to 0$. Dividing (5.114) by ε/p gives, uniformly for y near one,

$$1 = y^p + \frac{p}{2} q y + O_b(\varepsilon q^2).$$

The mean value theorem first yields $y - 1 = O_b(q)$. Taylor expansion then yields $p(y-1) + (p/2)q = O_b(q^2)$, so $y = 1 - q/2 + O_b(q^2)$. This proves (5.111). Formula (5.113) follows from continuity of the gamma factors and $1 - e^{-T} \sim T$. □

At the symmetric boundary point $b = p = 1/2$, the analytic correction from the gamma quotient and the nonlinear root correction have the same relative order. Using $\psi(1/2) = -\gamma - 2\log 2$ gives the explicit consequence

$$T_\varepsilon = \frac{4\pi}{\zeta(1/2)^2}\varepsilon^2\left[1 - \left(4\log 2 + \frac{2\pi}{\zeta(1/2)^2}\right)\varepsilon + O(\varepsilon^2)\right]. \tag{5.115}$$

For other b, retaining the exact gamma quotient in τ_ε keeps the algebraic and fractional-power corrections in (5.111) unambiguous.

Theorem 5.57 (Condensation and the exponential positive-mass law)**.** *Let $\nu_\varepsilon = \nu_{p+\varepsilon,b}$, and denote its positive and negative parts by ν_ε^+ and ν_ε^-, so $\nu_\varepsilon = \nu_\varepsilon^+ - \nu_\varepsilon^-$. Then*

$$\nu_\varepsilon^- \longrightarrow -\kappa_{p,b}(u)\,du \quad \text{in total variation}, \qquad \nu_\varepsilon^+ \Longrightarrow \frac{1}{p}\delta_1. \tag{5.116}$$

In particular $\nu_\varepsilon \Longrightarrow \nu_{p,b}$.

Let λ_ε be the pushforward of the finite positive measure $p\,K_\varepsilon(T)^+ e^{-T}\,dT$ under $Y_\varepsilon = -\varepsilon\log T$. For all sufficiently small ε it is supported in $[0,\infty)$ and

$$\left\|\lambda_\varepsilon - e^{-y}\mathbf{1}_{y>0}\,dy\right\|_{\mathrm{TV}} = O_b(\varepsilon|\log\varepsilon|)\,. \tag{5.117}$$

The same convergence and rate hold after normalizing λ_ε to a probability measure.

Proof. For $T \le T_0 < 1$, Lemma 5.55 gives

$$K_\varepsilon(T) = \frac{\varepsilon}{p\Gamma(1+\varepsilon)}T^{\varepsilon-1} + R_\varepsilon(T), \qquad |R_\varepsilon(T)| \le C_b T^{p+\varepsilon-1}. \tag{5.118}$$

The first term is positive, so the negative part is bounded by $C_b T^{p-1}$, an integrable bound independent of ε. For $T \ge T_0$, formula (5.52) and $0 < C_{a,b}(T) < k_{a+b}(T)$ give a common polynomial majorant as a ranges over $[p, p+\varepsilon_0]$; hence $e^{-T}|K_\varepsilon(T)| \le C_b e^{-T}(1+T^{\varepsilon_0})$. Pointwise convergence and dominated convergence prove the negative-part assertion in (5.116). Its limiting mass is $1/p$ by Theorem 5.17. Each ν_ε has total mass zero, so the positive masses have the same limit. Their supports lie in $(e^{-T_\varepsilon}, 1)$, which shrinks to 1; this proves the positive weak limit.

For the quantitative distributional statement set $y_\varepsilon = -\varepsilon\log T_\varepsilon$. Theorem 5.56 gives $y_\varepsilon = O_b(\varepsilon|\log\varepsilon|)$ and $y_\varepsilon > 0$ for small ε. On the positive support $0 < T < T_\varepsilon$, replace the density in (5.118) by its leading term and replace e^{-T} by one. The total variation error is at most

$$C_b T_\varepsilon^{p+\varepsilon} + C_b\varepsilon T_\varepsilon^{1+\varepsilon} = O_b(\varepsilon),$$

because $T_\varepsilon^p = O_b(\varepsilon)$. Total variation cannot increase under pushforward. With $T = e^{-y/\varepsilon}$, the pushforward of the resulting leading measure is exactly

$$\frac{e^{-y}}{\Gamma(1+\varepsilon)}\mathbf{1}_{y>y_\varepsilon}\,dy.$$

Its total variation difference from the unit exponential density is at most $O(\varepsilon) + \int_0^{y_\varepsilon} e^{-y}\,dy$. This proves (5.117). Its mass consequently differs from one by the same order; normalizing changes the total variation by at most that mass discrepancy. □

Thus the power-law crossing width does not describe the typical positive-mass location. Under the normalized positive part, $-\varepsilon \log(-\log u)$ tends to an exponential random variable of mean one. A typical logarithmic distance $T = -\log u$ is therefore exponentially small in $1/\varepsilon$, even though the outer edge of the positive region is of order $\varepsilon^{1/(1-b)}$.

The accompanying boundary diagnostics compare the two-term root formula at $b = 1/4, 1/2, 3/4$ and decreasing ε, and compare positive-mass distribution functions with the exponential law. The convergent series (5.109) provides a stable evaluator with a recorded analytic truncation bound. Numerical rounding is still separate from that bound; these are diagnostics, not new proof assumptions.

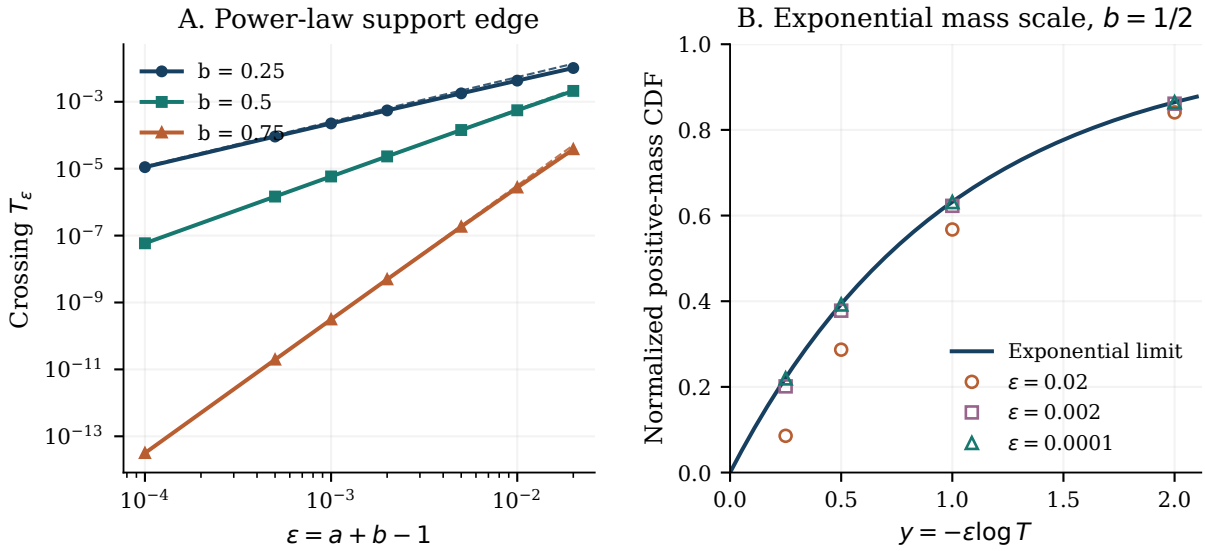


Figure 5.3: The two scales of endpoint concentration. Left: numerical crossings (markers) and the leading power laws (dashed curves) at three fixed inner orders. Right: four sampled distribution-function values for each indicated ε at $b = 1/2$, compared with the proved unit-exponential limit. These are diagnostics of the analytic results.

All limits in this section keep b fixed. As $b \downarrow 0$, the small parameter $\tau_\varepsilon/\varepsilon$ has exponent $b/(1-b) \downarrow 0$, so the stated hierarchy is not uniform at that endpoint. As $b \uparrow 1$, the exponent $1/(1-b)$ and the constants become singular. Joint limits involving either endpoint require additional analysis and are not consequences of these theorems.

5.19 Euler enclosures and the failure of an unrestricted constant one

Let $a, b > 0$ with $w \geq 1$. Define

$$f(n) = c_{2n+1} = \frac{H_{2n}^{(b)}}{(2n+1)^a}, \qquad d_k = \sum_{n=0}^{k} (-1)^n \binom{k}{n} f(n), \qquad E_N = \sum_{k=0}^{N-1} \frac{d_k}{2^{k+1}}. \tag{5.119}$$

The error is measured against $g_{a,b} = \operatorname{Im} F_{a,b}(\mathrm{i})$. On the critical line this is an Abel value; the ordinary alternating series with terms $(-1)^n f(n)$ diverges by the term test.

Theorem 5.58 (Extended one-sided Euler bounds)**.** *For every $k \geq 1$ one has $d_k < 0$, while $d_0 = 0$. The E_N decrease strictly to $g_{a,b}$, and*

$$E_N - g_{a,b} = -2^{-N} \int_{[0,1]} \frac{(1-u^2)^N}{1+u^2} \, \mathrm{d}\nu_{a,b}(u) > 0. \tag{5.120}$$

Consequently

$$E_N - M(a,b)2^{-N} < g_{a,b} < E_N \qquad (N \geq 1), \tag{5.121}$$

with the following elementary choices:

$$M(a,b) = \begin{cases} 1, & a \geq 1, \\ (1-b)^{-1}, & 0 < a < 1,\ 0 < b < 1,\ w \geq 1, \\ b/(b-1), & 0 < a < 1,\ b > 1, \\ 1 + 1/a, & 0 < a < 1,\ b = 1. \end{cases} \tag{5.122}$$

On the critical line the sharper uniform choice is proved in Corollary 5.33; its matching subcritical Euler theorem uses the positive difference measure. The choice one in the first line is the prior optimal universal constant on that restricted domain; it is not asserted to work in the other lines.

Proof. The binomial theorem and (5.44) give $d_k = \int (1-u^2)^k \, \mathrm{d}\nu(u)$. Thus $d_0 = 0$, while the strictly decreasing weight gives $d_k < 0$ for $k \geq 1$. With $m = m(\nu)$, we have $|d_k| \leq m$. Therefore the Euler series is absolutely convergent and may be summed under the measure integral:

$$\sum_{k \geq 0} \frac{d_k}{2^{k+1}} = \int \frac{\mathrm{d}\nu(u)}{1+u^2} = \operatorname{Im} F(\mathrm{i}).$$

Summing the geometric tail gives (5.120). Its weight is strictly decreasing, bounded by one, and strictly less than one for every $u > 0$. Since the negative Jordan part has no atom at zero, $0 < E_N - g < m2^{-N}$. It remains to bound m.

For $b < 1, w > 1$, rewrite (5.52) as

$$K_{a,b} = \frac{k_{w-1}}{1-b} - \big[-\zeta(b)k_a + C_{a,b}\big].$$

The two displayed densities are positive, and each has mass $1/(1-b)$ in the u measure, by zero total mass. Hence $m < 1/(1-b)$. On $w = 1$ the exact Jordan mass is $1/a = 1/(1-b)$. For $b > 1$, (5.52) is a difference of positive densities $\zeta(b)k_a$ and $k_{w-1}/(b-1) + C_{a,b}$, so $m < \zeta(b) < 1 + 1/(b-1) = b/(b-1)$.

For $b = 1$ and $0 < a < 1$, use a translation estimate rather than the singular limit of either of these bounds. Let σ_a be the probability law of $u = e^{-Y}$ with $Y \sim \mathrm{Gamma}(a,1)$. Its logarithmic density $p_a(s) = e^{-s}s^{a-1}/\Gamma(a)$ decreases strictly. If $D_{e^{-t}}$ denotes pushforward under $u \mapsto e^{-t}u$, then direct comparison of $p_a(s)$ and $p_a(s-t)$ gives

$$\|\sigma_a - D_{e^{-t}}\sigma_a\|_{\mathrm{TV}} = 2\int_0^t p_a(s) \, \mathrm{d}s \leq \frac{2t^a}{\Gamma(a+1)}.$$

The signed-measure integral

$$\widetilde{\nu} = \int_0^\infty \frac{\sigma_a - D_{e^{-t}}\sigma_a}{e^t - 1} \, \mathrm{d}t$$

converges in total variation. Its $(n-1)$st moment is $n^{-a}\int_0^\infty (1-e^{-(n-1)t})(e^t-1)^{-1}\,\mathrm{d}t = c_n$. By uniqueness it is $\nu_{a,1}$, and

$$m \le \frac{1}{\Gamma(a+1)}\int_0^\infty \frac{t^a}{e^t-1}\,\mathrm{d}t = \zeta(a+1) < 1 + 1/a.$$

Finally, for $a \ge 1$ the earlier sharper bound can be recovered without any fractional-order gap. At $a = 1$ the equivalent base density is

$$K_{1,b}(T) = -\frac{T^b}{\Gamma(b+1)} + \frac{1}{\Gamma(b)}\int_T^\infty \frac{t^{b-1}}{e^t-1}\,\mathrm{d}t.$$

Its nth Laplace value is $-n^{-b-1} + n^{-1}\sum_{j=1}^n j^{-b} = H_{n-1}^{(b)}/n$, by Tonelli and a finite geometric sum. Moment uniqueness therefore identifies this base density with (5.52) or (5.54). Both positive terms in this difference have $e^{-T}\,\mathrm{d}T$ integral one. For the second term, interchange the integrals and use $(1-e^{-t})/(e^t-1) = e^{-t}$; the first is a gamma integral. Their positive overlap makes the Jordan mass strictly less than one. For $a > 1$, convolution by k_{a-1} in the T coordinate preserves moments after multiplication by $n^{-(a-1)}$ and contracts the $L^1(e^{-T}\,\mathrm{d}T)$ norm, because $\int_0^\infty e^{-T}k_{a-1}(T)\,\mathrm{d}T = 1$. Moment uniqueness identifies the result with the present kernel, so $m < 1$ for every real $a \ge 1$. This proves all four choices. □

The same translation proof gives, when $0 < a \le 1$ and $a + b > 1$, the optional non-elementary bound

$$m \le \frac{\Gamma(a+b)\zeta(a+b)}{\Gamma(a+1)\Gamma(b)}.$$

It may improve a particular estimate, but diverges as $a + b \downarrow 1$. The critical atomic construction is therefore not replaced by this translation argument.

Proposition 5.59 (Exact finite Euler weights)**.** *For $N \ge 1$,*

$$E_N = 2^{-N}\sum_{n=0}^{N-1}(-1)^n f(n)\sum_{j=n+1}^{N}\binom{N}{j}. \tag{5.123}$$

Thus the finite sum can be formed without constructing a triangular difference table.

Proof. Substitute the definition of d_k and interchange the finite sums. The coefficient identity is

$$\sum_{k=n}^{N-1} 2^{N-k-1}\binom{k}{n} = \sum_{j=n+1}^{N}\binom{N}{j},$$

which follows by induction from Pascal's identity. Harmonic numbers and binomial tails can then each be updated successively. This rearrangement is already used in the integer-order manuscript; the new feature in the implementation here is certified rational interval arithmetic for fractional powers. □

Theorem 5.60 (A certified scope obstruction)**.** *For $a = 1/10$ and $b = 9/10$,*

$$\begin{aligned} -0.52976193228577126309462936 55 &< g_{1/10,9/10} \\ &< -0.52976193228577126309462936 53. \end{aligned} \tag{5.124}$$

In particular, $g_{1/10,9/10} < -13/25 < -1/2$. *Since* $E_1 = 0$, *the inequality* $E_N - 2^{-N} < g_{a,b}$ *fails at* $N = 1$ *for this pair. A uniform constant one cannot be extended to the entire region* $a + b \geq 1$. *The failure also occurs strictly above the threshold, with ordinary boundary convergence:*

$$-0.530844252966159326172071260 2 < g_{1/10,1} < -0.530844252966159326172071260 0 < -0.53. \tag{5.125}$$

Proof. The accompanying standard-library program `code/certify.py` evaluates (5.123) with $N = 96$ by outward rational interval arithmetic. Every fractional power is bounded by integer tenth-root inequalities, as detailed in Section 5.19.1. The critical mass bound is the exact rational $M = 10$. Subtracting $10\,2^{-96}$ from the lower endpoint of the finite Euler enclosure and leaving its upper endpoint unchanged gives the interval recorded in `data/exact_certificates.json`. Outward decimal rounding yields (5.124); exact rational comparisons verify its upper endpoint is less than $-13/25$. No numerical polylogarithm, floating-point root, or unproved identity enters this certificate. For $(a, b) = (1/10, 1)$ the same computation uses $M = 1 + 1/a = 11$ and proves (5.125). Its ordinary boundary convergence follows from $a + b = 11/10 > 1$ and Corollary 5.19. □

This is *not* a counterexample to the baseline theorem, which explicitly assumes $a \geq 1$. It is a counterexample to dropping that hypothesis while retaining the same universal error constant.

Proposition 5.61 (The exact critical constant for fixed parameters)**.** *On* $a + b = 1$, *put* $d\sigma = -K_{1-b,b}(-\log u)\,du$. *The scaled errors* $2^N(E_N - g_{a,b})$ *decrease strictly to zero for* $N \geq 1$. *The smallest constant in the non-strict estimate* $E_N - g_{a,b} \leq C2^{-N}$, *valid for all integers* $N \geq 1$ *at that fixed pair, is* $C = -2g_{a,b}$.

Proof. The endpoint atom contributes zero to (5.120) when $N \geq 1$. Hence

$$2^N(E_N - g) = \int_0^1 \frac{(1-u^2)^N}{1+u^2}\,\mathrm{d}\sigma(u).$$

Strict pointwise decrease and dominated convergence prove the first claim. Its largest value occurs at $N = 1$ and is $2(E_1 - g) = -2g$. For the corresponding strict inequality the infimum of admissible constants is the same, but equality at $N = 1$ prevents attaining it. □

5.19.1 Exact arithmetic and independent native evidence

The new standard-library verifier encloses every rational fractional power by integer root inequalities on a 10^{100} grid. Products, harmonic sums and finite Euler weights are then combined as exact rational intervals. It certifies 616 strictly positive Hausdorff-difference signs, 112 negative Euler increments and seven Gaussian analytic-value enclosures at 192 Euler terms. The six triangle cases use the rational budget 57/50; the remaining axis case uses 5/4. The axis case $b = 3/2$ also proves $-2g_{0,3/2} > 227/200$, supplying the finite arithmetic premise in the axis-maximum theorem.

A separate native Wolfram calculation at 100 working digits evaluates the outer Gamma integral of a depth-one polylogarithm, or the direct axis value. All seven values lie inside their rational Euler enclosures. The native quadrature is a numerical diagnostic, not interval quadrature. The sign and remainder theorems depend on the written positive-measure proofs. The original threshold report's tenth-root certificates are preserved unchanged with its delivered code and data.

5.20 A sharp Euler constant at every positive real order

The Gaussian maximum does not bound arbitrary Euler errors merely because it bounds the value of the function. A contraction inequality for the Euler kernel supplies the missing implication. It removes the divergent parameter-dependent budgets near the signed-measure threshold, while retaining the stronger constant one on the earlier domain $a \geq 1$.

Lemma 5.62 (A difference-of-powers bound)**.** *For $m \geq 2$, $0 \leq r \leq 1$ and $0 \leq v \leq 1$,*

$$0 \leq (1 - rv)^m - (1 - r)^m \leq \frac{1 - v}{1 + v}.$$

For $0 < v < 1$, equality on the right is possible only when $m = 2$, and then it holds precisely at $r = 1/(1 + v)$.

Proof. The endpoints $v = 0, 1$ are immediate. Fix $0 < v < 1$ and put $q = v^{1/(m-1)}$. Differentiation shows that the unique maximum over $r \in [0, 1]$ is at $r = (1 - q)/(1 - vq)$, where its value is

$$\frac{(1 - v)^m}{(1 - v^{m/(m-1)})^{m-1}}.$$

Let $p = m/(m - 1) \in (1, 2]$. The function $p \mapsto \log(1 - v^p)$ has strictly negative second derivative $-(\log v)^2 v^p/(1 - v^p)^2$. Its chord between $p = 1$ and $p = 2$ therefore gives

$$1 - v^p \geq (1 - v)(1 + v)^{p-1},$$

strictly when $1 < p < 2$. Substitution in the maximum proves the bound and both equality assertions. □

Lemma 5.63 (Contraction of the Euler tail kernel)**.** *For an integer $N \geq 1$, set $H_N(t) = (1 - t)^N/(1 + t)$ on $[0, 1]$. For $0 \leq r, v \leq 1$,*

$$0 \leq H_N(rv) - H_N(r) \leq \frac{1 - v}{1 + v}. \tag{5.126}$$

If $0 < v < 1$, equality on the right holds precisely when $N = 1$ and $r = 1$.

Proof. For $N = 1$, the difference between the right side and the middle expression factors exactly as

$$\frac{(1 - v)(1 - r)(1 - rv)}{(1 + v)(1 + rv)(1 + r)}.$$

For $N \geq 2$, the uniformly convergent positive expansion

$$H_N(t) = \sum_{k \geq 0} \frac{(1 - t)^{N+k}}{2^{k+1}}$$

expresses its difference as a probability-weighted sum of the preceding power differences. Each is bounded as claimed, and the terms with $N + k > 2$ make the upper inequality strict for $0 < v < 1$. Decrease of H_N proves the lower inequality. □

Let ω_b be the probability measure

$$d\omega_b(v) = \frac{(-\log v)^{b-1}}{\Gamma(b)}\,dv, \qquad 0 < v < 1.$$

Let π_a be the law of $X = e^{-T_a}$ with $T_a \sim \mathrm{Gamma}(a,1)$ when $a > 0$, and put $\pi_0 = \delta_1$. These measures are independent. They retain ordinary probability mass throughout the parameter domain, including the critical and subcritical regions where the unweighted one-variable moment measure is not finite.

Theorem 5.64 (All-order Euler representation and the sharp universal constant)**.** *For every $a \geq 0, b > 0$, define d_k and E_N from the coefficients c_{2n+1} as in (5.119), and use the analytic value $g_{a,b} = \operatorname{Im} F_{a,b}(i)$. Then $d_0 = 0$, $d_k < 0$ for $k \geq 1$, and E_N decreases strictly to $g_{a,b}$. Its exact error is*

$$2^N(E_N - g_{a,b}) = \iint \frac{H_N(x^2v^2) - H_N(x^2)}{1-v}\,d\pi_a(x)\,d\omega_b(v). \tag{5.127}$$

Consequently, for every $N \geq 1$,

$$0 < 2^N(E_N - g_{a,b}) \leq C(b), \qquad C(b) = \beta(b) + 2^{-b}\eta(b). \tag{5.128}$$

Equality in the upper bound holds exactly when $a = 0$ and $N = 1$. The sharp uniform constant over all nonnegative outer orders is

$$C_* = C(b_*),$$

where b_* is the unique nondegenerate maximum from Theorem 5.32. It satisfies

$$\frac{227}{200} < C_* < \frac{1+\sqrt{2}}{2} < \frac{5}{4}.$$

For positive a, b the strict inequality $0 < E_N - g_{a,b} < C_* 2^{-N}$ holds, and C_* is its least possible uniform constant. When $a = 0$ is included, the non-strict upper bound is attained only at $a = 0, b = b_*, N = 1$.

Proof. The elementary moment laws give, for $n \geq 0$,

$$c_{2n+1} = \iint x^{2n}\frac{1-v^{2n}}{1-v}\,d\pi_a(x)\,d\omega_b(v).$$

Indeed the two factors integrate to $(2n+1)^{-a}$ and $H_{2n}^{(b)}$. The quotient is a finite geometric sum, so this identity includes $n = 0$ without an endpoint singularity. Finite binomial expansion now gives

$$d_k = \iint \frac{(1-x^2)^k - (1-x^2v^2)^k}{1-v}\,d\pi_a(x)\,d\omega_b(v).$$

For $k \geq 1$ its integrand is strictly negative and has absolute value at most $2k$, by the mean-value theorem. Finite summation is therefore legitimate for every pair of parameters.

The Gaussian double kernel, or the axis formula when $a = 0$, gives

$$g_{a,b} = \iint \frac{(1+x^2)^{-1} - (1+x^2v^2)^{-1}}{1-v}\,d\pi_a(x)\,d\omega_b(v).$$

Its integrand is $-x^2(1+v)/((1+x^2)(1+x^2v^2))$, hence bounded. Subtract the finite geometric sum defining E_N from this identity to obtain (5.127); no divergent boundary series is used.

Apply Lemma 5.63 with $r = x^2$ and its variable v replaced by v^2. The resulting majorant is

$$\frac{1 - v^2}{(1 - v)(1 + v^2)} = \frac{1 + v}{1 + v^2}.$$

Its ω_b integral is $C(b)$ by the axis identity. It is bounded by $(1 + \sqrt{2})/2$, so the exact error tends to zero at least geometrically. The strict signs of d_k give the strict decrease of E_N. The contraction equality criterion, together with $\pi_0 = \delta_1$ and the interior densities for $a, b > 0$, proves the fixed-b equality claim.

The unique axis maximum supplies the uniform constant and its bounds. Sharpness over $a, b > 0$ follows by fixing $b = b_*$, taking $a \downarrow 0$ at $N = 1$, and using bounded convergence in the Gaussian probability integral. Thus the supremum is approached at the axis, while inclusion of the axis gives the stated unique attaining triple. □

This theorem closes the previously open universal real-order Euler-budget question. The separate contraction argument is essential: Gaussian magnitude alone would not control later scaled errors. In particular, this result does not say that their maximum at each positive parameter pair is the first error, nor that the scaled errors always decrease. The constant one remains sharper on $a \geq 1$, and the sharp critical constant $\pi/4 + \log 2/2$ remains sharper on $a + b \leq 1$. The decimal diagnostic $C_* \approx 1.13656110333950956095$ inherits the axis experiment's status in the first proof. The independent rational enclosure in Theorem 5.36 now certifies this constant to the displayed width and proves that 57/50 is a valid uniform rational Euler budget at all positive real orders. The exact definition and universal contraction proof do not rely on a numerical stationary-point solver.

5.21 A uniform rational kernel inequality

The independent continuations [104, 105] separate the first Euler step from every later one. Retain the residual function $H_N(t)$ above, and define the continuously extended kernel

$$K_N(p, y) = \frac{H_N(py^2) - H_N(p)}{1 - y}, \qquad K_N(p, 1) = -2pH_N'(p).$$

It is nonnegative. The following finite polynomial certificate and analytic mixture reduction prove a stronger uniform budget for $N \geq 2$.

Theorem 5.65 (Kernel bound for every later truncation)**.** *For every integer $N \geq 2$ and $0 \leq p, y \leq 1$,*

$$0 \leq K_N(p, y) < \frac{9}{8}. \tag{5.129}$$

The lower bound is strict when $p > 0$ and $y < 1$. The strict upper bound includes the continuous boundary values.

The proof has two finite base cases and an all-N envelope. We give enough information to reconstruct every rational certificate without a symbolic algebra system.

5.21.1 Two explicit polynomial base cases

Multiplication by the positive denominator shows that $K_N < 9/8$ is equivalent to positivity of

$$P_N(p, y) = 8(1 + p)(1 + py^2)\{9/8 - K_N(p, y)\}.$$

For $N = 2, 3$ these are the following polynomials:

$$\begin{aligned} P_2 &= 8y^2(1+y)p^3 + (8y^3 + 17y^2 + 8y + 8)p^2 \\ &\quad + (9y^2 - 24y - 15)p + 9, \end{aligned} \tag{5.130}$$

$$\begin{aligned} P_3 = &- 8y^2(y+1)(y^2+1)p^4 \\ &+ (-8y^5 - 8y^4 + 16y^3 + 16y^2 - 8y - 8)p^3 \\ &+ (24y^3 + 33y^2 + 24y + 24)p^2 + (9y^2 - 32y - 23)p + 9. \end{aligned} \tag{5.131}$$

An identity check can avoid rational-function division altogether:

$$\begin{aligned} (1-y)P_N = &\, 9(1-y)(1+p)(1+py^2) \\ &- 8\{(1-py^2)^N(1+p) - (1-p)^N(1+py^2)\}. \end{aligned} \tag{5.132}$$

Both verifiers check this identity or independently reconstruct its quotient.

For a polynomial $P(x, y) = \sum c_{kl} x^k y^l$ of bidegree at most (r, s), its tensor Bernstein coefficients on the unit square are

$$B_{ij} = \sum_{k \le i,\, l \le j} c_{kl} \frac{\binom{i}{k}}{\binom{r}{k}} \frac{\binom{j}{l}}{\binom{s}{l}}. \tag{5.133}$$

The Bernstein basis functions are nonnegative and sum to one. Hence a strictly positive minimum of these coefficients proves strict positivity of P on the whole square, including its boundary.

For a mesh of denominator d, first substitute $p = (u + i)/d$, $y = (v + j)/d$ on each cell. Formula (5.133) then gives an exact certificate. The data are summarized by

Polynomial	Bidegree	Mesh	Coefficients	Smallest coefficient
P_2	$(3, 3)$	8×8	1024	631/12288
P_3	$(4, 5)$	4×4	480	11337/40960

Every coefficient is stored as a rational number in the accompanying JSON file. More importantly, the standard-library verifier regenerates all of them from (5.130)–(5.133); it does not trust a stored minimum. These finitely many exact additions and multiplications prove the two base cases.

5.21.2 An envelope for binomial differences

Introduce

$$J_m(p, y) = \frac{(1 - py^2)^m - (1-p)^m}{1-y}, \qquad m \ge 2.$$

For fixed $0 < y < 1$, differentiation in p gives a unique interior maximum at

$$p_m(y) = \frac{1 - y^{2/(m-1)}}{1 - y^{2m/(m-1)}}. \tag{5.134}$$

Indeed the derivative changes sign exactly when $1 - p = y^{2/(m-1)}(1 - py^2)$. Its sign is positive at $p = 0$ and negative at $p = 1$. Substitution yields the exact envelope

$$J_m(p, y) \le B_m(y) := (1+y) \left(\frac{1 - y^2}{1 - y^{2m/(m-1)}} \right)^{m-1}. \tag{5.135}$$

The endpoint values follow by continuity.

Lemma 5.66 (The binomial envelope decreases in its index)**.** *For every* $0 < y < 1$, $B_{m+1}(y) < B_m(y)$ *for* $m \geq 2$.

Proof. Put $k = m - 1$, $t = -2\log y > 0$ and $h(v) = \log(1 - e^{-(1+v)t})$. Direct differentiation gives $h''(v) < 0$ for $v \geq 0$. Apart from $\log(1 + y)$, the logarithm of (5.135) is

$$k\{h(0) - h(1/k)\} = -\frac{h(1/k) - h(0)}{1/k}.$$

The secant slope of a strictly concave function decreases as the right endpoint increases. Thus the displayed negative secant slope decreases strictly when k increases. Exponentiation proves the claim. □

5.21.3 The fourth envelope is below nine eighths

Set $y = t^3$, $0 \leq t \leq 1$. The inequality $B_4(y) < 9/8$ is equivalent away from $t = 1$ to

$$9(1 - t^8)^3 - 8(1 + t^3)(1 - t^6)^3 = (1 - t^2)^3 Q(t) > 0, \tag{5.136}$$

where

$$\begin{aligned} Q(t) = {} & 9t^{18} + 27t^{16} - 8t^{15} + 54t^{14} - 24t^{13} + 82t^{12} - 48t^{11} \\ & + 84t^{10} - 56t^9 + 60t^8 - 48t^7 + 34t^6 - 24t^5 + 6t^4 - 8t^3 + 3t^2 + 1. \end{aligned} \tag{5.137}$$

The degree-18 Bernstein coefficients after $t = u/2$ and $t = (1 + u)/2$ have respectively the strictly positive minima

$$\frac{192789}{262144}, \qquad \frac{690189}{1949696}. \tag{5.138}$$

There are only 38 coefficients. Formula (5.133) with one variable regenerates them. At $y = 1$ the envelope limit is $B_4(1) = 2(3/4)^3 = 27/32 < 9/8$, so the endpoint is covered as well.

5.21.4 The infinite family follows without an infinite computation

Since $1/(1 + v) = \sum_{j \geq 0}(1 - v)^j/2^{j+1}$ on $[0, 1]$,

$$K_N(p, y) = \sum_{j=0}^{\infty} 2^{-j-1} J_{N+j}(p, y). \tag{5.139}$$

For $y < 1$ this follows by subtracting the two convergent geometric series. If $N \geq 4$, (5.135), Lemma 5.66 and (5.136) give

$$J_{N+j}(p, y) \leq B_{N+j}(y) \leq B_4(y) < 9/8.$$

Averaging proves $K_N \leq B_4 < 9/8$ for every $N \geq 4$. Continuity covers $y = 1$. The polynomial base cases cover $N = 2, 3$, completing Theorem 5.65.

Corollary 5.67 (A smaller uniform budget after the first step)**.** *For every* $a \geq 0, b > 0$ *and integer* $N \geq 2$*,*

$$0 < 2^N(E_N - g_{a,b}) < \frac{9}{8}.$$

Proof. Take expectations of the strict pointwise inequality in (5.127), with $p = x^2$ and $y = v$. The measures are probability measures, and their kernel is strictly positive almost everywhere, including the $a = 0$ axis. □

The first-step maximum C_* is strictly larger than $9/8$, already by $C_* > 227/200$ in Theorem 5.64. This does not imply that scaled errors decrease at each fixed pair: the exact incoming replay certifies $R_2(4,1) > R_1(4,1)$, where $R_N = 2^N(E_N - g)$. The unscaled E_N still decrease by the theorem above.

Remark 5.68 (Nature of the finite computation). The inequality is a computer-assisted ordinary proof: three stated polynomial identities, a stated rational basis conversion, and 1542 strictly positive rational numbers close its finite component. The all-index reduction is analytic. The check is reproducible under Python's optimized mode and is not a test of finitely many real sample points. The supplied SymPy replay independently derives the two rational kernels rather than importing their coefficient arrays.

The fresh standard-library replay regenerates all 1,542 coefficients on 82 cells, checks the three defining polynomial identities, and rejects a corrupted polynomial. The same verifier succeeds with optimized Python, whose disabled assertions do not bypass its explicit validations. A separate SymPy program reconstructs the base rational kernels. Receipts are preserved in `verificati on/sixth-replay/sharp/`. The independently replayed golden/Cayley delivery uses another positive-mixture route and its own finite base certificates; it is supporting evidence for the same bound, not a second definition of the sharp all-index constant.

5.22 Strict depth, signed oscillation and minimal positive compensation

The real-order depth-two factorization suggests a structural question at higher strict depth: can the initial zero moments account for every sign change of a representing density? The continuation [106] answers this when the leading indices are positive integers and the last index is positive real. Its construction gives a minimal polynomial compensator, a complete upper-arc count and exact integral identities. The ordinary positive transform supplies the geometry only after the signed-density zeros and initial moment cancellation have been established. The proof chain follows those dependencies here.

5.22.1 Strict single-variable notation

For an index of depth d, define

$$L_{\boldsymbol{s}}(z) = \mathrm{Li}_{s_1,\dots,s_d}(z,1,\dots,1) = \sum_{n_1>\cdots>n_d\geq 1} \frac{z^{n_1}}{n_1^{s_1}\cdots n_d^{s_d}} = \sum_{n\geq 1} c_{\boldsymbol{s}}(n) z^n, \qquad |z|<1. \tag{5.140}$$

Our parameter domain is

$$s_1,\dots,s_{d-1} \in \mathbb{N}_{\geq 1}, \qquad s_d = b > 0. \tag{5.141}$$

At depth one, this means any $b > 0$. Indices are strictly decreasing, not weakly decreasing. Only the outer summation variable receives z^{n_1}. This convention agrees with the single-variable convention in the manuscript and in the multiple-polylogarithm literature. The principal branch conventions are those fixed at the start of the book.

Write

$$C_{\boldsymbol{s}} = c_{\boldsymbol{s}}(d) = \frac{1}{d^{s_1}(d-1)^{s_2}\cdots 2^{s_{d-1}}} > 0. \tag{5.142}$$

The empty product at depth one is one. The value of $C_{\boldsymbol{s}}$ is independent of b, because the innermost index of the first admissible chain is one. Also $c_{\boldsymbol{s}}(1) = \cdots = c_{\boldsymbol{s}}(d-1) = 0$. Throughout, principal continuation means continuation to

$$\Omega = \mathbb{C} \setminus [1, \infty).$$

The word "zero" refers to a zero of a holomorphic function, unless the imaginary part or a signed density is explicitly specified.

5.22.2 Main result

Theorem 5.69 (All-depth compensation and zero geometry)**.** *Let* (5.141) *hold. There is a smooth real density* $\kappa_{\boldsymbol{s}} \in L^1(0,1)$ *such that*

$$\int_0^1 u^{n-1} \kappa_{\boldsymbol{s}}(u)\, \mathrm{d}u = c_{\boldsymbol{s}}(n), \qquad \|\kappa_{\boldsymbol{s}}\|_1 \le 2^{d-1}.$$

It has exactly $d-1$ *simple zeros* $0 < r_1 < \cdots < r_{d-1} < 1$*, with alternating signs and positive sign near one. Set*

$$P_{\boldsymbol{s}}(u) = \prod_{j=1}^{d-1} (u - r_j), \qquad D_{\boldsymbol{s}}(z) = \prod_{j=1}^{d-1} (1 - r_j z).$$

Then $\mathrm{d}\mu_{\boldsymbol{s}} = P_{\boldsymbol{s}} \kappa_{\boldsymbol{s}}\, \mathrm{d}u$ *is a positive measure of mass* $C_{\boldsymbol{s}}$*, and*

$$L_{\boldsymbol{s}}(z) = \frac{z^d}{D_{\boldsymbol{s}}(z)} H_{\boldsymbol{s}}(z), \qquad H_{\boldsymbol{s}}(z) = \int_0^1 \frac{\mathrm{d}\mu_{\boldsymbol{s}}(u)}{1 - zu}. \tag{5.143}$$

The polynomial $D_{\boldsymbol{s}}$ *is the unique normalized polynomial of smallest degree making* $D_{\boldsymbol{s}} L_{\boldsymbol{s}} / z^d$ *a positive Hausdorff–Stieltjes transform. In particular,* $L_{\boldsymbol{s}}$ *has no zero in* Ω *other than its zero of multiplicity* d *at zero.*

For every $0 < \rho \le 1$*, there are exactly* $d-1$ *simple zeros*

$$0 < \theta_1(\rho) < \cdots < \theta_{d-1}(\rho) < \pi$$

of $\theta \mapsto \operatorname{Im} L_{\boldsymbol{s}}(\rho e^{\mathrm{i}\theta})$*. With* $\alpha_k = k\pi/d$ *they satisfy*

$$\alpha_k - \arcsin(\rho \sin \alpha_k) < \theta_k(\rho) < \alpha_k. \tag{5.144}$$

Every branch satisfies $\theta_k'(\rho) < 0$ *for* $0 < \rho \le 1$ *(with the one-sided derivative at one). The imaginary part starts positive and alternates sign at these zeros; the value of* $L_{\boldsymbol{s}}$ *at the* k*th crossing is real with sign* $(-1)^k$*.*

The proof is divided into operator, oscillation, compensation, and angular-count steps below. None of those steps is inferred from the finite verification suite.

5.23 Two bounded operators on signed moment densities

For $f \in L^1(0,1)$ define its moments by

$$m_n(f) = \int_0^1 u^{n-1} f(u)\, \mathrm{d}u, \qquad n \ge 1.$$

We use the two operators

$$(\mathsf{A}f)(u) = \int_u^1 \frac{f(v)}{v}\,\mathrm{d}v, \tag{5.145}$$

$$(\mathsf{B}f)(u) = \int_0^u \frac{f(v)}{1-v}\,\mathrm{d}v - \int_u^1 \frac{f(v)}{v}\,\mathrm{d}v. \tag{5.146}$$

For every fixed $0 < u < 1$, both integrals in (5.146) are absolutely convergent. In particular, this definition never subtracts two divergent endpoint integrals. The operator A is the outer-index-raising operator already present at depth two in the manuscript [106]. The companion B encodes a strict prefix sum of moments.

Lemma 5.70 (Norm, derivative, and moment identities)**.** *Both operators map $L^1(0,1)$ to itself and satisfy*

$$\|\mathsf{A}f\|_1 \le \|f\|_1, \qquad \|\mathsf{B}f\|_1 \le 2\|f\|_1. \tag{5.147}$$

Their moments are

$$m_n(\mathsf{A}f) = \frac{m_n(f)}{n}, \qquad m_n(\mathsf{B}f) = \frac{1}{n}\sum_{j=1}^{n-1} m_j(f). \tag{5.148}$$

If f is continuous in the open interval, then

$$(\mathsf{A}f)'(u) = -\frac{f(u)}{u}, \qquad (\mathsf{B}f)'(u) = \frac{f(u)}{u(1-u)}. \tag{5.149}$$

Moreover $\mathsf{A}f$ extends continuously to one with value zero, and $m_1(\mathsf{B}f) = 0$.

Proof. Tonelli's theorem applied to absolute values gives

$$\int_0^1 \int_u^1 \frac{|f(v)|}{v}\,\mathrm{d}v\,\mathrm{d}u = \int_0^1 |f(v)|\,\mathrm{d}v.$$

Similarly,

$$\int_0^1 \int_0^u \frac{|f(v)|}{1-v}\,\mathrm{d}v\,\mathrm{d}u = \int_0^1 |f(v)|\,\mathrm{d}v.$$

These identities prove the two operator bounds and justify every following application of Fubini to a signed integrand. For the first operator,

$$m_n(\mathsf{A}f) = \int_0^1 \frac{f(v)}{v}\left(\int_0^v u^{n-1}\,\mathrm{d}u\right)\mathrm{d}v = \frac{m_n(f)}{n}.$$

The moment of the first term of $\mathsf{B}f$ is

$$\frac{1}{n}\int_0^1 \frac{1-v^n}{1-v} f(v)\,\mathrm{d}v = \frac{1}{n}\sum_{j=1}^{n} m_j(f).$$

Subtracting $m_n(f)/n$ yields the strict prefix in (5.148). The derivative formulas follow locally from the fundamental theorem of calculus. Finally, $\int_u^1 f(v)/v\,\mathrm{d}v \to 0$ as $u \uparrow 1$, because $1/v$ is bounded near one and f is integrable there. □

5.23.1 Construction at arbitrary depth

The positive starting density is

$$f_b(u) = \frac{(-\log u)^{b-1}}{\Gamma(b)}, \qquad b > 0. \tag{5.150}$$

Its mass is one and the Gamma integral gives $m_n(f_b) = n^{-b}$. Define, with composition read from right to left,

$$\kappa_{\boldsymbol{s}} = \mathsf{A}^{s_1-1}\mathsf{B}\mathsf{A}^{s_2-1}\mathsf{B}\cdots\mathsf{A}^{s_{d-1}-1}\mathsf{B}f_b. \tag{5.151}$$

There are $d-1$ occurrences of B; all powers of A are nonnegative integer powers. At depth one the right side is simply f_b.

Proposition 5.71 (Moment realization and continuation)**.** *The density* (5.151) *is smooth on* $(0,1)$*, belongs to* $L^1(0,1)$*, and satisfies*

$$m_n(\kappa_{\boldsymbol{s}}) = c_{\boldsymbol{s}}(n), \qquad \|\kappa_{\boldsymbol{s}}\|_1 \le 2^{d-1}. \tag{5.152}$$

The principal continuation is

$$L_{\boldsymbol{s}}(z) = z\int_0^1 \frac{\kappa_{\boldsymbol{s}}(u)}{1-zu}\,\mathrm{d}u, \qquad z \in \Omega. \tag{5.153}$$

The density is the unique finite signed measure realizing these moments.

Proof. The coefficient recurrence for a nonempty tail $\boldsymbol{t}$ is

$$c_{(a,\boldsymbol{t})}(n) = \frac{1}{n^a}\sum_{j=1}^{n-1} c_{\boldsymbol{t}}(j).$$

It is exactly the effect of $\mathsf{A}^{a-1}\mathsf{B}$ in (5.148). Induction proves the moments and the norm bound. The derivative identities prove local smoothness, starting from (5.150). Expanding the denominator in the open disk is justified by the integrable bound $|\kappa|/(1-|z|)$, and gives the series (5.140).

On a compact subset of Ω, $|1-zu|$ is bounded away from zero uniformly in $u \in [0,1]$. The signed integral is consequently holomorphic there and continues the same germ. If two finite signed measures have these moments, their difference integrates every polynomial to zero. Uniform polynomial approximation on $[0,1]$, followed by uniqueness of a finite measure from its continuous-function integrals, proves equality. □

Corollary 5.72 (Ordinary convergence on the nontrivial unit circle)**.** *For* (5.141)*,* $c_{\boldsymbol{s}}(n) \to 0$ *and*

$$\sum_{n\ge 1} |c_{\boldsymbol{s}}(n+1) - c_{\boldsymbol{s}}(n)| \le \|\kappa_{\boldsymbol{s}}\|_1 \le 2^{d-1}. \tag{5.154}$$

Thus the original series (5.140) *converges at every* $|z| = 1$*,* $z \ne 1$*, and agrees there with* (5.153)*.*

Proof. Dominated convergence proves the coefficient limit. Also

$$|c(n+1) - c(n)| \le \int_0^1 u^{n-1}(1-u)|\kappa(u)|\,\mathrm{d}u.$$

Summing the positive majorants gives (5.154). Summation by parts against the bounded partial sums of z^n then proves convergence when $|z| = 1$, $z \ne 1$. Abel's limiting theorem identifies its sum with the radial limit of the disk series, which in turn equals the continuous signed integral at that point. □

This conclusion concerns the parameter domain of this article. It is not a claim that all of the manuscript's more general fractional-order boundary series converge ordinarily.

5.24 Exact oscillation and two interlacing laws

The central mechanism is a balance between moment orthogonality and Rolle's theorem. Vanishing moments force sign changes; the two derivative identities prevent extra zeros. This proof does not assume a variation-diminishing theorem for fractional integration.

Lemma 5.73 (Vanishing moments force sign changes)**.** *Suppose a nonzero continuous real $f \in L^1(0,1)$ has finitely many zeros and*

$$\int_0^1 u^j f(u)\,\mathrm{d}u = 0 \qquad (0 \le j \le m-1).$$

Then f has at least m sign changes in $(0,1)$.

Proof. If its sign changes were $t_1, \dots, t_k$ with $k < m$, the polynomial $Q(u) = \prod_{j=1}^{k}(u - t_j)$ would have degree at most $m-1$ and Qf would have one fixed nonzero sign except at finitely many points. Its integral would be nonzero, contradicting orthogonality. A zero at which no sign change occurs is not included in Q. □

Theorem 5.74 (Exact density zeros)**.** *For (5.141), $\kappa_{\boldsymbol{s}}$ has exactly $d-1$ distinct simple zeros in $(0,1)$. Its sign alternates at those zeros, and it is positive near one.*

Proof. At depth one, f_b is strictly positive and has no zero. Suppose first that a density f at depth $d-1$ has exactly $d-2$ simple zeros. By (5.149), the derivative of $\mathsf{B}f$ has exactly those zeros. Rolle's theorem therefore bounds the number of distinct zeros of $\mathsf{B}f$ by $d-1$. This also rules out an infinite set of zeros: any d distinct zeros would already force too many zeros of the derivative.

The first $d-1$ moments of the new density vanish, by (5.148). Lemma 5.73 supplies at least $d-1$ sign changes, so the upper bound is attained. There is one derivative zero between each pair of consecutive new zeros, using all $d-2$ derivative zeros. None remains at a new zero. Hence every new zero is simple.

Now suppose f is a depth-d density with exactly $d-1$ simple zeros, and put $g = \mathsf{A}f$. We have $g(1) = 0$. If g had k interior zeros, Rolle's theorem between successive interior zeros and between the largest one and the endpoint one would give at least k zeros of $g' = -f/u$. Thus $k \le d-1$. Its first $d-1$ moments still vanish, so the same lower bound gives exactly $d-1$ sign changes. The derivative zeros are now all allocated to those successive intervals, including the last interval ending at one. Consequently the interior zeros of g are simple. Repeating this argument for each occurrence of A and B proves the zero count.

Let $P(u)$ be the monic polynomial whose roots are the resulting $d-1$ zeros. The product $P\kappa_{\boldsymbol{s}}$ has a fixed strict sign off its roots. Orthogonality to degrees at most $d-2$ gives

$$\int_0^1 P(u)\kappa_{\boldsymbol{s}}(u)\,\mathrm{d}u = \int_0^1 u^{d-1}\kappa_{\boldsymbol{s}}(u)\,\mathrm{d}u = C_{\boldsymbol{s}} > 0.$$

Thus $P\kappa_{\boldsymbol{s}}$ is positive, and since P is positive to the right of its largest root, so is $\kappa_{\boldsymbol{s}}$. □

Corollary 5.75 (Index raising and depth raising interlace)**.** *For $d \ge 2$, if $f = \kappa_{\boldsymbol{s}}$ has roots $r_1 < \cdots < r_{d-1}$ and $\mathsf{A}f$ has roots $q_1 < \cdots < q_{d-1}$, then*

$$q_1 < r_1 < q_2 < r_2 < \cdots < q_{d-1} < r_{d-1} < 1. \tag{5.155}$$

Thus raising the first index by one moves every density zero strictly toward zero. For $d \ge 3$, if a depth-$(d-1)$ density has roots $t_1 < \cdots < t_{d-2}$ and $\mathsf{B}f$ has roots $r_1 < \cdots < r_{d-1}$, then

$$r_1 < t_1 < r_2 < \cdots < t_{d-2} < r_{d-1}. \tag{5.156}$$

Proof. At $d = 2$, depth raising creates one interior zero and there is no previous zero to interlace. For the stated ranges, these are precisely the derivative-zero allocations in the preceding proof. In the A case the last interval is $(q_{d-1}, 1)$; in the B case all intervals are between consecutive interior zeros. The number of derivative zeros equals the number of allocated intervals, so the inequalities are strict and exhaustive. □

5.24.1 Positive seeds, terminal mixtures, and a Lerch extension

The preceding proof used only positivity, smoothness, integrability, and the positive mass of the initial density. The explicit form (5.150) was needed to identify its moments as n^{-b}.

Theorem 5.76 (Positive-seed extension)**.** *Let f be a smooth strictly positive density in $L^1(0,1)$, of mass $M_0 > 0$. Replace f_b by f in (5.151). All density-zero, interlacing, compensation, slit-plane nonvanishing, and angular-count conclusions of this article remain valid. The norm bound becomes $2^{d-1}M_0$, and the mass of the positive compensated measure is*

$$\frac{M_0}{d^{s_1}(d-1)^{s_2}\cdots 2^{s_{d-1}}}.$$

Proof. The operator identities still hold. After $d-1$ strict prefix steps, the first $d-1$ moments vanish and the first nonzero moment is the stated positive number. The proof of Theorem 5.74 therefore applies unchanged. The arguments in the next two sections use only these moment and oscillation properties, and so establish the remaining assertions as well. □

In particular, for fixed leading indices, every finite positive linear combination

$$\sum_{j=1}^{N} a_j L_{s_1,\dots,s_{d-1},b_j}(z), \qquad a_j > 0, \quad b_j > 0, \tag{5.157}$$

has the same nonvanishing and exact angular-count properties. Its seed is $\sum a_j f_{b_j}$. This is a special convex-closure statement; sums of arbitrary zero-free holomorphic functions need not be zero-free.

For a second application, take $\lambda > -1$ and

$$f_{b,\lambda}(u) = u^{\lambda}\frac{(-\log u)^{b-1}}{\Gamma(b)}.$$

Its moments are $(n+\lambda)^{-b}$ and its mass is $(1+\lambda)^{-b}$. The resulting strict shifted-terminal family is

$$L^{(\lambda)}_{s_1,\dots,s_{d-1};b}(z) = \sum_{n_1>\cdots>n_d\ge 1} \frac{z^{n_1}}{n_1^{s_1}\cdots n_{d-1}^{s_{d-1}}(n_d+\lambda)^b}. \tag{5.158}$$

It has the same minimal compensator degree $d-1$ and exactly $d-1$ angular zeros for $0 < \rho \le 1$. At depth one it is the usual z-multiple of a Lerch series. No assertion is made here about shifts in all denominators simultaneously.

5.25 Canonical positive compensation and slit-plane nonvanishing

Fix $\boldsymbol{s}$ and abbreviate $\kappa = \kappa_{\boldsymbol{s}}$, $P = P_{\boldsymbol{s}}$, $D = D_{\boldsymbol{s}}$, and $C = C_{\boldsymbol{s}}$. Empty products at depth one are one. The preceding section proves that

$$P(u)\kappa(u) > 0 \quad \text{for all } u \in (0,1) \setminus \{r_1, \dots, r_{d-1}\}. \tag{5.159}$$

Since P is bounded on $[0,1]$, the positive measure $\mathrm{d}\mu = P\kappa\,\mathrm{d}u$ is finite.

Lemma 5.77 (Polynomial cancellation)**.** *Suppose $m_1(\kappa) = \cdots = m_{d-1}(\kappa) = 0$. For any real polynomial E of degree at most $d-1$, the removable quotient at zero satisfies*

$$E(z)\frac{L(z)}{z^d} = \int_0^1 \frac{u^{d-1}E(1/u)\kappa(u)}{1-zu}\,\mathrm{d}u. \tag{5.160}$$

Here $u^{d-1}E(1/u)$ is a polynomial, including at $u = 0$.

Proof. First subtract the initial $d-1$ zero moments in (5.153) to obtain

$$L(z) = z^d \int_0^1 \frac{u^{d-1}\kappa(u)}{1-zu}\,\mathrm{d}u.$$

For $0 \le j \le d-1$, multiplying this last transform by z^j and moving the factor z^j to the density changes it to $u^{d-1-j}\kappa(u)$. The difference is a finite linear combination of the moments of degrees $d-1-j, \ldots, d-2$, all zero. Applying this monomial identity to every coefficient of E proves (5.160) in the disk and hence throughout Ω. □

Theorem 5.78 (Positive factorization and nonvanishing)**.** *The measure μ has mass C, and*

$$D(z)L(z) = z^d \int_0^1 \frac{\mathrm{d}\mu(u)}{1-zu}. \tag{5.161}$$

Its transform H has strictly positive imaginary part when $\operatorname{Im} z > 0$, strictly negative imaginary part when $\operatorname{Im} z < 0$, and $H(x) > 0$ for every real $x < 1$. The only zero of L on Ω is its zero of multiplicity d at zero.

Proof. We have $u^{d-1}D(1/u) = P(u)$, so the preceding lemma gives (5.161). Its value after division by z^d at zero is $\int P\kappa = C$, as already shown in the oscillation proof. For nonreal z,

$$\operatorname{Im} H(z) = (\operatorname{Im} z)\int_0^1 \frac{u\,\mathrm{d}\mu(u)}{|1-zu|^2}.$$

The integral is strictly positive because the measure has positive density except at finitely many interior points. For real $x < 1$, the integrand defining $H(x)$ is positive. Thus H has no zero on Ω. All zeros of D are $1/r_j > 1$, outside Ω, while $D(0) = 1$ and $H(0) = C > 0$. The factorization proves the claimed multiplicity and the global nonvanishing assertion. □

Theorem 5.79 (Minimality and uniqueness of the compensator)**.** *Among real polynomials E with $E(0) > 0$ for which $E(z)L(z)/z^d$ is the transform of a finite positive measure on $[0,1]$, the smallest possible degree is $d-1$. Every such polynomial of degree $d-1$ is $E(0)D(z)$. In particular the normalization $E(0) = 1$ singles out D.*

Proof. Existence in degree $d-1$ follows from Theorem 5.78. Suppose that $\deg E \le d-1$. Lemma 5.77 already realizes the transform as the finite signed measure with density

$$Q_E(u)\kappa(u), \qquad Q_E(u) = u^{d-1}E(1/u).$$

Uniqueness of finite Hausdorff moments implies that any purported positive representing measure must be this signed measure. Its density must therefore be nonnegative almost everywhere. Since κ changes sign at every r_j, continuity forces $Q_E(r_j) = 0$. As $r_j \neq 0$, these are $d-1$ distinct roots $1/r_j$ of E. Consequently $\deg E \ge d-1$. At equality those roots determine E up to its constant factor, which is fixed by $E(0)$. □

This is minimality of *polynomial compensation for an ordinary positive Stieltjes representation.* It is not minimal depth in an algebra of numerical periods and not a minimal generalized Stieltjes order. Those are different questions.

5.25.1 Quantitative consequences

For $x > 0$, positivity and $0 < u < 1$ give

$$\frac{Cx^d}{(1+x)D(-x)} < (-1)^d L(-x) < \frac{Cx^d}{D(-x)}. \tag{5.162}$$

Indeed $C/(1+x) < H(-x) < C$ and $D(-x) > 0$. Without computing the compensator roots one still has the coarser bounds

$$C\left(\frac{x}{1+x}\right)^d < (-1)^d L(-x) < Cx^d, \tag{5.163}$$

because $1 \le D(-x) \le (1+x)^{d-1}$, with the relevant strictness supplied by H even at depth one.

Write $P(u) = \sum_{j=0}^{d-1} p_j u^j$ and

$$h_n = \int_0^1 u^n \, \mathrm{d}\mu(u) = \sum_{j=0}^{d-1} p_j c_{\boldsymbol{s}}(n+j+1), \qquad n \ge 0. \tag{5.164}$$

For all $n, k \ge 0$,

$$(-1)^k \Delta^k h_n = \int_0^1 u^n (1-u)^k \, \mathrm{d}\mu(u) > 0, \qquad \Delta h_n = h_{n+1} - h_n. \tag{5.165}$$

Every finite Hankel matrix $(h_{i+j})_{0 \le i,j \le m}$ is positive definite. The same is true of (h_{i+j+1}) and $(h_{i+j} - h_{i+j+1})$. To see this, pair each matrix with a nonzero real vector and integrate the square of its polynomial against, respectively, μ, $u\mu$, or $(1-u)\mu$. Each integral is strictly positive. In particular,

$$h_n h_{n+2} - h_{n+1}^2 > 0.$$

These are exact nonlinear positivity constraints on the original nested harmonic coefficients and the compensator roots. They should not be mistaken for rational-coefficient identities among special values.

5.26 Exactly $d-1$ angular zeros on every upper semicircle

The signed transform gives, for $0 < \rho \le 1$ and $0 < \theta < \pi$,

$$\operatorname{Im} L(\rho e^{\mathrm{i}\theta}) = \rho \sin\theta \, J_\rho(\cos\theta), \qquad J_\rho(c) = \int_0^1 \frac{\kappa(u)}{1 - 2\rho c u + \rho^2 u^2} \, \mathrm{d}u. \tag{5.166}$$

A bound based only on simple sign-change arguments would leave open the possibility of tangencies. We therefore prove a multiplicity-sensitive upper bound before proving existence.

5.26.1 An elementary confluent Cauchy-kernel bound

Lemma 5.80 (Angular variation bound with multiplicities)**.** *Suppose $\kappa \in L^1(0,1)$ is continuous, has exactly m sign changes at $r_1 < \cdots < r_m$, and has a nonzero fixed sign on each complementary interval. Then J_ρ has at most m zeros in $(-1,1)$, counted with analytic multiplicity, for every $0 < \rho \le 1$.*

Proof. The function

$$t(u) = \frac{1+\rho^2u^2}{2\rho u} \tag{5.167}$$

is strictly decreasing on $(0,1)$, since $t'(u) = (\rho^2u^2-1)/(2\rho u^2) < 0$. Its values are at least one, and it tends to infinity as $u \downarrow 0$. Thus

$$J_\rho(c) = \int_0^1 \frac{\kappa(u)}{2\rho u}\frac{1}{t(u)-c}\,\mathrm{d}u.$$

The factor $\kappa(u)/u$ need not be integrable by itself; this causes no difficulty because the displayed Cauchy factor is $O(u)$ at zero. All the integrals below are interpreted with their complete kernels.

Suppose there are distinct zeros $c_1,\dots,c_q \in (-1,1)$ with positive multiplicities $\ell_1,\dots,\ell_q$ summing to $m+1$. If more zeros are available, retain any subcollection of total multiplicity $m+1$. Form the proper rational function

$$R(t) = \frac{\prod_{j=1}^m (t-t(r_j))}{\prod_{a=1}^q (t-c_a)^{\ell_a}}. \tag{5.168}$$

Its partial fractions are linear combinations of $(t-c_a)^{-h}$ for $1 \le h \le \ell_a$. Differentiation of J_ρ under the integral is valid locally for $c \in (-1,1)$; explicitly,

$$J_\rho^{(h-1)}(c) = (h-1)!\int_0^1 \frac{\kappa(u)}{2\rho u}\frac{1}{(t(u)-c)^h}\,\mathrm{d}u.$$

The selected zero multiplicities therefore imply

$$\int_0^1 \frac{\kappa(u)}{2\rho u} R(t(u))\,\mathrm{d}u = 0. \tag{5.169}$$

This integral is absolutely convergent: near zero $R(t) = O(1/t) = O(u)$, and near one its denominators stay positive and bounded away from zero.

But each factor $t(u)-t(r_j)$ changes sign exactly where κ changes sign. The denominator of $R(t(u))$ is strictly positive, because $t(u) \ge 1 > c_a$. Hence the integrand in (5.169) has one fixed nonzero sign except at the r_j. Its integral cannot vanish. This contradiction proves the bound and also excludes an identically zero J_ρ. □

The proof is a finite partial-fraction argument, including repeated poles. No unproved strict-total-positivity assertion is being used.

5.26.2 Positive compensation supplies the missing zeros

For $0 < \theta < \pi$ and $0 \le u \le 1$, put

$$\beta_u(\theta) = -\operatorname{Arg}(1-\rho u e^{\mathrm{i}\theta}).$$

For $0 < u < 1$ one has

$$0 < \beta_u(\theta) < \beta_1(\theta) < \frac{\pi}{2}. \tag{5.170}$$

The angle is strictly increasing in u, since differentiation with respect to u gives $\rho\sin\theta/|1-\rho u e^{\mathrm{i}\theta}|^2 > 0$. A positive mixture of the resolvents $(1-zu)^{-1}$ stays strictly inside this sector. Consequently, with H as in (5.161),

$$0 < \operatorname{Arg} H(\rho e^{\mathrm{i}\theta}) < \beta_1(\theta). \tag{5.171}$$

For example, the second inequality follows by rotating each resolvent through $-\beta_1$ and observing that its imaginary part is strictly negative for $0 < u < 1$. All rays lie in an angle of width less than $\pi/2$, so no argument ambiguity arises.

Theorem 5.81 (Complete angular count and signs)**.** *For* (5.141) *and every* $0 < \rho \le 1$, $\operatorname{Im} L(\rho e^{\mathrm{i}\theta})$ *has exactly* $d-1$ *simple zeros in* $0 < \theta < \pi$*. They have the signs and bounds stated in Theorem 5.69.*

Proof. A continuous unwrapped argument of L along the upper arc is

$$\Phi(\theta) = d\theta + \sum_{j=1}^{d-1} \beta_{r_j}(\theta) + \operatorname{Arg} H(\rho e^{\mathrm{i}\theta}). \tag{5.172}$$

The factorization proves this formula exactly; Φ is not reduced modulo 2π. The sector bounds imply

$$d\theta < \Phi(\theta) < d\big(\theta + \beta_1(\theta)\big) < d\pi. \tag{5.173}$$

At the other end of the arc, $\Phi(\theta) \to d\pi$ as $\theta \uparrow \pi$, since all resolvents approach positive real values and $H(-\rho) > 0$.

If $\rho < 1$, then $\Phi(\theta) \to 0$ as $\theta \downarrow 0$. The case $\rho = 1$ requires more care: $L(1)$ may be singular, and there is no reason to assign it argument zero. Instead, each fixed $r_j < 1$ gives $\beta_{r_j}(\theta) \to 0$, while (5.171) gives

$$\limsup_{\theta \downarrow 0} \operatorname{Arg} H(e^{\mathrm{i}\theta}) \le \pi/2.$$

It follows that $0 < \Phi(\theta) < \pi$ for sufficiently small positive θ. This weaker endpoint statement is all that is needed.

By continuity, the phase crosses each of $\pi, 2\pi, \dots, (d-1)\pi$. These yield at least $d-1$ distinct zeros of the imaginary part. On the other hand, Lemma 5.80 with $m = d-1$, together with (5.166), gives at most $d-1$ zeros counted with multiplicities. The change of variable $c = \cos\theta$ has nonzero derivative in the open arc, and the prefactor $\rho \sin\theta$ is positive. Thus multiplicities are preserved. Every zero is simple, and there is exactly one crossing of each listed phase level. The levels must occur in their stated order, since a reversal would force a repeated crossing of an earlier level.

The initial phase lies in $(0, \pi)$, so the imaginary part starts positive. Simplicity forces alternating signs thereafter. At the kth zero, the phase is $k\pi$ and the modulus is nonzero, so the real value has sign $(-1)^k$.

For the location bounds, write $\alpha_k = k\pi/d$. At a zero, (5.173) gives

$$\theta_k < \alpha_k < \theta_k + \beta_1(\theta_k).$$

The function $h(\theta) = \theta + \beta_1(\theta)$ is strictly increasing, because

$$h'(\theta) = \frac{1 - \rho\cos\theta}{1 - 2\rho\cos\theta + \rho^2} > 0.$$

For $\rho < 1$, solving $h(\theta) = \alpha$ gives $\theta = \alpha - \arcsin(\rho \sin\alpha)$. At $\rho = 1$ the same expression is the solution when $\alpha > \pi/2$; when $\alpha \le \pi/2$ it is zero, and the strict lower bound follows from $\theta_k > 0$. This proves (5.144). □

The argument proves the needed crossings without asserting that Φ is globally monotone. Such an assertion is unnecessary here.

Corollary 5.82 (Analytic motion and no interior collisions)**.** *For fixed integer leading indices, each angular branch is real analytic in* $(\rho, b) \in (0,1) \times (0,\infty)$*, and extends real analytically across* $\rho = 1$ *locally at each of its nontrivial upper-arc zeros. The density roots* r_j *are real analytic in* $b > 0$*. No two of the angular branches collide for* $0 < \rho \le 1$*.*

Proof. On compact subsets of $\operatorname{Re} b > 0$, the seed f_b is holomorphic as an L^1-valued function; factors $|\log(-\log u)|^j$ arising from parameter differentiation have an integrable common majorant on smaller compact subsets. The bounded operators preserve this holomorphic parameter dependence. The signed transform is locally holomorphic in $z \in \Omega$ and in b. Simplicity of its angular zeros permits the real analytic implicit-function theorem. The unit-radius zeros remain away from the cut, so the same local argument crosses $\rho = 1$. The density derivative formulas give joint real analyticity in u, b in the open interval, and their simple roots obey the same theorem. Distinctness follows from Theorem 5.81. □

The local extension across radius one does not imply that the global count persists at all larger radii. For $L_{1,1} = \log^2(1-z)/2$, an upper-half-plane imaginary zero requires $|1 - z| = 1$, so its angle obeys $\cos\theta = \rho/2$. There are no open-upper-arc zeros when $\rho > 2$.

5.26.3 Strict radial motion of every angular branch

The compensation yields more than a zero count. It also turns a monotonicity property of ordinary positive Stieltjes transforms into strict motion of all the strict-depth angular zeros.

Lemma 5.83 (Radial argument monotonicity for positive transforms)**.** *Let* ν *be a nonzero finite positive measure on* $[0,1]$ *and* $Q(z) = \int (1-uz)^{-1}\,\mathrm{d}\nu(u)$*. For* $0 < \theta < \pi$ *and* $0 < \rho \le 1$*,*

$$\frac{\partial}{\partial \rho} \operatorname{Arg} Q(\rho e^{\mathrm{i}\theta}) \ge 0. \tag{5.174}$$

The argument is chosen continuously from $Q(0) > 0$*.*

Proof. First take a finite measure with positive weights at distinct points $0 < u_1 < \cdots < u_n < 1$. The rational function $\sum_j w_j/(t-u_j)$ is strictly decreasing between consecutive poles, with limits $+\infty$ and $-\infty$ at the two ends. Its numerator therefore has exactly one root $v_j \in (u_j, u_{j+1})$ in each such interval. Factoring the numerator gives

$$Q(z) = Q(0)\frac{\prod_{j=1}^{n-1}(1 - v_j z)}{\prod_{j=1}^{n}(1 - u_j z)}.$$

Consequently the radial derivative of its argument is

$$\sin\theta \left\{ f(u_1) + \sum_{j=1}^{n-1} \big(f(u_{j+1}) - f(v_j)\big) \right\}, \qquad f(t) = \frac{t}{1 - 2\rho t\cos\theta + \rho^2 t^2}.$$

Every summand is nonnegative, because

$$f'(t) = \frac{1 - \rho^2 t^2}{(1 - 2\rho t\cos\theta + \rho^2 t^2)^2} \ge 0 \qquad (0 \le t \le 1).$$

Approximate an arbitrary finite positive ν weakly by such finite measures, with their masses converging to $\nu([0,1])$. For each compact subset of Ω, the integrands defining the transform and

its first complex derivative are bounded and continuous uniformly in $u \in [0,1]$. The transforms and derivatives converge locally uniformly. Also $Q(\rho e^{\mathrm{i}\theta}) \neq 0$: it has positive imaginary part unless the measure is supported at zero, in which case it is a positive constant. Passing to the limit in $\mathrm{Im}(e^{\mathrm{i}\theta}Q'(z)/Q(z))$ proves the nonnegative derivative. The same argument works at $\rho = 1$ because θ is strictly between zero and π. □

Theorem 5.84 (Strict inward angular motion)**.** *For $d \geq 2$ and (5.141), every angular branch satisfies*

$$\frac{\mathrm{d}\theta_k}{\mathrm{d}\rho} < 0, \qquad 0 < \rho \leq 1, \quad 1 \leq k \leq d-1. \tag{5.175}$$

The derivative at one is the one-sided derivative, or equivalently the derivative of the local continuation. The same result holds for the positive-seed and shifted-terminal families of Theorem 5.76.

Proof. At fixed θ, differentiate the unwrapped phase (5.172) with respect to the radius:

$$\Phi_\rho = \sum_{j=1}^{d-1} \frac{r_j \sin\theta}{1 - 2\rho r_j \cos\theta + \rho^2 r_j^2} + \partial_\rho \operatorname{Arg} H(\rho e^{\mathrm{i}\theta}) > 0.$$

The first sum is strictly positive since $d \geq 2$ and $r_j > 0$; the last term is nonnegative by Lemma 5.83. At the unique simple crossing of the phase level $k\pi$, one has $\Phi_\theta > 0$. Indeed the phase starts below that level, ends above it, and crosses it exactly once; simplicity excludes a zero phase derivative there. Implicit differentiation of $\Phi(\theta_k(\rho), \rho) = k\pi$ now gives

$$\theta_k'(\rho) = -\frac{\Phi_\rho}{\Phi_\theta} < 0.$$

All ingredients also hold for a general positive seed. □

For depth two, angular decrease alone does *not* settle the higher-leading-order question about $\eta(\rho) = \cos\theta(\rho)/\rho$: the derivative of that quotient has a second term, and its sign is not implied by $\theta'(\rho) < 0$. The two kinds of radial monotonicity must remain separate in the integration ledger.

5.27 Small-radius and large-outer-index expansions

The global count removes an ambiguity from local expansions: the zeros found near the first Fourier-mode zeros are the entire list of angular zeros, not just selected branches.

Theorem 5.85 (Two-term radial expansion)**.** *Fix an index satisfying (5.141) with $d \geq 2$, and put*

$$p = \frac{c_{\boldsymbol{s}}(d+1)}{C_{\boldsymbol{s}}}, \qquad q = \frac{c_{\boldsymbol{s}}(d+2)}{C_{\boldsymbol{s}}}, \qquad \alpha_k = \frac{k\pi}{d}.$$

The kth angular branch extends real analytically to $\rho = 0$, and

$$\begin{aligned}\theta_k(\rho) = \alpha_k &- \frac{p}{d}\sin\alpha_k\,\rho \\ &+ \left(\frac{d+1}{d^2}p^2 - \frac{2q}{d}\right)\sin\alpha_k\cos\alpha_k\,\rho^2 + O_{\boldsymbol{s}}(\rho^3).\end{aligned} \tag{5.176}$$

In particular, every branch initially moves strictly toward smaller angles as the radius increases from zero.

Proof. Divide the imaginary part of the disk series by $C_{\boldsymbol{s}}\rho^d$. The resulting analytic equation is

$$0 = \sin(d\theta) + p\rho\sin((d+1)\theta) + q\rho^2\sin((d+2)\theta) + O(\rho^3).$$

At $(\rho,\theta) = (0,\alpha_k)$ its θ derivative is $d(-1)^k \neq 0$, so the analytic implicit-function theorem applies. There are $d-1$ such branches; for small positive radius they exhaust the global list from Theorem 5.81.

Write $\theta = \alpha_k + A\rho + B\rho^2 + O(\rho^3)$ and divide by $(-1)^k$. The coefficients of ρ and ρ^2 are respectively

$$dA + p\sin\alpha_k, \qquad dB + (d+1)pA\cos\alpha_k + q\sin(2\alpha_k).$$

Solving gives (5.176). The first derivative is negative because $p > 0$ and $0 < \alpha_k < \pi$. □

Higher Taylor coefficients are obtained by the same finite coefficient recursion. At each order the new unknown occurs with coefficient $d(-1)^k$, while all remaining terms depend only on earlier coefficients and finitely many moments. This is an exact algebraic recurrence, not a proof of any global radial sign.

5.27.1 Uniformity as the first index increases

Fix $d \geq 2$ and positive integer middle indices $s_2, \ldots, s_{d-1}$. Let the first index be $a \in \mathbb{N}$ and the last be $b > 0$. Define

$$T_N(b) = \sum_{N \geq n_2 > \cdots > n_d \geq 1} \frac{1}{n_2^{s_2}\cdots n_{d-1}^{s_{d-1}} n_d^b}, \qquad T_* = T_{d-1}(b) > 0.$$

The latter number is a single-chain value and is independent of b. At depth two, $T_N(b) = H_N^{(b)}$ and $T_* = 1$.

Theorem 5.86 (First displacement, uniformly in the final order)**.** *Set $\lambda = d/(d+1)$ and $\mu = d/(d+2)$. As $a \to \infty$ through positive integers, uniformly for $b > 0$, $0 \leq \rho \leq 1$, and $1 \leq k \leq d-1$,*

$$\theta_k(\rho) = \alpha_k - \frac{T_d(b)}{dT_*}\sin\alpha_k\,\rho\lambda^a + O\big(\rho^2\mu^a\big). \tag{5.177}$$

At $\rho = 0$, the branch is interpreted by its analytic extension. The implied constant and the starting threshold for a may depend on the fixed depth and middle indices, but not on b, ρ, or k.

Proof. The normalized imaginary part is

$$\sin(d\theta) + \frac{T_d(b)}{T_*}\rho\lambda^a\sin((d+1)\theta) + R(\theta), \tag{5.178}$$

where

$$R(\theta) = \sum_{n \geq d+2} \frac{T_{n-1}(b)}{T_*}\rho^{n-d}\left(\frac{d}{n}\right)^a \sin(n\theta).$$

All inner denominators are at least one, so $0 \leq T_{n-1}(b) \leq n^{d-1}$ uniformly for $b > 0$. Choose the fixed integer $a_0 = d+3$. For $a \geq a_0$, termwise differentiation is absolutely justified and

$$|R(\theta)| + |R'(\theta)| \leq K\rho^2\mu^a, \quad K = \frac{2}{T_*}\sum_{n \geq d+2} n^d\left(\frac{d+2}{n}\right)^{a_0} < \infty. \tag{5.179}$$

Also $T_d(b)/T_* \leq d^{d-1}/T_*$, independently of b.

On disjoint fixed small neighborhoods of the α_k, the derivative of the first term of (5.178) is bounded away from zero. Its perturbation tends to zero uniformly by (5.179). Evaluating at $\alpha_k \pm K_1\rho\lambda^a$, for a sufficiently large fixed K_1, gives opposite signs for all sufficiently large a. This follows from the linear term $d(\theta - \alpha_k)$ and the uniform bounds on the perturbations; $\mu/\lambda < 1$. Thus each neighborhood contains a unique root with displacement $\delta = O(\rho\lambda^a)$. The global count identifies these roots as all the branches in their established order.

Expand (5.178) at α_k and divide by $(-1)^k$. The root equation becomes

$$0 = d\delta + \frac{T_d(b)}{T_*}\rho\lambda^a \sin\alpha_k + O\big(\delta^2 + \rho\lambda^a|\delta| + \rho^2\mu^a\big).$$

Because $\lambda^2 < \mu$, the previously obtained displacement bound makes the error $O(\rho^2\mu^a)$. Division by d proves (5.177). All constants used above are independent of the final order. □

At depth two and unit radius this recovers the leading scale $(2/3)^a$ already present in the manuscript's more detailed depth-two expansion [106]. The added content is the simultaneous all-depth statement, the exact branch indexing, and uniformity over the entire range $b > 0$ and $0 \le \rho \le 1$.

5.28 Explicit logistic kernels and positive integral identities

The compensator is generally implicit: its roots are the zeros of a signed density. The all-ones family makes every part explicit and gives an infinite collection of exact positive integral identities.

5.28.1 The all-ones density and its roots

Let $\mathbf{1}_d = (1, \dots, 1)$ and define

$$\ell(u) = \log\frac{u}{1-u}, \qquad 0 < u < 1.$$

The familiar identity

$$L_{\mathbf{1}_d}(z) = \frac{(-\log(1-z))^d}{d!} \tag{5.180}$$

follows recursively from $(1-z)L'_{\mathbf{1}_d}(z) = L_{\mathbf{1}_{d-1}}(z)$ and the vanishing initial value at zero. We use it as a classical identity with a short proof, not as a new reduction of multiple polylogarithms.

Theorem 5.87 (Explicit signed kernels and logistic roots)**.** *For every $d \ge 1$,*

$$\begin{aligned}\kappa_{\mathbf{1}_d}(u) &= \frac{\operatorname{Im}(\ell(u) + \mathrm{i}\pi)^d}{\pi d!} \\ &= \sum_{j=0}^{\lfloor (d-1)/2 \rfloor} \frac{(-1)^j \pi^{2j} \ell(u)^{d-1-2j}}{(2j+1)!(d-1-2j)!}.\end{aligned} \tag{5.181}$$

Its increasing list of roots is

$$r_j = \frac{1}{1 + \exp(\pi\cot(j\pi/d))}, \qquad 1 \le j \le d-1. \tag{5.182}$$

In particular, $r_j + r_{d-j} = 1$.

Proof. Consider the generating function

$$G(t,u) = e^{t\ell(u)}\frac{\sin \pi t}{\pi t} = \sum_{d\geq 1} K_d(u) t^{d-1}. \tag{5.183}$$

For $|\operatorname{Re} t| < 1$, its integral at moment one is the beta integral

$$\int_0^1 G(t,u)\,\mathrm{d}u = \frac{\sin \pi t}{\pi t}\Gamma(1+t)\Gamma(1-t) = 1.$$

The reflection formula supplies the last equality, with removable values understood at zero. Thus $\int K_1 = 1$ and $\int K_d = 0$ for $d \geq 2$. Coefficient extraction is legitimate on a smaller complex neighborhood of zero, since $u^{\operatorname{Re} t}(1-u)^{-\operatorname{Re} t}$ and its logarithmic parameter derivatives have an integrable majorant.

Now $K_1 = 1$ and

$$\frac{\mathrm{d}}{\mathrm{d}u}G(t,u) = \frac{t}{u(1-u)}G(t,u),$$

so $K_d' = K_{d-1}/(u(1-u))$. This is exactly the derivative of $\mathsf{B}K_{d-1}$. Their difference is constant, and both have zero integral for $d \geq 2$, so $K_d = \mathsf{B}K_{d-1}$. Hence these are the constructed all-ones kernels. Expanding the exponential and sine gives (5.181).

The imaginary part of $(\ell + \mathrm{i}\pi)^d$ vanishes when $\operatorname{Arg}(\ell + \mathrm{i}\pi)$ is one of $j\pi/d$. This gives $\ell = \pi\cot(j\pi/d)$. Reversing the order of these real values and solving $u/(1-u) = e^{\ell}$ yields (5.182). The symmetry follows from $\cot(\pi - x) = -\cot x$. □

Proposition 5.88 (A continuous Mellin identity and Stirling moments)**.** *For real $x > 0$ and complex t satisfying $-x < \operatorname{Re} t < 1$,*

$$\int_0^1 u^{x-1}e^{t\ell(u)}\frac{\sin \pi t}{\pi t}\,\mathrm{d}u = \frac{\Gamma(x+t)}{\Gamma(x+1)\Gamma(1+t)}. \tag{5.184}$$

The values at removable singularities are obtained by continuation. For every integer $n \geq 1$,

$$\int_0^1 u^{n-1}\kappa_{\mathbf{1}_d}(u)\,\mathrm{d}u = \frac{\left[{n \atop d}\right]}{n!}, \tag{5.185}$$

where $\left[{n \atop d}\right]$ is the unsigned Stirling number of the first kind, zero for $d > n$.

Proof. The integral equals $\sin(\pi t)B(x+t, 1-t)/(\pi t)$. The beta formula and Gamma reflection simplify it to (5.184). At integer $x = n$, the right side is

$$\frac{(t+1)(t+2)\cdots(t+n-1)}{n!}.$$

Its coefficient of t^{d-1} is $\left[{n \atop d}\right]/n!$, proving (5.185) by (5.183). □

These beta and Stirling identities provide an independent normalization check on the general signed-moment construction.

5.28.2 An explicit infinite family of compensated identities

For the roots (5.182), define

$$P_d(u) = \prod_{j=1}^{d-1}(u - r_j), \qquad D_d(z) = \prod_{j=1}^{d-1}(1 - r_j z).$$

The integrable function $P_d\kappa_{\mathbf{1}_d}$ is strictly positive away from its finitely many zeros. Combining (5.161) and (5.180) gives the exact identity

$$\boxed{\int_0^1 \frac{P_d(u)\kappa_{\mathbf{1}_d}(u)}{1 - zu}\,\mathrm{d}u = \frac{D_d(z)}{d!}\left(\frac{-\log(1-z)}{z}\right)^d, \qquad z \in \Omega.} \tag{5.186}$$

At $z = 0$ the removable value is $1/d!$. At $z = -1$ this becomes

$$\int_0^1 \frac{P_d(u)\kappa_{\mathbf{1}_d}(u)}{1 + u}\,\mathrm{d}u = \frac{D_d(-1)}{d!}\log^d 2. \tag{5.187}$$

The content is not a new formula for $L_{\mathbf{1}_d}$; it is the explicit positive density, its canonical compensation, and the resulting positive-integrand identities. They remain valid at every depth without a numerical relation search.

5.28.3 Depth three

Put

$$h_3 = \frac{\pi}{\sqrt{3}}, \qquad q_3 = \frac{1}{2 + 2\cosh h_3}.$$

Then

$$P_3(u) = u^2 - u + q_3, \qquad D_3(z) = 1 - z + q_3 z^2, \qquad \kappa_{\mathbf{1}_3}(u) = \frac{\ell(u)^2 - \pi^2/3}{2}.$$

Both factors P_3 and $\ell^2 - \pi^2/3$ are negative between the same two roots and positive outside them. Hence their product is nonnegative throughout the interval. Formula (5.187) gives

$$\boxed{\int_0^1 \frac{(u^2 - u + q_3)(\ell(u)^2 - \pi^2/3)}{1 + u}\,\mathrm{d}u = \frac{2 + q_3}{3}\log^3 2.} \tag{5.188}$$

The companion normalization is

$$\int_0^1 (u^2 - u + q_3)(\ell(u)^2 - \pi^2/3)\,\mathrm{d}u = \frac{1}{3}. \tag{5.189}$$

These identities are proved statements, not high-precision conjectures.

5.28.4 Depth four

Set

$$q_4 = \frac{1}{2 + 2\cosh \pi}, \qquad P_4(u) = \left(u - \frac{1}{2}\right)(u^2 - u + q_4).$$

Now

$$D_4(z) = \left(1 - \frac{z}{2}\right)(1 - z + q_4 z^2), \qquad \kappa_{\mathbf{1}_4}(u) = \frac{\ell(u)^3 - \pi^2\ell(u)}{6}.$$

The roots are the logistic images of $-\pi, 0, \pi$, so again the product $P_4\kappa_{\mathbf{1}_4}$ is nonnegative. The corresponding pair of identities is

$$\int_0^1 \frac{P_4(u)(\ell(u)^3 - \pi^2\ell(u))}{1+u}\,\mathrm{d}u = \frac{3(2+q_4)}{8}\log^4 2, \tag{5.190}$$

$$\int_0^1 P_4(u)(\ell(u)^3 - \pi^2\ell(u))\,\mathrm{d}u = \frac{1}{4}. \tag{5.191}$$

All endpoint singularities are logarithmic powers and are integrable. Thus no principal-value convention is implicit in these formulas.

5.28.5 An explicit density-edge asymptotic

The smallest all-ones density root gives a simple large-depth benchmark:

$$\log r_1 = -d + \frac{\pi^2}{3d} + \frac{\pi^4}{45d^3} + O(d^{-5}) + O(e^{-d}). \tag{5.192}$$

Indeed $\log r_1 = -\pi\cot(\pi/d) - \log(1 + e^{-\pi\cot(\pi/d)})$, and the Taylor expansion of cot at zero gives the displayed terms. This concerns a density root, not an angular zero. Symmetry supplies $1 - r_{d-1} = r_1$.

5.29 A divided-difference identity across depths and colors

The explicit logistic density suggests an identity search that keeps the depth parameter symbolic instead of recognizing one numerical value at a time. Its beta moments already determine the one-resolvent transform. Partial fractions then introduce several independent colors, while coalescing those colors produces finite formulas at every resolvent order. The resulting Gaussian logarithmic moments are exact polynomials in π and $\log 2$, with no integer-relation premise.

For $-1 < \operatorname{Re} t < 1$ put

$$W_t(u) = \left(\frac{u}{1-u}\right)^t \frac{\sin \pi t}{\pi t}, \qquad 0 < u < 1,$$

using the removable value at $t = 0$. This is the generating density (5.183). Its integral is one, even for complex t; for real $-1 < t < 1$ it is positive. Denote divided differences by $[z_1, \dots, z_m]f$, with the continuous confluent interpretation when nodes coincide.

Theorem 5.89 (Several resolvents and their confluent limits)**.** *Let $z_1, \dots, z_m \in \Omega$, where $\Omega = \mathbb{C} \setminus [1, \infty)$. For distinct nodes and $m \geq 2$,*

$$\int_0^1 \frac{W_t(u)}{\prod_{j=1}^m (1 - z_j u)}\,du = \frac{1}{t}[z_1, \dots, z_m]\{z^{m-2}((1-z)^{-t} - 1)\}. \tag{5.193}$$

For $m = 1$ the transform is

$$\int_0^1 \frac{W_t(u)}{1 - zu}\,du = \frac{(1-z)^{-t} - 1}{tz}. \tag{5.194}$$

All powers use the principal logarithm; the apparent singularities at $t = 0$, $z = 0$, and coincident nodes are removable. In particular, for $m \geq 2$,

$$\int_0^1 \frac{W_t(u)}{(1-zu)^m}\,du = (1-z)^{-t-m+1}\mathcal{P}_m(t, z),$$

$$\mathcal{P}_m(t, z) = \sum_{j=0}^{m-2} \binom{m-2}{j} \frac{(t+1)_j}{(j+1)!} z^j (1-z)^{m-2-j}, \tag{5.195}$$

where $(t+1)_j$ *is the rising factorial and* $(t+1)_0 = 1$.

Proof. The beta identity (5.184) gives

$$\int_0^1 u^n W_t(u)\,du = \frac{(1+t)_n}{(n+1)!}.$$

For $|z| < 1$, expand the geometric kernel and sum the binomial series. Its result is (5.194). The integrable endpoint majorants $u^{-1+\epsilon}(1-u)^{-1+\epsilon}$, on smaller parameter strips, justify local holomorphy and continuation to the principal slit plane. For distinct nonzero nodes, the exact partial fractions are

$$\frac{1}{\prod_j (1-z_j u)} = \sum_{j=1}^{m} \frac{z_j^{m-1}}{\prod_{k\neq j}(z_j - z_k)} \frac{1}{1-z_j u}.$$

Integrating and using (5.194) gives the Lagrange formula for (5.193). Continuity allows zero nodes.

When every node tends to z, the divided difference becomes the $(m-1)$st derivative divided by $(m-1)!$. Apply the product rule to $z^{m-2}((1-z)^{-t}-1)/t$. The term differentiating the second factor zero times vanishes, because the polynomial has degree $m-2$. For $j+1 \geq 1$ derivatives of the second factor one has

$$\frac{d^{j+1}}{dz^{j+1}} \frac{(1-z)^{-t}-1}{t} = (t+1)_j (1-z)^{-t-j-1}.$$

Collecting the binomial factors proves (5.195). All confluent limits also follow directly from domination of the original integral on compact subsets of Ω. □

For example, the first three confluent cases are

$$\begin{aligned}
\mathcal{P}_2 &= 1,\\
\mathcal{P}_3 &= 1 + (t-1)z/2,\\
\mathcal{P}_4 &= (1-z)^2 + (t+1)z(1-z) + (t+1)(t+2)z^2/6.
\end{aligned}$$

Thus the squared-resolvent identity has the particularly simple form

$$\int_0^1 \frac{W_t(u)}{(1-zu)^2}\,du = (1-z)^{-t-1}.$$

The several-color identity is symmetric in its nodes; repeated colors are derivatives, rather than a separate guessed formula.

Corollary 5.90 (Every Gaussian logarithmic moment)**.** *For integers* $m \geq 2$ *and* $k \geq 0$*, define*

$$I_{m,k} = \int_0^1 \frac{\log^k(u/(1-u))}{(1-\mathrm{i}u)^m}\,du, \qquad \lambda = \tfrac{1}{2}\log 2 - \tfrac{\mathrm{i}\pi}{4}.$$

These are ordinary absolutely convergent integrals, and

$$I_{m,k} = \frac{k!}{(1-\mathrm{i})^{m-1}}[t^k]\left\{\mathcal{P}_m(t,\mathrm{i})\exp\left(-\lambda t + \sum_{j\geq 1}\frac{\zeta(2j)}{j}t^{2j}\right)\right\}. \tag{5.196}$$

In particular $I_{m,k} \in \mathbb{Q}[\mathrm{i}, \pi, \log 2]$ *for every* m, k. *For the squared resolvent,*

$$I_{2,0} = \frac{1}{1-\mathrm{i}}, \qquad I_{2,1} = -\frac{\lambda}{1-\mathrm{i}},$$
$$I_{2,2} = \frac{\lambda^2 + \pi^2/3}{1-\mathrm{i}}, \qquad I_{2,3} = -\frac{\lambda^3 + \pi^2\lambda}{1-\mathrm{i}},$$
$$I_{2,4} = \frac{\lambda^4 + 2\pi^2\lambda^2 + 7\pi^4/15}{1-\mathrm{i}}.$$

Proof. Multiply (5.195) by $\pi t/\sin \pi t$ and differentiate k times at zero. Powers of the endpoint logarithms are integrable, so this differentiation is legitimate. The reflection formula and its local logarithmic expansion give

$$\frac{\pi t}{\sin \pi t} = \Gamma(1+t)\Gamma(1-t) = \exp\left(\sum_{j \geq 1} \frac{\zeta(2j)}{j} t^{2j}\right), \qquad |t| < 1.$$

At $z = \mathrm{i}$, the principal logarithm is $\log(1-\mathrm{i}) = \lambda$. This proves the coefficient formula and the listed values. Only finitely many terms enter a coefficient of order k, and every even zeta value is a rational multiple of π^{2j}. □

For a concrete weight-two pair, the real and imaginary parts of $I_{2,2}$ give the ordinary integral identities

$$\int_0^1 \frac{(1-u^2)\log^2(u/(1-u))}{(1+u^2)^2}\, du = \frac{\log^2 2}{8} + \frac{13\pi^2}{96} + \frac{\pi \log 2}{8},$$
$$\int_0^1 \frac{2u\log^2(u/(1-u))}{(1+u^2)^2}\, du = \frac{\log^2 2}{8} + \frac{13\pi^2}{96} - \frac{\pi \log 2}{8}. \tag{5.197}$$

Both follow by expanding λ^2 in the proved formula, so their constants are derived rather than recognized numerically.

Corollary 5.91 (Two real Gaussian generating identities)**.** *For real* $-1 < t < 1$,

$$\int_0^1 \frac{(u/(1-u))^t}{1+u^2}\, du = \frac{\pi\, 2^{-t/2}\sin(\pi t/4)}{\sin \pi t},$$
$$\int_0^1 \frac{u(u/(1-u))^t}{1+u^2}\, du = \frac{\pi\{1 - 2^{-t/2}\cos(\pi t/4)\}}{\sin \pi t}. \tag{5.198}$$

At $t = 0$ *the removable values are respectively* $\pi/4$ *and* $\log 2/2$. *Differentiating at zero supplies every real logarithmic moment of these two kernels.*

Proof. Use (5.194) at $z = \mathrm{i}$ and $z = -\mathrm{i}$ and the identities

$$\frac{1}{1+u^2} = \frac{1}{2}\left(\frac{1}{1-\mathrm{i}u} + \frac{1}{1+\mathrm{i}u}\right), \qquad \frac{u}{1+u^2} = \frac{1}{2\mathrm{i}}\left(\frac{1}{1-\mathrm{i}u} - \frac{1}{1+\mathrm{i}u}\right).$$

Multiply by $\pi t/\sin \pi t$ and simplify the conjugate principal powers. Continuity gives the zero-parameter values. □

The finite confluent polynomials and moment recurrence are checked exactly in `verification/certify_logistic_resolvents.py`. Independent logistic-coordinate quadratures are labeled as numerical diagnostics; the arbitrary-order identities follow from the beta moments, partial fractions and analytic parameter differentiation. These formulas add a several-color integral family to the development; they do not settle the separate mixed harmonic sums S_6 or S_8, and no literature-priority claim is made for this classical beta-integral mechanism.

5.30 Exact rational evaluation and finite certificates

The density zeros need not be computed in order to obtain numerical certificates. The unsigned norm bound $\|\kappa\|_1 \le 2^{d-1}$ already gives an explicit geometric remainder for a centered resolvent expansion. The proof works with real final order; the delivered exact implementation accepts positive integer indices so that all required moments are rational.

5.30.1 A centered-moment expansion

Define

$$B_k = \int_0^1 (2u-1)^k \kappa(u)\,\mathrm{d}u = \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} 2^j c_{\boldsymbol{s}}(j+1). \tag{5.199}$$

Then $|B_k| \le M$, where $M = 2^{d-1}$. For $\operatorname{Re} z < 1$, put

$$w = \frac{z}{2-z}, \qquad |w| < 1.$$

The inequality is equivalent to $|z| < |2-z|$. The identity $1 - zu = (1 - z/2)(1 - w(2u-1))$ gives

Theorem 5.92 (Geometric exact-evaluation certificate)**.** *For every integer $N \ge 1$,*

$$V_N(z) = \frac{z}{1-z/2} \sum_{k=0}^{N-1} B_k w^k, \tag{5.200}$$

$$|L(z) - V_N(z)| \le \left|\frac{z}{1-z/2}\right| \frac{M|w|^N}{1-|w|}. \tag{5.201}$$

At $z = \mathrm{i}$ and even $N = 2m$, a purely rational bound is

$$|L(\mathrm{i}) - V_{2m}(\mathrm{i})| < \frac{2^d}{5^m}. \tag{5.202}$$

For $|z| \le 1/2$, the uniform bound is

$$|L(z) - V_N(z)| \le \frac{2^{d-1}}{3^N}. \tag{5.203}$$

Proof. The centered geometric series converges uniformly for $u \in [0,1]$, and its tail is bounded by $|w|^N/(1-|w|)$. Integrating against $|\kappa|$ proves (5.201). At $z = \mathrm{i}$, $|w| = 1/\sqrt{5}$ and $|z/(1-z/2)| = 2/\sqrt{5}$. The multiplicative constant is therefore $2M/(\sqrt{5}-1) < 2M = 2^d$ after extracting 5^{-m}. On $|z| \le 1/2$, one has $|w| \le 1/3$ and $|z/(1-z/2)| \le 2/3$, proving (5.203). □

For integer indices, (5.199) and (5.200) use only rational arithmetic at rational complex z. An enclosing complex disk gives enclosing intervals for each coordinate by adding and subtracting its radius. The evaluator never treats a small residual as a proof of an identity.

5.30.2 Implementation choices and replay

The file `code/exact_evaluator.py` first obtains the moments by dynamic strict prefix sums. It computes B_k with a common integer denominator and a repeated-difference transform, rather than a floating-point alternating binomial sum. Horner evaluation then gives the exact rational center. The generic branch can upper-bound square roots by dyadic rationals; the Gaussian and half-disk

cases use the closed rational bounds above. Near $\operatorname{Re} z = 1$, the fixed norm-bound precision in the generic branch can be insufficient, in which case the implementation raises an error rather than returns an uncertified value. It is not advertised as a complete optimal evaluator on the entire slit plane.

From the package root, the principal replay is

```
python code/verify.py
```

It reads frozen rational certificates and independently recomputes them. The optional command

```
python code/verify.py --regenerate
```

uses floating point to *propose* angular brackets, then requires rational arithmetic to certify both endpoint signs before saving them. The proposal stage uses a strict-disk series and does not decide any accepted sign. This discovery/replay separation is deliberate.

5.30.3 Gaussian values

The delivered Gaussian certificate file contains eight mixed-depth examples, all evaluated with $N = 240$ centered terms. Each complex-disk radius is smaller than 10^{-81}. The decimal entries below are readable midpoint diagnostics; the certificates are the rational endpoints in `data/gaussian_enclosures.json`.

Index	$\operatorname{Re} L_{\boldsymbol{s}}(\mathrm{i})$, approximately	$\operatorname{Im} L_{\boldsymbol{s}}(\mathrm{i})$, approximately
$(2,1)$	-0.174715661158	-0.113648917600
$(1,1,1)$	0.099953993665	-0.033577147847
$(2,1,2)$	0.032383901801	-0.026348629072
$(1,2,1,1)$	0.001158016810	0.008343470492
$(2,1,1,2)$	0.001825088603	0.006510815484
$(1,1,1,1,1)$	-0.003396506582	-0.001886779708
$(2,1,2,1,1)$	-0.000326599679	-0.000010475684
$(1,1,1,1,1,1)$	0.000443168800	-0.000335617002

In addition, the eight all-ones Gaussian values at depths one through eight are checked against *independent exact* intervals for

$$L_{\mathbf{1}_d}(\mathrm{i}) = \frac{(-\frac{1}{2}\log 2 + \frac{\mathrm{i}\pi}{4})^d}{d!}.$$

The logarithm and π intervals are derived from different rational series, as detailed in Section 5.31. These eight normalization controls are distinct from the eight stored mixed-depth examples above.

5.30.4 Certified angular brackets

On the semicircle of radius $1/2$, set

$$z(t) = \frac{1-t^2}{2(1+t^2)} + \mathrm{i}\frac{t}{1+t^2}, \qquad \theta = 2\arctan t, \quad t > 0. \tag{5.204}$$

Rational t gives rational real and imaginary coordinates. Every saved root bracket has t-width at most 10^{-12}; the corresponding angle-width is less than $2 \cdot 10^{-12}$. With $N = 104$, the error bound (5.203) proves the two endpoint signs by strict rational comparisons. For illustration:

Index	k	$10^{12}t_{\mathrm{left}}$	$10^{12}t_{\mathrm{right}}$
$(2,1,2)$	1	472099547153	472099547154
$(2,1,2)$	2	1487963538987	1487963538988
$(1,2,1,1)$	1	314972569528	314972569529
$(1,2,1,1)$	2	806524527583	806524527584
$(1,2,1,1)$	3	2019771345242	2019771345243
$(1,1,1,1,1,1)$	3	774596669241	774596669242

The file `data/angular_brackets.json` contains 24 brackets in eight complete index families of depths two through six, together with all exact coordinate intervals. The existence of a zero in each bracket follows from the opposite signs. The theorem, rather than a scan, proves uniqueness in every bracket and excludes all additional angular zeros. For the displayed middle branch at depth six, the all-ones formula independently gives $\theta_3(1/2) = \arccos(1/4)$ and $t = \sqrt{3/5}$.

5.30.5 What the exact suite checks

The fresh replay reports 5,713 finite check groups:

Check family	Count
Strict coefficients against direct finite nested enumeration	1080
Initial zero moments and leading coefficients	120
Centered moments: independent binomial formula and norm bound	2880
Polynomial compensation identities for arbitrary test roots	1200
All-ones moments against unsigned Stirling recurrence	240
Logistic-kernel derivatives through depth twelve	11
Depth-three/four symbolic kernel formulas	2
Exact generalized-kernel counterexample checks	2
Rejected negative controls	2
Rational atomic radial-argument derivative signs	80
Independent exact all-ones Gaussian controls	8
Replayed stored Gaussian enclosures	8
Replayed angular endpoint signs	48
Replayed root brackets and complete families	$24 + 8$
Total	5713

The polynomial compensation tests deliberately use arbitrary rational roots: they test the cancellation identity, *not* positivity for those roots. Positivity requires the actual density zeros and is proved analytically. Similarly, finite atomic derivative checks do not replace the interlacing-and-approximation proof of Lemma 5.83.

5.31 Independent rational logarithm and Gaussian controls

The exact controls use only rational arithmetic and elementary remainder bounds. They do not share the centered-moment evaluator's recurrence. For $0 < x < 1$, let

$$A_N(x) = \sum_{j=0}^{N-1} \frac{(-1)^j x^{2j+1}}{2j+1}.$$

The alternating-series estimate gives

$$\arctan x \in \begin{cases} [A_N(x),\ A_N(x) + x^{2N+1}/(2N+1)], & N \text{ even}, \\ [A_N(x) - x^{2N+1}/(2N+1),\ A_N(x)], & N \text{ odd}. \end{cases}$$

The identity

$$\pi = 16\arctan(1/5) - 4\arctan(1/239) \tag{5.205}$$

is verified by tangent addition: if $a = \arctan(1/5)$, then $\tan(4a) = 120/119$, and $\tan(4a - \arctan(1/239)) = 1$. Elementary bounds put the angle in $(0, \pi/2)$, so it equals $\pi/4$ and not another tangent branch. The implementation uses $N = 120$ and $N = 40$ for the two arctangent series.

For the logarithm, set

$$B_N = 2\sum_{j=0}^{N-1} \frac{(1/3)^{2j+1}}{2j+1}.$$

Integrating the geometric series for $2/(1-x^2)$, or using the power series of $2\operatorname{arctanh} x$, gives

$$B_N < \log 2 < B_N + \frac{2(1/3)^{2N+1}}{(2N+1)(1-1/9)}. \tag{5.206}$$

The code uses $N = 160$. Both interval endpoints are exact rational numbers. Cartesian complex interval multiplication then encloses $(-\frac{1}{2}\log 2 + \frac{\mathrm{i}\pi}{4})^d/d!$.

For each depth $1 \le d \le 8$, the independently generated real and imaginary intervals lie wholly inside the corresponding coordinate intervals returned by the centered evaluator with 240 terms. This is an exact containment check, not a decimal comparison. It is a strong normalization and branch control, while the identity (5.180) still rests on its differential-equation proof.

The exact evaluator also supplies a dyadic upper bound on $\sqrt{x}$ for a nonnegative rational $x = a/b$. With an integer precision p, set

$$q = \frac{\lfloor\sqrt{\lfloor 2^{2p}a/b\rfloor}\rfloor + 1}{2^p}.$$

Then $q^2 > x$. This is computed by integer square root and exact comparison. If its use in the generic geometric-tail formula fails to produce an upper bound smaller than one for $|w|$, the code stops rather than accepting a possibly divergent bound.

The complete original source and delivered rational certificates remain under `reports/angular-zero-continuation/continuations/all-depth-signed-oscillation/`. The isolated replay record in `verification/sixth-replay/all-depth/` checks the finite arithmetic independently of the analytic zero-count proof. The manuscript uses the stronger positive-transform radial theorem proved next, while retaining the incoming interlacing argument as an independent route. Fractional leading indices at higher depth still require a new construction: iterating an integer number of moment operators does not establish a fractional variation theorem.

5.32 A strict radial theorem for every positive Stieltjes transform

The preceding interlacing proof gives a nonnegative radial argument derivative. A direct symmetrization strengthens it to strict positivity, gives a quantitative bound, and identifies the sharp universal radius range. This also removes the integer-leading restriction from angular motion at depth two when combined with the earlier real-order factorization.

Theorem 5.93 (Strict radial phase and its quantitative bound)**.** *Let ν be a nonzero finite positive measure on $[0,1]$, and put*

$$Q(z) = \int_{[0,1]} \frac{d\nu(u)}{1-uz}, \qquad M_j = \int_{[0,1]} u^j\, d\nu(u).$$

For $0 < \theta < \pi$ and $0 < \rho \le 1$, the continuous phase satisfies

$$\partial_\rho \operatorname{Arg} Q(\rho e^{\mathrm{i}\theta}) \ge \frac{4\rho^2 \sin^3\theta (1-\cos\theta)^2}{(1+\rho)^8} \frac{M_1 M_2}{M_0^2}. \tag{5.207}$$

In particular, this derivative is strictly positive unless ν is supported at zero, in which case Q is constant. The range $\rho \le 1$ is sharp as a universal assertion over positive measures on $[0,1]$: at every prescribed radius greater than one a two-atom measure gives a strictly negative radial phase derivative on the entire open upper arc.

Proof. Write $c = \cos\theta$, $h = \sin\theta$, $x = \rho u$, $y = \rho v$ and $D_x = 1 - 2cx + x^2$. These denominators are positive on the arc, and Q is nonzero: its imaginary part is positive unless ν is supported at zero, where it is a positive constant. Local domination justifies differentiation and the double integral for $\operatorname{Im}(e^{\mathrm{i}\theta} Q'(z)\overline{Q(z)})$. Rationalizing and symmetrizing in u, v gives

$$\partial_\rho \operatorname{Arg} Q(\rho e^{\mathrm{i}\theta}) = \frac{h}{\rho |Q(\rho e^{\mathrm{i}\theta})|^2} \iint \frac{\mathcal{P}(x,y,c)}{D_x^2 D_y^2}\, d\nu(u)\, d\nu(v), \tag{5.208}$$

where

$$\mathcal{P} = \frac{1}{2}\{x D_y^2 + y D_x^2 - (x-y)^2 (x D_y + y D_x)\}.$$

The exact polynomial identity that determines the sign is

$$\begin{aligned} \mathcal{P} = {} & \frac{(1-x)(1-y)}{2}\{(x-y)^2 + x(1-y)^2 + y(1-x)^2\} \\ & + 4xy(1-x)(1-y)(1-c) + 2xy(x+y)(1-c)^2. \end{aligned} \tag{5.209}$$

Every term is nonnegative for $0 \le x, y \le 1$ and $-1 < c < 1$. The last term is strictly positive when $x, y > 0$, including $x = y = 1$. Thus a positive mass on $(0,1]$ gives a strictly positive integral in (5.208); no approximation argument is needed for strictness.

For the explicit lower bound, retain only the last term and use $D_x, D_y \le (1+\rho)^2$. Then

$$\partial_\rho \operatorname{Arg} Q \ge \frac{4\rho^2 h (1-c)^2 M_1 M_2}{(1+\rho)^8 |Q|^2},$$

since $\iint uv(u+v)\, d\nu(u)\, d\nu(v) = 2M_1 M_2$. Also $|1 - te^{\mathrm{i}\theta}|^2 = (t-c)^2 + h^2 \ge h^2$, so $|Q| \le M_0/h$. Substitution proves (5.207). All operations also apply at $\rho = 1$, because the chosen arc point is away from the cut.

For sharpness, let $R > 1$, choose $R^{-2} < v < 1$, and take $\nu = v\delta_0 + (1-v)\delta_1$. Its transform is

$$Q_v(z) = \frac{1 - vz}{1 - z}.$$

Direct differentiation gives

$$\partial_\rho \operatorname{Arg} Q_v(\rho e^{\mathrm{i}\theta}) = \frac{(1-v)(1-\rho^2 v)\sin\theta}{(1 - 2\rho\cos\theta + \rho^2)(1 - 2\rho v\cos\theta + \rho^2 v^2)}. \tag{5.210}$$

It is strictly negative at $\rho = R$ for every $0 < \theta < \pi$. This proves sharpness of the uniform disk range, without asserting that every individual transform loses monotonicity just outside it. □

Corollary 5.94 (All-real-order inward angular motion at depth two)**.** *For every $a \geq 0, b > 0$, let $\theta_{a,b}(\rho)$ be the unique upper-semicircle imaginary zero of the principal continuation of $F_{a,b}$. Then*

$$\theta'_{a,b}(\rho) < 0, \qquad 0 < \rho \leq 1.$$

The derivative at one is the derivative of the local analytic branch, equivalently its left derivative. For $0 < \rho < \sigma \leq 1$ one also has

$$\frac{\sigma \cos \theta_{a,b}(\sigma)}{\rho \cos \theta_{a,b}(\rho)} > \frac{\sigma}{\rho},$$

so the Cartesian abscissa grows faster than the radius in this precise ratio sense. This is distinct from the sign of $\partial_\rho(\cos\theta/\rho)$.

Proof. Theorems 5.18 and 5.26 supply, throughout the parameter domain, a factorization

$$F_{a,b}(z) = \frac{z^2 Q_{a,b}(z)}{1 - rz}, \qquad 0 < r \leq 1,$$

with $Q_{a,b}$ a finite positive Stieltjes transform. In the supercritical region with $a > 0$, r is the interior density crossing; on the critical/subcritical region and the entire $a = 0$ axis one may take $r = 1$. The positive measures have mass 2^{-a} and positive density except possibly at the crossing. For an upper-arc point their continuous unwrapped phase is

$$\Phi = 2\theta - \operatorname{Arg}(1 - r\rho e^{\mathrm{i}\theta}) + \operatorname{Arg} Q_{a,b}(\rho e^{\mathrm{i}\theta}).$$

The resolvent sector bounds give $0 < \Phi < 2\pi$. Hence at the unique imaginary zero, $\Phi = \pi$ and $F_{a,b}$ is strictly negative real. The preceding real-order zero theorems prove $\partial_\theta \operatorname{Im} F_{a,b} < 0$ there. Since $\operatorname{Im} F = |F| \sin \Phi$, this is equivalent to $\Phi_\theta > 0$. At fixed angle,

$$\Phi_\rho = \frac{r \sin\theta}{1 - 2r\rho\cos\theta + r^2\rho^2} + \partial_\rho \operatorname{Arg} Q_{a,b} > 0.$$

Implicit differentiation of $\Phi(\theta_{a,b}(\rho), \rho) = \pi$ therefore gives $\theta' = -\Phi_\rho/\Phi_\theta < 0$. The analytic continuation and angular simplicity justify this argument at the endpoint and on parameter regions whose boundary series fail the ordinary term test. Finally the unique angle lies in $(0, \pi/2)$, so its strict decrease makes $\cos\theta$ strictly increase. The displayed Cartesian ratio follows immediately. □

The direct pair identity and positive decomposition are independently re-expanded symbolically and tested by exact rational arithmetic in `verification/certify_stieltjes_radial.py`. Separate controls check one-atom normalization and two-atom examples outside the disk, including a prescribed-radius counterexample approaching radius one. These finite checks catch algebraic mistakes; the measure theorem follows from the sign decomposition and dominated integral argument, and the all-real-order consequence uses the previously established positive factorization.

5.33 A geometric evaluation formula that survives $xy = 1$

The preceding sections concern finite identities. We now give an evaluation method for the full double polylogarithm, including components for which no finite classical reduction has been established.

Lemma 5.95 (One-dimensional integral)**.** *For $a, b \geq 1$, $|x| \leq 1$, $x \neq 1$, and $|y| \leq 1$,*

$$\operatorname{Li}_{a,b}(x, y) = \frac{x}{(a-1)!} \int_0^1 \frac{(-\log t)^{a-1} \operatorname{Li}_b(xyt)}{1 - xt}\, dt. \tag{5.211}$$

Proof. For $|x|<1$, the Mellin identity $n^{-a}=\int_0^1 t^{n-1}(-\log t)^{a-1}\,dt/(a-1)!$ and absolute convergence give

$$\sum_{n>m\ge 1}(xt)^n y^m/m^b=\frac{xt}{1-xt}\operatorname{Li}_b(xyt).$$

This proves (5.211). For a nontrivial unit outer argument, $1-rxt$ stays uniformly away from zero for r sufficiently close to one. Also $|\operatorname{Li}_b(rxyt)|\le \operatorname{Li}_b(t)$, and

$$\frac{1}{(a-1)!}\int_0^1(-\log t)^{a-1}\operatorname{Li}_b(t)\,dt=\sum_{m\ge 1}\frac{1}{m^b(m+1)^a}\le\zeta(a+b). \tag{5.212}$$

Dominated convergence and summation by parts in the nested series therefore give the boundary case. Notice that $xy=1$ causes no problem: the logarithmic endpoint at $b=1$ is integrable in (5.212). □

Let T_n denote the Chebyshev polynomial defined by $T_n(\cos u)=\cos(nu)$. Put

$$d=\sqrt{1-x},\qquad r=\frac{1-d}{1+d},\qquad R=|r|, \tag{5.213}$$

where the square root has positive real part. For $|x|\le 1$, $x\ne 1$, we have $\operatorname{Re} d>0$ and therefore $R<1$.

Lemma 5.96 (The kernel expansion)**.** *Uniformly for $0\le t\le 1$,*

$$\frac{1}{1-xt}=\frac{1}{d}\left(1+2\sum_{n\ge 1}r^nT_n(2t-1)\right). \tag{5.214}$$

Proof. For $s\in[-1,1]$, summing the two geometric series associated with $s=\cos u$ gives the classical identity

$$1+2\sum_{n\ge 1}r^nT_n(s)=\frac{1-r^2}{1-2rs+r^2}.$$

This is also a direct consequence of the generating function in [49]. Substituting (5.213) shows

$$1-xt=d\frac{1-2r(2t-1)+r^2}{1-r^2}.$$

The series is uniform because $|T_n(s)|\le 1$ and $R<1$. □

Definition 5.97. *For $v=xy$ and $n\ge 0$, define the Chebyshev moment*

$$\mathcal{M}_n^{a,b}(v)=\frac{1}{(a-1)!}\int_0^1(-\log t)^{a-1}T_n(2t-1)\operatorname{Li}_b(vt)\,dt. \tag{5.215}$$

Theorem 5.98 (Certified Chebyshev–moment expansion)**.** *Under the hypotheses of Lemma 5.95, with $x\ne 0$,*

$$\operatorname{Li}_{a,b}(x,y)=\frac{x}{d}\left(\mathcal{M}_0^{a,b}(v)+2\sum_{n\ge 1}r^n\mathcal{M}_n^{a,b}(v)\right). \tag{5.216}$$

For $K \geq 1$, let E_K be the difference between the left side and the truncation with indices $0 \leq n < K$. Then

$$|E_K| \leq \frac{2|x|\zeta(a+b)}{|d|(1-R)} R^K. \tag{5.217}$$

This estimate is uniform over $|y| \leq 1$, including $xy = 1$. For nontrivial level-three or level-four outer roots,

$$|E_K| \leq \frac{16}{3\,4^K}. \tag{5.218}$$

Proof. Insert (5.214) into (5.211). The integrable majorant in (5.212) justifies termwise integration and gives $|\mathcal{M}_n^{a,b}(v)| \leq \zeta(a+b)$ for all n. The tail of the resulting geometric series is (5.217). For the specified roots, direct substitution in (5.213) gives $R < 1/4$ and $|d| > 1$; the code also proves these inequalities by interval arithmetic before using them. Since $\zeta(a+b) \leq \zeta(2) < 2$, the bound becomes $4(1/4)^K/(1-1/4) = 16/(3\,4^K)$. The zero outer argument, omitted only to avoid unnecessary notation, gives $\mathrm{Li}_{a,b}(0,y) = 0$ directly. □

The proof does not appeal to cancellation in a sampled sequence, a fitted asymptotic expansion, or an empirical convergence diagnostic. In particular, when the product of two roots is one, the same argument and the same bound apply. The only new feature is the elementary cancellation in the exact moment formula, to which we now turn.

5.34 Every moment reduces to classical polylogarithms

Write the shifted Chebyshev polynomial as

$$Q_n(t) := T_n(2t-1) = \sum_{j=0}^{n} c_{n,j} t^j. \tag{5.219}$$

Its coefficients are integers and can be generated without symbolic integration:

$$Q_0 = 1, \quad Q_1 = 2t-1, \quad Q_{n+1} = 2(2t-1)Q_n - Q_{n-1}. \tag{5.220}$$

Thus

$$\mathcal{M}_n^{a,b}(v) = \sum_{j=0}^{n} c_{n,j} J_{a,b}(j+1; v), \tag{5.221}$$

where

$$J_{a,b}(c; v) = \sum_{m \geq 1} \frac{v^m}{m^b (m+c)^a}, \qquad c \in \mathbb{Z}_{>0}. \tag{5.222}$$

The power series in (5.222) is absolutely convergent for $|v| \leq 1$.

Lemma 5.99 (Two-pole partial fractions)**.** *Put*

$$\begin{aligned} A_s &= (-1)^{b-s} \binom{a+b-s-1}{a-1} c^{-(a+b-s)} && (1 \leq s \leq b), \\ C_s &= (-1)^{b} \binom{a+b-s-1}{b-1} c^{-(a+b-s)} && (1 \leq s \leq a). \end{aligned} \tag{5.223}$$

Then, for $u \neq 0, -c$,

$$\frac{1}{u^b (u+c)^a} = \sum_{s=1}^{b} \frac{A_s}{u^s} + \sum_{s=1}^{a} \frac{C_s}{(u+c)^s}. \tag{5.224}$$

In particular, $A_1 + C_1 = 0$.

Proof. The principal part at $u = 0$ follows by expanding $(u + c)^{-a} = c^{-a}(1 + u/c)^{-a}$. Its coefficient of u^{-s} after multiplication by u^{-b} is exactly A_s. At $u = -c$, write $w = u + c$ and expand $u^{-b} = (-1)^b c^{-b}(1 - w/c)^{-b}$; this gives C_s. The difference between the two sides has no finite poles and tends to zero at infinity, so it is identically zero. The equality $A_1 = -C_1$ follows from $\binom{a+b-2}{a-1} = \binom{a+b-2}{b-1}$. □

For $s \geq 1$, let $H_c^{(s)}(v) = \sum_{k=1}^{c} v^k/k^s$ and $H_c^{(s)} = H_c^{(s)}(1)$. A coefficient outside the ranges in (5.223) is understood to be zero in the following theorem.

Theorem 5.100 (Finite classical moment formula)**.** *For $0 < |v| \leq 1$, $v \neq 1$,*

$$J_{a,b}(c; v) = \sum_{s=1}^{b} A_s \operatorname{Li}_s(v) + v^{-c} \sum_{s=1}^{a} C_s\big(\operatorname{Li}_s(v) - H_c^{(s)}(v)\big). \tag{5.225}$$

At the resonant endpoint $v = 1$ the finite expression is instead

$$J_{a,b}(c; 1) = -C_1 H_c^{(1)} + \sum_{s=2}^{\max(a,b)} (A_s + C_s)\zeta(s) - \sum_{s=2}^{a} C_s H_c^{(s)}. \tag{5.226}$$

No regularized zeta value is needed. At $v = 0$, $J_{a,b}(c; 0) = 0$.

Proof. Multiply (5.224) by v^m and sum over $m \geq 1$. For $|v| < 1$, all rearrangements are absolute, and

$$\sum_{m\geq 1} \frac{v^m}{(m + c)^s} = v^{-c}\big(\operatorname{Li}_s(v) - H_c^{(s)}(v)\big).$$

Radial limits give (5.225) at $|v| = 1$, $v \neq 1$. At $v = 1$, combine the two $s = 1$ pieces at the level of partial sums:

$$A_1 \sum_{m=1}^{N} \frac{1}{m} + C_1 \sum_{m=1}^{N} \frac{1}{m + c} = -C_1 H_c^{(1)} + C_1 \sum_{m=N+1}^{N+c} \frac{1}{m}.$$

The last term tends to zero. Every $s \geq 2$ term converges separately, which gives (5.226). □

For example, when $a = b = 1$, (5.226) reduces to

$$J_{1,1}(c; 1) = \frac{H_c}{c}. \tag{5.227}$$

Consequently the resonant depth-two function has the explicit expansion

$$\operatorname{Li}_{1,1}(x, x^{-1}) = \frac{x}{\sqrt{1 - x}} \left[1 + 2 \sum_{n\geq 1} r^n \sum_{j=0}^{n} c_{n,j} \frac{H_{j+1}}{j + 1}\right], \tag{5.228}$$

valid for a nontrivial unit outer argument. Its coefficients are rational. The error after K terms is bounded by (5.217), and the uniform level-three/four bound (5.218) applies. This is an exact infinite identity for a conditionally convergent depth-two case; it is not a finite rational expression in classical constants.

5.34.1 A resonant example beyond the first weight

For $(a,b)=(2,1)$, the coefficients are $A_1=c^{-2}$, $C_1=-c^{-2}$, and $C_2=-c^{-1}$, giving

$$J_{2,1}(c;1)=\frac{H_c}{c^2}+\frac{H_c^{(2)}-\zeta(2)}{c}. \tag{5.229}$$

Substitution in (5.221) and (5.216) yields a geometrically convergent formula for $\mathrm{Li}_{2,1}(\rho,\rho^{-1})$, $\rho=e^{2\pi\mathrm{i}/3}$, whose coefficient data involve only rational numbers and $\zeta(2)$. The location of the singular product $v=1$ has become an explicit, finite harmonic-sum correction rather than a numerical failure mode.

5.35 Optimal geometric rate and the conductor dependence

The rate in Theorem 5.98 is not merely an artifact of an arbitrary expansion point. It is the best exponential rate obtainable by uniformly approximating its outer rational kernel with polynomials. This statement concerns the kernel, not the minimal complexity of all possible algorithms for evaluating a special value.

Theorem 5.101 (Sharp exponential rate for polynomial kernels)**.** *Fix $0<|x|\le 1$, $x\neq 1$, and let d,r,R be as in (5.213). For $K\ge 1$, define*

$$e_{K-1}(x)=\inf_{\deg p<K}\max_{0\le t\le 1}\left|\frac{1}{1-xt}-p(t)\right|,$$

where the infimum allows complex polynomial coefficients. Then

$$\frac{R^K}{|d|}\le e_{K-1}(x)\le\frac{2R^K}{|d|(1-R)}. \tag{5.230}$$

In particular, $\lim_{K\to\infty}e_{K-1}(x)^{1/K}=R$.

Proof. The upper bound is the uniform tail bound in (5.214). For the lower bound, use the Kth Chebyshev coefficient of $f(s)=(1-x(s+1)/2)^{-1}$. By (5.214), it is $2r^K/d$. The corresponding coefficient of a polynomial of degree less than K vanishes. For any continuous complex function g on $[-1,1]$,

$$\left|\frac{2}{\pi}\int_0^\pi g(\cos u)\cos(Ku)\,du\right|\le 2\|g\|_\infty.$$

Apply this with $g=f-p$ to obtain $2R^K/|d|\le 2\|f-p\|_\infty$ and take the infimum. Taking Kth roots proves the final statement. □

Corollary 5.102 (Adjacent roots)**.** *For $x_q=e^{2\pi\mathrm{i}/q}$, as $q\to\infty$,*

$$-\log R_q=2\sqrt{\frac{\pi}{q}}+O(q^{-3/2}). \tag{5.231}$$

For fixed a,b, the bound (5.217) guarantees an absolute error at most 10^{-P} using

$$K=O\big(\sqrt{q}(P+\log q)\big) \tag{5.232}$$

outer-kernel terms, uniformly in $|y|\le 1$.

Proof. With $\theta = 2\pi/q$, $d = \sqrt{1-e^{\mathrm{i}\theta}} = e^{-\mathrm{i}\pi/4}\sqrt{\theta}\,(1+O(\theta))$. Hence

$$-\log|r| = \operatorname{Re}\left(\log(1+d) - \log(1-d)\right) = 2\operatorname{Re}(d + d^3/3 + \cdots) = \sqrt{2\theta} + O(\theta^{3/2}),$$

which is (5.231). Also $|d| \asymp q^{-1/2}$ and $1 - R \asymp q^{-1/2}$, so the prefactor in (5.217) is $O(q)$. Solving the resulting geometric inequality gives (5.232). □

For comparison, expansion around $t = 1/2$ gives

$$\frac{1}{1-xt} = \frac{1}{1-x/2}\sum_{n\geq 0}\left(\frac{x(t-1/2)}{1-x/2}\right)^n.$$

At a unit outer argument its uniform ratio is $R_{\mathrm{mid}} = 1/|2-x|$. For $x = x_q$,

$$-\log R_{\mathrm{mid}} = \tfrac{1}{2}\log(5 - 4\cos(2\pi/q)) = \frac{4\pi^2}{q^2} + O(q^{-4}). \tag{5.233}$$

Thus its corresponding sufficient degree is $O(q^2(P + \log q))$. This is a comparison between two proved expansions, not a claim to improve every previously published polylogarithm algorithm.

Arithmetic cost versus approximation degree. With weights fixed, the monomial moments for $1 \leq c \leq K$ can be assembled from finitely many single values and incrementally updated $H_c^{(s)}(v)$ in $O((a+b)K)$ arithmetic operations. The triangular transform (5.221) costs $O(K^2)$ operations in the direct implementation. Thus, after the necessary classical values are available, $O(K^2 + (a+b)K)$ arithmetic operations suffice for the spectral truncation. These are unit-cost arithmetic counts, not bit complexity bounds. At general large conductor, computing the inner single values and representing the algebraic arguments have their own costs. The implementation supplied here certifies levels three and four; the analytic theorem has the wider stated domain.

5.36 Outward-rounded evaluation and the treatment of cancellation

A theorem about a truncation error is not by itself a certificate for a floating-point implementation. This section specifies how finite arithmetic errors are enclosed in the accompanying executable code.

5.36.1 An independent geometric expansion for the single values

To avoid using the desired parity identities inside their own numerical checks, the certified evaluator computes its classical constants from a separate Euler expansion.

Lemma 5.103 (Euler expansion with a simple rigorous tail)**.** *Let $s \geq 1$ be an integer and let $z \neq 1$, with $\tau = z/(z-1)$ satisfying $|\tau| < 1$. For the arguments used below,*

$$\mathrm{Li}_s(z) = -\sum_{N\geq 1}\tau^N\frac{h^\star_{s-1}(N)}{N}, \tag{5.234}$$

where $h^\star_d(N)$ is the complete homogeneous symmetric polynomial of degree d in $1, 1/2, \ldots, 1/N$, and $h^\star_0(N) = 1$. If $|\tau| \leq r_0 < 1$, then the error after $N \leq M$ is at most

$$\frac{r_0^{M+1}}{1-r_0}. \tag{5.235}$$

Proof. In the standard Mellin integral for $\mathrm{Li}_s(z)$, expand

$$\frac{z}{1-zt} = -\sum_{N\ge 1} \tau^N (1-t)^{N-1}.$$

The coefficient integral is

$$\frac{1}{(s-1)!}\int_0^1 (-\log t)^{s-1}(1-t)^{N-1}\,dt = \frac{h^\star_{s-1}(N)}{N}.$$

Indeed, summing its generating series in u^{s-1} gives

$$\int_0^1 t^{-u}(1-t)^{N-1}\,dt = \frac{1}{N}\prod_{j=1}^{N}(1-u/j)^{-1},$$

first for u close to zero. This can be proved by the beta integral or by $N-1$ integrations by parts. Each coefficient integral is between zero and one, since $(1-t)^{N-1} \le 1$. The geometric tail then proves (5.235). □

For the nontrivial roots of levels three and four, and for $z=-1$, $|\tau| \le 1/\sqrt{2} < 3/4$. The implementation uses the rational choice $r_0 = 3/4$ and computes the coefficients incrementally from

$$h^\star_d(N) = h^\star_d(N-1) + \frac{h^\star_{d-1}(N)}{N}. \tag{5.236}$$

It obtains $\zeta(s)$ for $s \ge 2$ from $-\mathrm{Li}_s(-1)/(1-2^{1-s})$, obtains π from $4\,\mathrm{Im}\,\mathrm{Li}_1(\mathrm{i})$, and obtains L from $-\mathrm{Li}_1(-1)$. No conjectural reduction enters these evaluations.

5.36.2 The interval arithmetic contract

Fix a decimal scale $Q = 10^D$. A real interval is stored as integers $[l,u]/Q$. Addition is exact at this scale. For multiplication, the four integer endpoint products are enclosed by a floor of their minimum and a ceiling of their maximum after division by Q. A reciprocal is permitted only when zero is excluded, and is again rounded outward. A nonnegative square root is enclosed with integer square roots of lQ and uQ. Complex values are rectangular pairs of such real intervals.

All these rules are elementary rational inequalities. By induction over the expression tree, the computed rectangle contains the exact finite expression. The analytic disk radius from (5.217) or (5.235) is then added to both real and imaginary coordinate intervals. Enclosing a disk by a square is conservative but valid. The square-root branch in (5.213) is fixed by its positive real part and the sign of the imaginary part of $1-x$.

The executable checks $R < 1/4$ and $|d| \ge 1$ by interval inequalities before using (5.218). It rejects unsupported outer arguments instead of silently using an unproved bound. In particular, the supplied certified implementation does not accept outer argument one.

5.36.3 Why extra working precision is necessary

The coefficients in (5.219) alternate in sign and satisfy

$$\sum_{j=0}^{n} |c_{n,j}| = T_n(3) = \frac{(3+2\sqrt{2})^n + (3-2\sqrt{2})^n}{2}. \tag{5.237}$$

The alternating signs follow, for example, from the explicit formula

$$c_{n,j} = (-1)^{n-j} \frac{n}{n+j} \binom{n+j}{2j} 4^j \quad (n \geq 1),$$

and evaluating Q_n at $t = -1$ gives (5.237). Thus computing the moments by monomial conversion can lose a number of digits proportional to K, even though the moments themselves are bounded.

The interval code does not assume that this cancellation is harmless. For target $P = 35$ it uses $K = 73$ spectral terms and $D = 246$ working decimal places. Each single value uses $M = 1997$ Euler terms and its own rigorous tail. These are conservative choices rather than tuned speed records. The actual final widths, not a heuristic guard-digit count, determine whether a certificate is accepted. A future stable recurrence or validated Clenshaw-style evaluation could reduce the guard overhead; that is not needed for the present certificates.

5.36.4 A positive-measure midpoint alternative

For inverse colors, the same integral defines the positive measure

$$d\mu_{a,b}(t) = \frac{(-\log t)^{a-1} \operatorname{Li}_b(t)}{(a-1)!}\, dt.$$

Its total mass is $J_{a,b}(1;1) \leq J_{1,1}(1;1) = 1$. Set $\alpha = z/(1-z/2)$ and $m_r = \int_0^1 (t-1/2)^r \, d\mu_{a,b}(t)$. The kernel's geometric series gives

$$\operatorname{Li}_{a,b}(z, z^{-1}) = \alpha \sum_{r \geq 0} \alpha^r m_r, \qquad q = \frac{|z|}{|2-z|} < 1.$$

The tail after $r < N$ is at most $|\alpha| J_{a,b}(1;1) q^N/(1-q)$. Each m_r is an exact binomial combination of $J_{a,b}(j+1;1)$ from Theorem 5.100. This is useful at Gaussian and Eisenstein points; near the outer root one, the Chebyshev expansion has the stronger kernel rate proved above. Neither rate statement is a lower bound for every evaluation algorithm.

5.37 Exact rational certificates for Gaussian values

We use the outermost-letter-first convention

$$G(a_1, \ldots, a_w; z) = \int_0^z \frac{G(a_2, \ldots, a_w; t)}{t - a_1}\, dt, \qquad G(;z) = 1.$$

Define the permitted alphabet region

$$\mathcal{A} = \{0, 1\} \cup \{a \in \mathbb{Q}(i) : |a| \geq 1,\ |1-a| \geq 1\}.$$

The two inequalities are exact rational comparisons after squaring the moduli. The Gaussian alphabet and the letter $1 - i$ both belong to this region.

Theorem 5.104 (Uniform exact certificate)**.** *Let $w \geq 1$, let $a_1, \ldots, a_w \in \mathcal{A}$, and assume $a_1 \neq 1$ and $a_w \neq 0$. Then $G(a_1, \ldots, a_w; 1)$ exists as an ordinary convergent iterated integral. It has the Hölder decomposition*

$$G(a_1, \ldots, a_w; 1) = \sum_{j=0}^{w} (-1)^j G(1-a_j, \ldots, 1-a_1; 1/2) G(a_{j+1}, \ldots, a_w; 1/2), \tag{5.238}$$

where a missing word denotes 1. *Replace each factor by its Taylor polynomial through degree* N, *evaluated at* $1/2$, *and call the resulting Gaussian rational number* C_N. *Then*

$$|G(a_1,\dots,a_w;1)-C_N|\le 2(w+1)\,T_w(N),\qquad T_d(N)=2^{-N}\sum_{k=0}^{d-1}\binom{N}{k}. \tag{5.239}$$

Here $\binom{N}{k}=0$ *for* $k>N$ *and* $T_0(N)=0$. *All the Taylor polynomials required by one word are computed using* $O(wN)$ *exact rational arithmetic operations. For an integer* $P\ge 0$, *the explicit choice*

$$N=3\big(P+w+\lceil\log_2(w+1)\rceil\big) \tag{5.240}$$

guarantees an error at most 2^{-P}. *In particular,* $N=O(P+w)$.

Proof. The hypotheses exclude singularities in the open integration segment. They also ensure integrability at its endpoints: the last letter is nonzero, and any logarithmic singularities at 1 are integrated against an outermost kernel regular there. Partition the ordered integration simplex according to the number j of variables above $1/2$. Replacing those variables by their complements reverses their word and contributes $(-1)^j$. This gives (5.238); this is the standard Hölder convolution, with its endpoint hypotheses made explicit.

Every nonzero letter in either factor of (5.238) has modulus at least 1. Moreover, every nonempty right factor ends at $a_w\neq 0$, and every nonempty left factor ends at $1-a_1\neq 0$. Thus these factors have Taylor series with zero constant coefficient.

Consider a word v of this kind, with d nonzero letters, and write $G(v;z)=\sum_{n\ge 0}c_nz^n$. Coefficientwise absolute values are bounded by replacing each nonzero kernel with $1/(1-z)$. Inserting a zero kernel divides the coefficient of z^n by $n\ge 1$, so deleting all zero letters can only increase this majorant. Consequently

$$|c_n|\le [z^n]\frac{(-\log(1-z))^d}{d!}\le\binom{n-1}{d-1}\quad(n\ge d),\qquad c_n=0\quad(n<d).$$

For the second inequality, expand the d logarithms; each composition of n into d positive parts contributes at most 1, and the additional factor $1/d!$ can be discarded. The empty word is handled separately. It follows that

$$|G(v;1/2)|\le 1,\qquad \left|\sum_{n=0}^{N}c_n2^{-n}\right|\le 1,$$

because $\sum_{n\ge d}\binom{n-1}{d-1}2^{-n}=1$. The tail of this majorant is exactly

$$\sum_{n>N}\binom{n-1}{d-1}2^{-n}=2^{-N}\sum_{k=0}^{d-1}\binom{N}{k}=T_d(N).$$

One proof interprets the left side as the probability that the dth success in fair independent Bernoulli trials occurs after trial N; the right side counts fewer than d successes among the first N trials. Equivalently, the equality follows by Pascal's identity.

For each product AB in the Hölder sum, with truncations A_N,B_N,

$$|AB-A_NB_N|\le|A-A_N|\,|B|+|A_N|\,|B-B_N|\le T_{d_A}(N)+T_{d_B}(N)\le 2T_w(N).$$

Summing over the $w+1$ cuts proves (5.239).

For the complexity claim, let b_n be the Taylor coefficients of a tail and c_n those of the word obtained by prepending a. For $n \geq 1$,

$$c_n = \begin{cases} b_n/n, & a = 0, \\ \big((n-1)c_{n-1} - b_{n-1}\big)/(an), & a \neq 0. \end{cases} \tag{5.241}$$

In both cases $c_0 = 0$, and the empty tail has $b_0 = 1$, $b_n = 0$ for $n \geq 1$. The second recurrence is coefficient comparison in $(z-a)G'(a,v;z) = G(v;z)$. It takes $O(N)$ operations per prepended letter. Compute the original suffix chain starting from the last letter; this gives every right factor. Compute the reflected reversed-prefix chain by successively prepending $1-a_1, 1-a_2, \ldots, 1-a_w$; this gives every left factor. Each chain requires $O(wN)$ operations in total. Summing the $w+1$ products adds $O(w)$ operations.

Finally, the binomial theorem gives the useful bound

$$T_w(N) \leq 2^{w-1}(3/4)^N.$$

Indeed, for $k \leq w-1$ one has $2^{-k} \geq 2^{-(w-1)}$, so

$$\sum_{k=0}^{w-1} \binom{N}{k} \leq 2^{w-1} \sum_{k=0}^{N} \binom{N}{k} 2^{-k} = 2^{w-1}(3/2)^N.$$

With $M = P + w + \lceil \log_2(w+1) \rceil$ and $N = 3M$,

$$2(w+1)T_w(N) \leq (w+1)2^w(3/4)^{3M} \leq (w+1)2^{w-M} \leq 2^{-P},$$

since $27/64 < 1/2$. This proves (5.240). □

Corollary 5.105 (Fixed imaginary quadratic fields)**.** *Let $K \subset \mathbb{C}$ be an imaginary quadratic field with its fixed complex embedding. Replace $\mathbb{Q}(i)$ in the alphabet definition by K. The same Hölder sum, tail bound, precision prescription, and $O(wN)$ arithmetic-operation count hold, with the center now in K.*

Proof. The proof of Theorem 5.104 uses the coefficient field only for exact additions, multiplications, and reciprocals. Each Taylor coefficient therefore lies in K, and the analytic majorant is unchanged. The tests $|a|^2 \geq 1$ and $|1-a|^2 \geq 1$ are exact rational comparisons because these squared moduli are field norms. Arithmetic in a fixed quadratic field requires a bounded number of rational operations per field operation. □

In particular, every sixth root of unity satisfies the alphabet condition: apart from 1, its distance from 1 is at least 1. Thus the theorem also supplies an exact-certificate route for the Eisenstein values. The supplied core implementation specializes to $\mathbb{Q}(i)$; the four mixed Eisenstein formulas are established analytically and checked numerically in the independent scripts.

A safe bit-complexity statement. The count $O(wN)$ is an arithmetic-operation count; it does not assign unit cost to arbitrarily large integers. Let Q be a common positive denominator for the real and imaginary components of the reciprocals of all nonzero letters used in both Hölder chains. Let $L_N = \operatorname{lcm}(1, \ldots, N)$. For a word of length $\ell \leq w$, each component of its degree-n Taylor coefficient has denominator dividing $Q^n L_n^\ell$. This follows from the convolution form of (5.241): a

term uses total reciprocal-letter degree n and one integer divisor at most n for each integration. Thus all evaluated partial sums have denominators dividing

$$D_N = (2Q)^N L_N^w.$$

The coefficient majorant bounds their numerator bit lengths by

$$B = O\big(N \log(2Q) + wN \log(N+1) + N\big),$$

using only the elementary estimate $L_N \le N!$. All product sums have denominator dividing D_N^2 and comparable operand lengths. Therefore standard exact rational arithmetic yields polynomial bit complexity in $w, N, \log Q$. For an entirely conservative explicit estimate, naive multiplication and division use $O(B^2)$ bit operations, the Euclidean algorithm uses at most $O(B)$ such divisions, and normalization after every operation yields $O(wNB^3)$ bit operations. This deliberately loose bound requires no fast-arithmetic assumptions. Standard fast multiplication and gcd algorithms improve it.

What is and is not new here. Hölder convolution and its use for numerical multiple-polylogarithm evaluation are established; see Borwein–Bradley–Broadhurst–Lisoněk [67] and Vollinga–Weinzierl [68]. The contribution of this specialization is the explicit rational tail certificate, the permitted rational Gaussian letter region, the reuse of both chains, and the uniform linear bound $N = O(P + w)$. This is an error bound for computing periods, not a proof of an identity inferred from a small residual.

Application to the manuscript's constants. The letters needed for π, $\log 2$, $\zeta(n)$ for $n \ge 2$, $\beta(n)$, and $\lambda_n = \Im \operatorname{Li}_n((1+i)/2)$ all satisfy the hypotheses:

$$\pi = 4\Im \operatorname{Li}_1(i), \quad \log 2 = -2\Re \operatorname{Li}_1(i), \quad \operatorname{Li}_n(z) = -G(0^{n-1}, z^{-1}; 1).$$

For the half point the nonzero letter is $1 - i$, whose reflected letter is i. Hence both sides of the Gaussian conjectures can be enclosed using one exact Gaussian-rational arithmetic framework.

5.37.1 Agreement with ordinary boundary series

For completeness, suppose $s_1, \dots, s_d$ are positive integers and $|z_j| = 1$. The outer-first colored multiple-polylogarithm series converges absolutely if $s_1 \ge 2$, and converges ordinarily if $s_1 = 1$ and $z_1 \ne 1$. These sufficient conditions include every colored value used in the Gaussian tables.

For $d \ge 2$, put

$$B_n = \sum_{n > n_2 > \cdots > n_d \ge 1} \prod_{j=2}^{d} \frac{z_j^{n_j}}{n_j^{s_j}}.$$

The elementary symmetric harmonic bound gives $|B_n| = O((1 + \log n)^{d-1})$ and

$$|B_{n+1} - B_n| = O\big((1 + \log n)^{d-2}/n\big).$$

The first estimate proves absolute convergence for $s_1 \ge 2$. For $s_1 = 1$ set $b_n = B_n/n$. Then $b_n \to 0$ and

$$|b_{n+1} - b_n| = O\big((1 + \log n)^{d-1}/n^2\big),$$

which is summable. The geometric partial sums of z_1^n are bounded when $z_1 \ne 1$, so summation by parts proves convergence of $\sum z_1^n b_n$. At $d = 1$ this is the usual Dirichlet test.

Introduce $0 < r < 1$ in the outer color, $z_1 \mapsto r z_1$. The absolutely convergent series/integral dictionary gives the GPL with the original word and endpoint r. Abel's theorem identifies the limit of the series with its ordinary boundary sum. On the integral side, the last letter is nonzero. Near zero every nonempty suffix is analytic and $O(t)$. Near one, recursive integration gives at worst powers of $|\log(1-t)|$ in an interior suffix. The outer letter is either 0 (when $s_1 \geq 2$) or $z_1^{-1} \neq 1$ (when $s_1 = 1$), so its kernel is bounded near one and these logarithmic powers are integrable. Thus the GPL has a continuous limit as $r \uparrow 1$ and agrees with the ordinary boundary series. No regularization is being silently used in these cases.

Chapter 12

Experimental discovery and the route to proof

The mathematical progression of the manuscript suggests a practical method. One begins with a value that is easy to compute but hard to recognize, finds an exact transformation that organizes its arguments, and then asks which coordinates remain after known laws are imposed. Numerical experiments are especially effective at proposing those coordinates. Their usefulness increases when the proposed identity is subsequently derived, and when a negative result is recorded with enough detail to explain its limits.

12.1 Constructing a meaningful integer-relation experiment

An experiment consists of a target, a finite ordered basket of constants, their coefficient field, a working precision, a coefficient bound and an acceptance rule. Weight and character parity give strong guidance for the basket. For instance a rational-angle Fourier transform can require $\sqrt{q}$-weighted atoms; searching only with rational coefficients in the unweighted values misses that structure. The $\pi^3\psi^{(-3)}(1/4)$ coordinate in Chapter 9 is another example: omitting the π^3 factor changes the question.

Before searching, remove known basket dependencies. A search containing both $\log\Gamma(1/2)$ and $\frac{1}{2}\log\pi$ may return their tautology instead of a relation involving the target. Reflecting arguments, distributing roots of unity and replacing elementary odd beta values reduce this problem before numerical linear algebra starts.

Working precision must exceed the cancellation loss as well as the digits needed to discriminate the proposed coefficient height. The heuristic $10^{P/d}$, for P available digits and d basket coordinates, estimates a scale at which accidental short numerical relations can appear. It is neither a theorem about a specific basket nor a rigorous exclusion bound. Coefficient sizes, conditioning and the number of searches all matter. The earlier false negative for S_5 illustrates this: a 51-coordinate basket at 150 digits obscured the height-3840 relation that a smaller, properly structured basket exposed.

12.2 Controls and independent recomputation

A planted positive control checks that the search can recover a known relation of comparable scale. A perturbation of one coefficient tests the resolution of the residual check. A negative control and an auxiliary “canary” constant can expose numerical coincidences. These controls are diagnostics.

In particular no independence theorem is known between the Champernowne constant and the periods in these baskets, and a zero canary coefficient cannot certify an identity.

A successful candidate should be frozen as exact rational or algebraic coefficients, then reevaluated at higher precision using an independent representation. For a nested sum that may be an iterated integral; for a CM row sum it may be a modular Fourier series; for Herglotz values it may be a defining integral compared with the finite dilogarithm sum. Agreement between two engines using the same incorrect formula is less informative than agreement between different mathematical representations. Error-controlled interval evaluation can establish a finite residual bound, but even an arbitrarily small nonzero numerical bound alone does not establish equality of general transcendental constants.

12.3 What exact linear algebra proves

An exact relation matrix proves an upper bound on the number of necessary coordinates in its presented quotient. Its row-combination matrix is a replayable finite certificate relative to the input identities. A free column is independent in that formal quotient, not automatically among the evaluated numerical periods. Additional functional equations or regularization can reduce the quotient further.

Likewise a motivic dimension theorem concerns motivic objects. To infer numerical independence one needs a period-injectivity statement, generally conjectural. A dimension for a chosen word-length grading need not be the dimension of the classical depth filtration. The Gaussian triples are a natural test case for this distinction.

Certificates require both exact atom cancellation and sound law instantiation. Substitution strings must be parsed within a restricted grammar; branches and domains must be established mathematically. A numerical branch guard is useful for rejecting errors but leaves a separate domain obligation. These issues motivate a future formal development rather than a current claim of proof-assistant verification.

12.4 Computational conventions

The manuscript uses decreasing nested indices. In the tested Wolfram Language 15 interface,

$$\mathrm{Li}_{a,b}(z_1,z_2) = \texttt{MultiplePolyLog[\{a,b\},\{z1,z2\}]}, \qquad \zeta(a,b) = \texttt{MultipleZeta[\{a,b\}]}.$$

Finite checks such as $\zeta(2,1,1) = \zeta(4)$ and $H(0,1;z) = \mathrm{Li}_2(z)$ calibrate the interface. A divergent call does not resolve ordering ambiguities on its own. The same calibration is needed for signed harmonic indices and generalized-polylogarithm letter order.

Negative-order polygamma means the integral based at zero: $\psi^{(-1)} = \log\Gamma$ and $\psi^{(-n)}(z) = \int_0^z \psi^{(-(n-1))}(t)\,dt$ for $n \geq 2$. Balanced conventions can differ by explicit polynomials. In Hurwitz jets, $\zeta^{(j)}$ always differentiates the spectral variable; the superscript (k) on γ_n differentiates its parameter. The CM hexagonal point $e^{i\pi/3}$ and the Eisenstein cube root $e^{2\pi i/3}$ are different arguments even when a local section denotes each by ρ.

12.5 The surviving research programme

Several concrete proof problems now fit into one framework. The odd-weight S_p directions need a precise choice of comparison algebra and a rigorous reduction or obstruction; the even-weight

direction has the generating-integral proof of Chapter 4. The Gaussian and Eisenstein inverse-color points now have an all-weight partial-fraction reduction, and the distribution ranks have a complete character normal form. A full regularized relation computation remains useful for the other colored points. Numerical independence of character derivative coordinates remains a separate arithmetic problem.

The gamma programme can transport its exact row certificates across the DFT and prove branch validity for each law instance. Its finite-matrix termination should remain separate from Rohrlich's conjecture. The CM programme can replace fitted coefficients by modular equations and class-field computations, while taking account of periods under Galois conjugation. The Herglotz programme needs complete explicit class-character and unit definitions for the regulator candidates before they become locally replayable, followed by the relevant Stark argument.

The cubic log-gamma moment now has the explicit Tornheim representation and certified enclosure of Section 8.5. Reduction of those derivative coordinates into a stated smaller algebra remains a precise question, alongside the opaque individual CM components. The Gaussian weight-five relation, weight-six doubles, weight-four triples and cubic/quartic ladders likewise now have exact proofs. The S_4 relation now has the exact certificate of Theorem 4.8. Higher golden ladder certificates and numerical period independence remain open; the S_6 candidate supplies a concrete next mixed-reduction problem. A richer representation is often more productive than a larger unstructured search. The scientific value of the experiments is that they tell us which representation to seek next.

12.6 Questions sharpened by the exact continuations

12.6.1 Real-order geometry after the positive-difference theorem

The finite signed-moment threshold no longer limits the zero-free theorem. The positive consecutive-difference measure of Theorem 5.24 supplies the missing subcritical representation, and Theorem 5.26 proves nonvanishing away from the double origin for every $a \geq 0, b > 0$ on the principal slit plane. Angular uniqueness holds throughout this domain. On and below $a + b = 1$, and on the entire $a = 0$ axis, the normalized angular radius strictly increases. The full-radius sign conjecture for integer $a \geq 2$ remains open; the fractional turning points show why local coefficient signs alone cannot settle a general global motion law.

The Gaussian experiments have also led to proofs rather than just narrower searches. Fixed-total monotonicity is proved in Theorem 5.31; the unique global Gaussian maximum has the exact stationary-point bracket $1 < b_* < 2$ in Theorem 5.32. Its location and height now have sharp rational enclosures in Theorem 5.36. Corollary 5.33 settles the optimal uniform Euler constant $\pi/4 + \log(2)/2$ for positive orders with $a + b \leq 1$. The contraction argument of Theorem 5.64 now settles the universal real-order Euler constant as the unique Gaussian axis maximum $C(b_*)$. The number of quartic transition points and the Bessel and uniform Lerch continuations remain further proof problems. The universal bound uses the separately proved tail-kernel contraction, rather than Gaussian magnitude alone.

12.6.2 Zero geometry beyond a fixed Laurent index

The complete classical counts are now known through Laurent index seven. The fifth-index transition has threshold three, and Theorem 10.25 proves sharp thresholds four and five at indices six and seven. Critical-point descent determines the deficits below saturation with exact signs, rather than extrapolating from a root scan. Small-order classification for higher indices remains open. Effective

thresholds uniform in the positive Lerch parameter require quantitative separation of the Appell roots and explicit concentration remainders. The spectral coefficient expansion for $n = O(\log k)$ does not by itself supply moving-zero asymptotics throughout that range, because the extreme root slopes can leave a compact scale. For higher zero branches, monotonicity in the deformation parameter is suggested by the first correction but is proved globally only at $n = 1$.

12.6.3 Certificates, integral structure and arithmetic evaluation

The complex distribution module is free even at resonant weights. What changes there is the usefulness of selected coordinates, not its rank. The division-free point basis and unit minor now settle integral unreflected Smith factors and arbitrary-ring base change. Reflection torsion, all-field rank loci and one-parameter Smith exponents are proved in Section 9.8; Theorem 9.43 also gives a closed product law for separable multivariable jets. General multivariable module extensions, sharper certificate-height bounds and arithmetic evaluation kernels remain separate questions. The same distinction applies to the exterior-symbol calculus: a nonzero formal obstruction rules out a certificate in that calculus, while the kernel of numerical period evaluation is a deeper arithmetic problem. Constructive reduction can now be benchmarked against an exact admissibility criterion rather than a denominator heuristic.

12.6.4 Further positive kernels and approximation regimes

The harmonic-index transition and Herglotz optimal cutoff illustrate two ways that positive kernels convert experiments into uniform asymptotics. Other roots of unity introduce complex weights and can destroy positivity; their extensions need a replacement argument rather than a formal reuse of the signed bounds. The general $J(p/q)$ symbol problem extends the proved $J(2/q)$ classification. At the optimal approximation scale, algebraic corrections and exponentially smaller arithmetic shifts require separately defined reference transitions. Their numerical diagnostics should retain that separation of scales.

12.6.5 A focused path toward formal verification

The finite algebraic interfaces are suitable first targets: character projectors, prime distribution edges, polynomial normal forms, exact partial fractions and the rational interval operations. Analytical formalization can then isolate the root-preserving operator, Laplace variation argument, positive-kernel remainder theorems and local normalized functional equation. These are explicit future tasks, not claims that the present ordinary proofs have already been checked by a proof assistant.

12.7 From decimal fits to short executable proofs

The Gaussian continuation illustrates a complete experimental cycle. A weight-five relation first found with large coefficients is the combination of two simple shuffle rows; the parity theorem then explains all five weight-six double reductions. The one-two integral supplies the three weight-four triples, while algebraic argument substitutions and rational row operations certify the cubic and quartic ladders. The cubic log-gamma moment requires a Tornheim derivative coordinate rather than a larger unstructured single-value basket.

The accompanying fresh replay re-expands all 511 ending-in-one binary words through weight nine and checks 45 bigraded sectors. It verifies 35 exact algebraic ladder substitutions and reproduces the

integer cubic moment certificate, whose endpoints share 112 decimal places. Independent defining-integral comparisons check 64 double and one-two formulas. The Chebyshev implementation supplies 89 interval residual enclosures, including one for S_4, which was conjectural at that checkpoint. That enclosure containing zero did not prove equality; the later octahedral certificate provides the exact argument. Six separate midpoint rectangles enclose the Gaussian and Eisenstein inverse-color values to widths below 10^{-70}. These finite replays test the implementations. The all-weight and asymptotic statements rest on the analytic and algebraic arguments in the chapters.

12.8 What the new reports change in the research programme

The exact S_4 proof adds a convergent octahedral word transformation and higher-depth lifted rows to a vocabulary that the finite obstructions had shown insufficient. The weight-three companion in Theorem 4.11 supplies a second exact mixed reduction. The product-matrix rank conjecture is now proved by Theorem 5.15. The next task is to compress the larger certificate and investigate the frozen S_6 relation with the same analytic control. Pure complementary seeds now have a complete collision classification and an algebraic-degree cutoff; arbitrary multiplicative unit relations and maximal ladder weights remain broader questions.

Primitive-grid determinants and Smith factors quantify coordinate failures without changing the free distribution quotient. Explicit finite descent and pole-cancelled Fourier jets give constructive all-index identities and trace zeros; integral lattices and certified high-order implementations remain distinct developments. The new sharp and reflected moment theories separate fixed-exponent asymptotics, optimal truncation and simultaneous large-exponent limits. Likewise, the Herglotz oscillation is proved for fixed positive argument and large derivative order; joint scaling requires further uniform estimates.

12.9 A certified rejection beyond the discovery precision

A new Cayley continuation supplies a useful counterexample to premature promotion of integer-relation output. At weight nine, define the ordered basket

$$\mathbf{b}_8 = (S_8, g_{8,1}, g_{7,2}, g_{6,3}, g_{5,4}, \pi^9, \beta(8)\log 2, \beta(2)\zeta(7), \beta(4)\zeta(5), \beta(6)\zeta(3)).$$

A search with 110 working digits, tolerance 10^{-90} and coefficient ceiling 10^{12} returned the integer vector

$$\begin{aligned}\mathbf{c}_8 = (&-6451250972, -4072849461, -710107844, -10984374763,\\ &-6393592639, 31875, -1955138863, 3459066102, -1034180101, -1477786143).\end{aligned}$$

Proposition 12.1 (The frozen weight-nine vector is false)**.** *With* $R_8 = \mathbf{c}_8 \cdot \mathbf{b}_8$*,*

$$-1.3274179020580459 \times 10^{-86} < R_8 < -1.3274179020580457 \times 10^{-86} < 0.$$

Proof. The delivered rational evaluator encloses each harmonic double and each single atom with 1000 Euler terms, using the explicit signed tail budgets. It forms all products and the frozen linear combination in exact interval arithmetic. The resulting rational residual interval has width less than 2.080×10^{-290} and lies strictly inside the displayed negative interval. The fresh isolated replay recomputes every rational atom endpoint and the residual, comparing exact fractions with the preserved record `reports/cayley-s4-continuation/data/S8_exclusion_N1000.json`. Both the negative sign and the containment are exact rational comparisons. This computation proves

the displayed exclusion; it uses no floating-point sign or integer-relation search as an acceptance criterion. □

This rejects precisely the frozen vector, not every possible relation in the basket. A later unsuccessful higher-precision search likewise proves no arithmetic independence. In contrast the new alternative S_6 vector still has an enclosure containing zero; it remains conjectural, and Proposition 4.41 proves by two convergent shuffles that it is the earlier S_6 conjecture in a different basket.

12.10 The complete Cayley ideal does not close the bounded weight-seven search

The pinned manuscript leaves the mixed Gaussian identities for S_6 and S_8 conjectural. Two recent source contributions sharpened the algebraic problem in complementary ways. One supplied an exact integer functional separating the S_6 target from a specified collection of 5131 relations of depth at most four [125]. The other constructed a depth-preserving retraction for the *full* shuffle ideal generated by convergent Cayley identities [126]. It explicitly proposed projecting the bounded rows by that retraction as a next proof attempt.

The exact resonant continuation [129] completes that proposed calculation. The inherited integer functional also annihilates the intersection of the *entire* Cayley shuffle ideal with the weight-seven space of depth at most four. This includes all Cayley rows and all their shuffle multiples, even when their construction uses higher-depth intermediate words. Thus completing the Cayley side alone cannot prove S_6 from the stated bounded additional relations. The numerical S_6 identity remains open. This is a result about an explicit formal presentation; it is not nonvanishing of its evaluated period residual, and it gives no period-independence assertion.

12.10.1 The target and the exact formal presentation

Use the convention

$$\mathrm{Li}_{r,s}(u,v)=\sum_{n>m\ge 1}\frac{u^n v^m}{n^r m^s},\qquad g_{r,s}=\operatorname{Im}\mathrm{Li}_{r,s}(i,1),\qquad S_p=\sum_{n\ge 0}\frac{(-1)^n H_n}{(2n+1)^p}.$$

Let $L=\log 2$, and write the conjectural residual in its already reconciled short coordinate form:

$$\begin{aligned}R_6 = S_6 &+ \frac{12334}{527}g_{6,1}+\frac{384}{31}g_{5,2}+\frac{2136}{527}g_{4,3}-\frac{3179\pi^7}{5713920}\\ &+\frac{801}{2108}\beta(4)\zeta(3)+2\beta(6)L.\end{aligned}\tag{12.1}$$

The conjecture is $R_6=0$.

For formal words take the positive forms

$$z=\frac{dt}{t},\quad c=\frac{dt}{1-t},\quad a=\frac{i\,dt}{1-it},\quad b=-\frac{dt}{1+t},\quad \bar a=-\frac{i\,dt}{1+it}.$$

Their iterated integrals $H(w)$ are written outer letter first on $[0,1]$. An admissible nonempty word begins with a letter other than c and ends with a letter other than z. Let $\mathcal{A}$ be the graded rational shuffle algebra on the admissible words and the empty word. Conjugation K interchanges $a,\bar a$ and

fixes z, c, b. The superscript $-$ denotes its negative eigenspace, and $F_D\mathcal{A}$ is the span of words with at most D nonzero letters. This is the usual series-depth filtration.

The map induced by $t \mapsto (1-t)/(1+t)$ is

$$C(v_1 \cdots v_n) = \tau(v_n) \cdots \tau(v_1), \qquad \begin{array}{c|ccccc} v & z & c & a & b & \bar{a} \\ \hline \tau v & c-b & z+b & b-\bar{a} & b & b-a. \end{array}$$

Here $\tau = -\phi^*$ incorporates the reversal of orientation. Every transformed admissible word is admissible, and change of variables gives $H(Cw) = H(w)$. Consequently the period map annihilates the graded, K-stable shuffle ideal

$$\mathcal{J} = \langle f - Cf : f \in \mathcal{A} \rangle_{\Pi}. \tag{12.2}$$

No restriction on weight, depth, or multiplier is made in this definition.

In the following polynomial, juxtaposition is concatenation and $\star$ is the usual series stuffle product, transferred to words by cumulative colors. Define

$$\begin{aligned} P_6 = {} & 128588 z^5 a\bar{a} + 3138084 z^5 aa + 1592832 z^4 aza + 521184 z^3 az^2 a - 216172 z^6 a \\ & + 48861\big((z^3 a) \star (z^2 c)\big) - 257176\big((z^5 a) \star b\big), \qquad T = \frac{P_6 - K P_6}{2}. \end{aligned} \tag{12.3}$$

This is the exact target used by the earlier bounded certificate. It belongs to $F_2\mathcal{A}_7^-$. The elementary harmonic split gives $S_6 = g_{6,1} + \operatorname{Im} \operatorname{Li}_{6,1}(i, -1)$, while $\beta(7) = 61\pi^7/184320$. Using also $H(b) = -L$ gives

$$H(T) = 128588\, i R_6. \tag{12.4}$$

Both products in (12.3) use convergent inputs, so their stuffle expansion requires no regularization.

For precision, $\mathcal{R}_4 \subset F_4\mathcal{A}_7^-$ denotes the *same* inherited finite row family as in [125]. Its construction is retained in the delivered independent enumerator:

1. All weight-seven shuffle-minus-stuffle products with sum of input depths at most four. Inputs are admissible except for the permitted single factor $\operatorname{Li}_1(1)$; the standard one-divergence comparison cancels its divergent term.

2. All convergent level-two-to-level-four distribution rows and their products with admissible complementary-weight words, provided the sum of depths is at most four. Products are expanded by stuffle.

3. The already proved imaginary S_2 and S_4 rows multiplied by real parts of all admissible complementary-weight words of depth at most two, again expanded by stuffle.

4. The inherited balanced Cayley rows and their specified products. In the adapted alphabet below, the seed u begins with a letter other than x, ends with a letter other than z, and satisfies $\min(n_z(u), n_x(u)) \geq 1$. A multiplier v is allowed when $|u| - \min(n_z(u), n_x(u)) + \operatorname{dep}(v) \leq 4$ and $|u| + |v| = 7$. Its product is expanded by stuffle.

Imaginary projection, rational normalization, and duplicate removal give, in this order, 3460, 1394, 40, and 237 new nonzero rows, respectively: 5131 rows in total. These counts specify a formal enumeration, not dimensions of periods. The fourth family may be retained even though the new assertion explicitly imposes the complete ideal (12.2).

Theorem 12.2 (Full-Cayley extension of the weight-seven obstruction)**.** *With the formal target and row family just specified,*

$$\boxed{T \notin \mathcal{J}_7^- + \operatorname{span}_{\mathbb{Q}} \mathcal{R}_4.} \tag{12.5}$$

In particular the inherited finite proof search remains unsuccessful even after all *convergent Cayley relations and arbitrary shuffle products of them have been included.*

The distinction between the full ideal and its bounded intersection is essential. We verify a finite basis in $F_4\mathcal{A}_7^-$, but the theorem allows $\mathcal{J}$ in the entire admissible word algebra. The next two subsections justify why those finite checks imply that stronger assertion.

12.10.2 Why the depth-preserving projector controls the entire ideal

We state and justify the precise prerequisite from [126], so the logic of the extension does not depend on a finite-rank extrapolation. Introduce

$$x = c - b, \qquad y = a - \bar{a}, \qquad h = a + \bar{a} - b, \qquad \mathcal{E} = \{z, y, h, b, x\}.$$

Then

$$c = x + b, \qquad a = (y + h + b)/2, \qquad \bar{a} = (-y + h + b)/2.$$

In this basis admissibility means first letter different from x and last letter different from z. Depth is $|w| - n_z(w)$, conjugation acts by $(-1)^{n_y(w)}$, and C reverses a word, interchanges z, x, and multiplies by $(-1)^{n_h(w)}$.

Let $\widetilde{\mathcal{A}}$ be the full shuffle algebra, including nonadmissible formal words. It is used here only algebraically. With the order $z < y < h < b < x$, every Lyndon word of length at least two is admissible. The shuffle-polynomial theorem [127] consequently gives

$$\widetilde{\mathcal{A}} = \mathcal{A}[z, x].$$

Let R_0 be the algebra retraction setting the two *shuffle polynomial* variables z, x to zero. It preserves both letter counts, commutes with C, K, and is explicitly

$$R_0(w) = \sum_{u=0}^{l} \sum_{v=0}^{r} (-1)^{u+v} x^u \sqcup w[u : |w| - v] \sqcup z^v, \tag{12.6}$$

where l, r are the initial x run and terminal z run. Deleting an initial x and deleting a terminal z are commuting shuffle derivations; their finite Taylor substitutions prove (12.6). The factorials cancel because shuffle powers of a single letter are factorial multiples of its concatenation powers. In particular this is not literal erasure of letters inside a word, and assigns no analytic value to divergent integrals.

For nonempty w put

$$e(w) = \sum_{k=1}^{|w|} \frac{(-1)^{k-1}}{k} \sum_{w = w_1 \cdots w_k} w_1 \sqcup \cdots \sqcup w_k, \qquad E = R_0 e, \tag{12.7}$$

where the blocks are consecutive and nonempty. The map e is the first Eulerian logarithm. It annihilates products of positive-degree factors and equals the identity modulo such products [128]. These facts also follow from the convolution logarithm of the identity character into the commutative shuffle algebra: its logarithm is an infinitesimal character, and the term $k = 1$ gives the congruence.

Thus $e^2 = e$. Reversal permutes consecutive cuts, so e commutes with C; it also commutes with K and preserves both counts.

The subspace $U = E(\mathcal{A})$ maps isomorphically onto the indecomposables of $\mathcal{A}$, and multiplication identifies $\mathcal{A}$ with $\mathrm{Sym}(U)$. To see this directly, E kills positive-degree products and is the identity modulo them. Any homogeneous lift of an indecomposable basis therefore gives polynomial generators, by a triangular change from the Lyndon generators. The lifts are bigraded by (n_z, n_x), and C exchanges these two degrees.

For a homogeneous $u \in U$ of degree (p, q) choose the orientation

$$Q_\uparrow u = \begin{cases} u, & p > q, \\ Cu, & p < q, \\ (u + Cu)/2, & p = q. \end{cases}$$

On each pair of different bidegrees this identifies the two transported generators. On the diagonal it retains the positive eigenspace of C and kills its negative eigenspace. Extending multiplicatively gives an algebra retraction whose kernel is exactly $\mathcal{J}$: on the quotient all $u - Cu$ vanish, hence C acts identically on every polynomial, and conversely $u - Cu$ are among the ideal generators. With $E_\uparrow = Q_\uparrow E$, the retraction is the finite formula

$$\Pi_\uparrow(w) = \sum_{k=1}^{|w|} \frac{1}{k!} \sum_{w = w_1 \cdots w_k} E_\uparrow(w_1) \,\Pi\, \cdots \,\Pi\, E_\uparrow(w_k), \qquad \Pi_\uparrow(1) = 1. \tag{12.8}$$

Convolution exponentiation of (12.7), followed by R_0 and the multiplicative orientation, proves this expression. For completeness, $E(w) \in U$ even if a block w is not admissible: in degree at least two, the polynomial decomposition $\mathcal{A}[z, x]$ makes $w - R_0 w$ a sum of positive-degree products, killed by e; in degree one, $R_0 e$ kills the two extra generators directly.

The orientation never decreases the number of z's in any nonzero output. Shuffle multiplication adds their counts. Consequently, on $\mathcal{A}$,

$$\Pi_\uparrow^2 = \Pi_\uparrow, \quad \ker \Pi_\uparrow = \mathcal{J}, \quad \Pi_\uparrow K = K \Pi_\uparrow, \quad \Pi_\uparrow(F_D \mathcal{A}) \subseteq F_D \mathcal{A}. \tag{12.9}$$

These are all-weight algebraic statements. Notice that $C\Pi_\uparrow = \Pi_\uparrow$ is neither required nor generally true.

The verifier evaluates (12.7) using a permutation formula; its equivalence with consecutive cuts is also elementary in every weight. For a permutation of the n positions let d be the number of descents of its inverse. A shuffle of consecutive blocks produces that permutation exactly when cuts are placed at those d required positions, with any additional cuts chosen freely. Its total coefficient in (12.7) is therefore

$$\sum_{k=d+1}^{n} \frac{(-1)^{k-1}}{k} \binom{n-1-d}{k-1-d} = (-1)^d \int_0^1 t^d (1-t)^{n-1-d}\, dt = \frac{(-1)^d}{n\binom{n-1}{d}}.$$

This is precisely the implemented coefficient. Repeated letters simply combine the integer multiplicities of equal output words. Thus the finite audits of the two formulas are convention checks; their equality at weight seven does not depend on extrapolating those audits.

Lemma 12.3 (The full ideal intersection has a finite spanning certificate)**.** *If $\mathcal{B}_4$ is any rational basis of $F_4\mathcal{A}_7^-$, then*

$$\mathcal{J}_7^- \cap F_4 \mathcal{A}_7 = \mathrm{span}_{\mathbb{Q}}\{w - \Pi_\uparrow(w) : w \in \mathcal{B}_4\}.$$

Proof. Every displayed difference belongs to the full ideal by the kernel property and belongs to the bounded space by depth preservation. Conversely, for f in the intersection, $\Pi_{\uparrow} f = 0$, so $f = (1 - \Pi_{\uparrow}) f$. Expanding f in the finite basis proves the opposite inclusion. No restriction on the construction of f inside $\mathcal{J}$ has been imposed. □

12.10.3 The integer certificate and the proof

Use the adapted-letter encoding $Z = 0, Y = 1, H = 2, B = 3, X = 4$. Enumerate all length-seven words lexicographically, retaining exactly those with first letter different from X, last letter different from Z, at least three Z's, and an odd number of Y's. This gives a basis $\mathcal{B}_4$ of $F_4\mathcal{A}_7^-$ with 2546 elements. The exact-depth counts are $1, 35, 390, 2120$. They can be checked directly from the enumeration, or by choosing the nonzero positions and projecting to odd Y parity.

The certificate's `separator.json` supplies an integer functional λ on this basis. It is obtained by expressing the inherited functional of [125] in adapted coordinates; the functional itself is not being claimed as a newly discovered vector. The new fact is the second row of the following exact certificate:

$$\lambda(R) = 0 \qquad (R \in \mathcal{R}_4), \tag{12.10}$$

$$\lambda(w - \Pi_{\uparrow} w) = 0 \qquad (w \in \mathcal{B}_4), \tag{12.11}$$

$$\lambda(T) = -\mathcal{N} \neq 0, \tag{12.12}$$

where the positive integer is

$$\begin{aligned}\mathcal{N} = {} & 18666062728923681262058369113049668 2 \cdot 10^{36} \\ & + 78843750753489297279472069700354208.\end{aligned}$$

The integer is an exact functional value, not a numerical residual of the polylogarithm conjecture. Its decimal expansion is also recorded in the certificate JSON.

Proof of Theorem 12.2. The standalone verifier, freshly replayed on an isolated copy, regenerates the complete inherited row family, checks (12.10), and independently reconstructs each $\Pi_{\uparrow}(w)$ by (12.6)–(12.8). It verifies (12.11) for all 2546 basis words, and reconstructs (12.3) before checking (12.12). All arithmetic is integral or rational. Thus Lemma 12.3 shows that λ annihilates $\mathcal{J}_7^- \cap F_4\mathcal{A}_7$.

If, contrary to the theorem, $T = j + r$ with $j \in \mathcal{J}_7^-$ and $r \in \operatorname{span} \mathcal{R}_4$, then $T, r \in F_4\mathcal{A}_7^-$ implies $j = T - r \in F_4\mathcal{A}_7^-$. Consequently $\lambda(j) = 0$; also $\lambda(r) = 0$ by (12.10). This contradicts $\lambda(T) \neq 0$ and proves (12.5). □

For reproducibility, execute

```
py certificates/cayley/code/verify_complete_cayley.py
```

from the package root. It requires only Python's standard library. The verifier checks SHA-256 digests of the two retained source modules and the functional data before use. It regenerates 5131 rows, verifies 2546 complete-ideal basis differences, and checks the nonzero target pairing. There are 2300 nonzero projector differences; no claim of their linear independence is made. Additional finite audits compare the Eulerian permutation formula against consecutive cuts on all 780 words through weight four and check 2150 shuffle products, as well as retraction, conjugation, and depth conventions. These audits support the implementation; the preceding algebraic proof supplies the all-weight projector properties and the passage to the full ideal.

12.10.4 Consequences for further research

The immediate computational recommendation is sharper than in the inspected sources. Completing Cayley symmetry while retaining precisely $\mathcal{R}_4$ does not settle S_6. A successful continuation must change the additional relation input: for example by admitting suitable higher-depth double-shuffle or distribution rows, by adding another functional equation, or by deriving a uniform generating identity that implies the target. A larger numerical-precision calculation does not address this missing algebraic information.

The theorem does not prove that every conceivable derivation must pass through depth five. It excludes exactly the full ideal plus the specified finite additional span; another depth-four family could carry information outside that span. Likewise, no conclusion for the revised S_8 candidate follows merely by analogy. Useful next questions are to identify the lowest-weight additional row whose projected class pairs nontrivially with λ, to test whether selected depth-five double-shuffle rows suffice, and to find a conceptual functional equation explaining the whole even-index family. These are exact proof-search questions, while $R_6 = 0$ itself remains the separate numerical-period conjecture.

12.10.5 Every comparison with one first letter X at depth five

The formal obstruction above motivates higher-depth input relations. A complete finite experiment can be stronger than sampling more such inputs. Let $\mathcal{V}$ be all admissible adapted words v of length six and depth four with odd conjugation parity. There are 1032 such words. With $X = C - B$, take the double-shuffle difference $DS(X, v)$, expand it in the adapted alphabet, and apply the proved full-Cayley retraction Π. Each input is a linear combination of a convergent comparison with B and a single-divergence comparison with C; its nonadmissible terms cancel.

Write F_5 for the odd admissible weight-seven space of depth at most five, and F_4 for its bounded subspace. Their dimensions are 8786 and 2546; the remaining 6240 coordinates have depth exactly five. Let

$$A : \mathbb{Q}^{\mathcal{V}} \longrightarrow F_5, \qquad A(e_v) = \Pi DS(X, v),$$

and let $p_5 : F_5 \to F_5/F_4$ retain every depth-five coordinate.

Theorem 12.4 (Exact completeness of the bounded contributions)**.** *The map p_5A has rank* 1008 *over $\mathbb{Q}$ and a* 24*-dimensional kernel. The* 24 *exact source combinations already obtained from the earlier* 190 *selected inputs form a basis of this complete kernel. Consequently*

$$\operatorname{im} A \cap F_4 = \operatorname{span}_{\mathbb{Q}}\{b_1, \dots, b_{24}\},$$

where the b_j are those previously reconstructed bounded relations. For any relation space $\mathcal{B} \subset F_4$,

$$(\mathcal{B} + \operatorname{im} A) \cap F_4 = \mathcal{B} + \operatorname{span}_{\mathbb{Q}}\{b_1, \dots, b_{24}\}.$$

This equality does not assert independence of the b_j after quotienting by $\mathcal{B}$, nor any arithmetic independence of periods.

Proof. The compressed exact record retains all 1032 projected rows, including every outside coordinate. The independently audited certificate `verification/S6-full-depth-five-kernel-certificate.json` first matches all 190 older rows to this complete record. It then reconstructs each of the 24 old rational source combinations using the full coefficients, verifying both the expected bounded result and cancellation of every depth-five coefficient. Their last nonzero input indices

are distinct, giving a triangular minor of the source combinations with nonzero diagonal. They are therefore independent, and $\operatorname{rank}_{\mathbb{Q}}(p_5 A) \leq 1032 - 24 = 1008$.

For the converse, the certificate selects a 1008-row, 1008-column minor of the outside matrix and reconstructs it directly from the rational projected rows. Clearing the common denominator 5760 gives an integer matrix whose determinant modulo 2147483647 is

$$221991308 \neq 0.$$

An independent integer implementation verifies this determinant; each modular product and subtraction fits signed 64-bit arithmetic, and no floating-point matrix operation is used. Thus the rational minor is nonzero and the rank is at least 1008. Equality proves that the 24 source combinations span the whole kernel. Finally $A(c)$ is bounded exactly when $p_5 A(c) = 0$, proving the first intersection identity. Subtracting a bounded element of $\mathcal{B}$ proves the second. □

The associated modular target search keeps all 8786 coordinates. Its 6187 rows consist of the 5131 inherited relations, the 24 previous bounded relations and the complete 1032 inputs. Its rank is 2905 modulo the stated prime and the target remains outside that modular span. This last observation is a search diagnostic: the exact outside-rank certificate does not, by itself, certify the rational rank of the bounded relation space or rational nonmembership of the target. It neither disproves the period identity nor proves a universal depth requirement. The exact kernel result explains why further sampling within this same first-letter family supplies no additional bounded information; other comparison families and functional relations remain appropriate next targets.

Appendix A

Literature problems and source provenance

The two source catalogs organize questions that motivated or could extend the programme. Their categories and answer status describe the June 2026 survey, not a current audit of the websites. A catalog entry is a research lead, not an accepted identity. Closely overlapping questions are retained under a single thread identifier. The external reports numbered one through four contributed leads to these catalogs; their unverified prose and claims are not used as mathematical authorities.

One erroneous external formula is corrected here with a proof. Euler reflection at $1/3$, together with Landen at $-1/3$ and duplication, gives

$$6\,\mathrm{Li}_2(1/3) - \mathrm{Li}_2(1/9) = \frac{\pi^2}{3} - \log^2 3.$$

For completeness, put $A = \mathrm{Li}_2(1/3)$ and $B = \mathrm{Li}_2(-1/3)$. The five-term relation at $x = y = 1/2$ and Landen at $-1/3$ give

$$2\,\mathrm{Li}_2(1/2) = \mathrm{Li}_2(1/4) + 2A + \log^2(2/3), \quad \mathrm{Li}_2(1/4) = -B - \frac{1}{2}\log^2(4/3).$$

With $2\,\mathrm{Li}_2(1/2) = \pi^2/6 - \log^2 2$, collecting the logarithms gives $2A - B = \pi^2/6 - \frac{1}{2}\log^2 3$. Duplication says $\mathrm{Li}_2(1/9) = 2A + 2B$, so subtracting it from $6A$ proves the corrected formula. This derivation fixes the factor-two error in the external report without relying on numerical recognition.

A.1 Polylogarithm questions

Thread	Research question	Historical category
MSE 1424600	Real part of $\mathrm{Li}_2(1/2 + i/6)$	Seed question
MSE 1430169	Dilogarithm values at algebraic complex arguments	Seed question
MSE 1373123	An identity for six rational tetralogarithm values	Seed question
MSE 945972	Relations among rational polylogarithm values	Seed question
MSE 1579355	Trilogarithms and the imaginary golden ratio	Seed question
MSE 918680	Closed Form for the Imaginary Part of Li_3((1+i)/2)	Primary answered matches on Math Stack Exchange
MSE 3290798	A conjectured value for Re Li_4(1+i)	Primary answered matches on Math Stack Exchange

Thread	Research question	Historical category
MSE 3447529	On the relationship between Re Li_n(1+i) and Li_n(1/2) when n >= 5	Primary answered matches on Math Stack Exchange
MSE 676124	Proving a Li_3(-1/3) - 2 Li_3(1/3) identity	Primary answered matches on Math Stack Exchange
MSE 1450917	A special value polylogarithm identity involving Li_3(-1/2), Li_3(-1/3), Li_3(2/3), Li_2(-1/3), Li_2(2/3)	Primary answered matches on Math Stack Exchange
MSE 935366	Simplification of an expression containing Li_3(x) terms	Primary answered matches on Math Stack Exchange
MSE 5092144	How to prove this zeta(3) form with Li_3 and Li_2	Primary answered matches on Math Stack Exchange
MSE 463682	Closed form of Li_2(phi) and Li_2(phi-1)	Primary answered matches on Math Stack Exchange
MSE 1410276	Closed-form of Li_2(1 +- i sqrt(3))	Primary answered matches on Math Stack Exchange
MSE 967398	Extract real and imaginary parts of Li_2(i(2 +- sqrt(3)))	Primary answered matches on Math Stack Exchange
MSE 1602647	Extract imaginary part of Li_3(2/3 - i 2 sqrt(2)/3) in closed form	Primary answered matches on Math Stack Exchange
MSE 980179	Closed-form of a special value dilogarithm identity	Primary answered matches on Math Stack Exchange
MSE 989704	Closed-forms of real parts of special value dilogarithm identities from inverse tangent integral function	Primary answered matches on Math Stack Exchange
MSE 1436493	Dilogarithm identity containing the tribonacci constant	Primary answered matches on Math Stack Exchange
MSE 481682	Polylogarithm ladders for the tribonacci and n-nacci constants	Primary answered matches on Math Stack Exchange
MSE 1064100	Why does the tribonacci constant have a trilogarithm ladder?	Primary answered matches on Math Stack Exchange
MSE 1959503	Proof of a dilogarithm identity	Primary answered matches on Math Stack Exchange
MSE 932932	Known exact values of the Li_3 function	Primary answered matches on Math Stack Exchange
MSE 1411312	Closed form of a series (dilogarithm)	Primary answered matches on Math Stack Exchange
MSE 2389425	A very odd-looking statement about zeta(3) and Li_2(1/phi^2)	Primary answered matches on Math Stack Exchange
MO 492746	Proving identity involving dilogarithms and pi/9	Primary answered matches on MathOverflow
MO 85986	A dilogarithm identity: known or new?	Primary answered matches on MathOverflow
MO 144322	The relationship between the dilogarithm and the golden ratio	Primary answered matches on MathOverflow
MO 261408	Several conjectured identities for polylogarithms	Primary answered matches on MathOverflow
MO 450252	An identity involving polylogarithms	Primary answered matches on MathOverflow
MO 511718	Two-term dilogarithmic identity simplification	Primary answered matches on MathOverflow
MO 511765	Daniel Shanks "simplest cubic fields" and dilogarithm identities	Primary answered matches on MathOverflow
MO 131111	Is this combination of generalized polygamma and dilogarithm actually zero?	Primary answered matches on MathOverflow

Thread	Research question	Historical category
MSE 5080786	Ramanujan type formulas for polylogarithm	Open prompts with usable identity examples but no meaningful answers
MSE 5023055	How to find closed form for a complex expression involving special functions and polylogarithms?	Open prompts with usable identity examples but no meaningful answers
MO 173492	Dilogarithm of -1/2?	Open prompts with usable identity examples but no meaningful answers
MO 141185	A second polylogarithm ladder for the tribonacci and n-nacci constants	Open prompts with usable identity examples but no meaningful answers
MO 490573	The imaginary part of two-term complex dilogarithms	Open prompts with usable identity examples but no meaningful answers
MO 496682	Deriving a 4-term functional dilogarithm identity by Kirillov	Open prompts with usable identity examples but no meaningful answers
MSE 3938336	Evaluate a series equal to Li_3((3-sqrt(5))/2)	Duplicate-like or low-new-information threads
MSE 5110303	Does the polylogarithm Li_k(x) solve a first or second order ODE/ADE?	Rejected or parked: no answer and no special-value identity example
MSE 5127863	Completing proof of formula for a hard polylogarithmic integral	Rejected or parked: no answer and no special-value identity example
MSE 4986171	Understanding polylogarithm branch points, monodromy and multivaluedness	Rejected or parked: no answer and no special-value identity example
MO 85774	Is the quantum dilogarithm related in any way to cohomology of quantum groups?	Rejected or parked: no answer and no special-value identity example
MO 160646	Inverse of polylogarithm	Rejected or parked: no answer and no special-value identity example
MO 333610	Limiting property of polylogarithm ratio	Rejected or parked: no answer and no special-value identity example
MO 279588	Approximations of polylogarithm and Lerch transcendent?	Rejected or parked: no answer and no special-value identity example
MO 218158	Root polylogarithm dominance questions	Rejected or parked: no answer and no special-value identity example
MSE 1210644	Proving a sum cos(n)/n^4 evaluation	Near misses and background threads
MSE 376180	Integral identity involving Li_2(-1-e^{ix})	Near misses and background threads
MSE 4284361	Closed form evaluation of a trigonometric integral in terms of polylogarithms	Near misses and background threads
MSE 5125161	Verify a polylogarithmic evaluation of a 5D integral over a positive orthant	Near misses and background threads

Thread	Research question	Historical category
MO 347747	Simplify the difference of two dilogarithms	Near misses and background threads
MO 2402	Abel's equation for the dilog	Near misses and background threads
MO 25428	What is special about polylogarithms that leads to so many interesting identities and applications?	Near misses and background threads
MO 242920	Polylogarithm sheaves	Near misses and background threads
MO 309945	A remarkable almost-identity	Near misses and background threads

A.2 Polygamma and Hurwitz questions

Thread	Research question	Historical category
MSE 897967	Trigamma identity 4 psi_1(1/5)+psi_1(2/5)-psi_1(1/10)=4 pi^2/(phi sqrt(5))	Primary answered matches on Math Stack Exchange
MSE 4290297	New trigamma identity for Psi_1(3/20)+6 Psi_1(1/5)+10 Psi_1(2/5)-Psi_1(1/20)	Primary answered matches on Math Stack Exchange
MSE 953976	Prove psi^(1)(5/6)-psi^(1)(1/6)=5(psi^(1)(2/3)-psi^(1)(1/3))	Primary answered matches on Math Stack Exchange
MSE 2802872	Trigamma identity psi_1(11/12)-psi_1(5/12)=4 sqrt(3) pi^2-80G	Primary answered matches on Math Stack Exchange
MSE 3063097	Alternative proof for zeta(2,1/4)=psi^(1)(1/4)=pi^2+8G	Primary answered matches on Math Stack Exchange
MSE 2952474	Is it possible to simplify psi^(2)(1/8) or psi^(2)(p/q)?	Primary answered matches on Math Stack Exchange
MSE 4040435	General summation formula for psi^(s)(1/2), including negative orders	Primary answered matches on Math Stack Exchange
MSE 4784920	Closed form for psi^(-2)(1/4)	Primary answered matches on Math Stack Exchange
MSE 893486	Identity connecting polylogarithm and polygamma functions of arguments 1/4 and 3/4	Primary answered matches on Math Stack Exchange
MSE 4504218	Identity relating the polygamma function and the Lerch transcendent	Primary answered matches on Math Stack Exchange
MSE 3653746	Decomposition of psi^(n)(1) in terms of psi^(n)(k)	Primary answered matches on Math Stack Exchange
MSE 1154373	An identity on Polygamma	Primary answered matches on Math Stack Exchange
MSE 597867	Multiplication formula for the Hurwitz zeta function	Primary answered matches on Math Stack Exchange
MSE 1335002	An infinite series in polygamma function	Primary answered matches on Math Stack Exchange
MSE 4441095	Closed form for a derivative limit involving psi^(0) and higher polygamma values at 1	Primary answered matches on Math Stack Exchange
MO 312479	pi in terms of polygamma	Primary answered matches on MathOverflow
MO 312550	psi(2,1/6), psi(4,1/6) in terms of zeta and pi only	Primary answered matches on MathOverflow

Thread	Research question	Historical category
MO 42696	Connection between Bernoulli polynomials and polygamma function	Primary answered matches on MathOverflow
MO 347520	Reference request: sums of rational functions and polygamma functions	Primary answered matches on MathOverflow
MO 251727	Closed form for sum involving digamma?	Primary answered matches on MathOverflow
MSE 1909595	Interesting connection between polylogarithms and polygamma functions	Open or no-meaningful-answer leads
MSE 4319642	Simplification of a difficult identity involving the digamma function	Open or no-meaningful-answer leads
MSE 3555737	Sum involving 2nd antiderivative of the digamma function	Open or no-meaningful-answer leads
MSE 7183	Connection between Bernoulli polynomials and polygamma function	Open or no-meaningful-answer leads
MSE 521203	Recurrence relation for polygamma reflection polynomials	Open or no-meaningful-answer leads
MSE 4845364	Closed form for psi^(1/k)(1), where k is an integer	Open or no-meaningful-answer leads
MO 152299	Recurrence formula for digamma function with rational number	Open or no-meaningful-answer leads
MO 372726	Can a digamma generating-function expression be simplified?	Open or no-meaningful-answer leads
MO 455136	Negativity of a determinant involving trigamma, tetragamma, and pentagamma	Open or no-meaningful-answer leads
MSE 1427206	Conjecture for an integral involving psi^(1)(1/3)	Adjacent answered leads
MSE 1346396	Logarithmic Integral I	Adjacent answered leads
MSE 3075324	Improper integral with trigamma identity in the solution path	Adjacent answered leads
MSE 4634827	Closed form for a fractional-part integral	Adjacent answered leads
MSE 2683542	Meaning and definition of psi^(-2)(x)	Adjacent answered leads
MSE 5080998	Integral of a trigamma difference at conjugate half-line arguments	Adjacent answered leads
MO 220759	Closed form for sum 1/(1+n+...+n^a)	Adjacent answered leads
MO 378665	Closed form of sum_{r>=2} zeta(r)/r^2	Adjacent answered leads

A.3 The provenance boundary

The original drafts retain their snapshot identifiers. The accompanying source inventory records the SHA-256 digest of all thirty-nine textual sources. The editorial ledger maps each source to the merged mathematical development, records which results later drafts supersede, and identifies claims that have been withdrawn or weakened. The old PDFs are source evidence and are superseded as reading artifacts by this manuscript.

The original identity stores, reduction tools, notebook parsers, certificate checker and many high-precision search scripts were not included in the import. A historical receipt in a source report is therefore not a fresh replay here. The new verification scripts exercise representative corrected formulas from independent definitions. Their receipt specifies engine, precision, tolerance and actual residuals; it does not assert a complete replay of every historical experiment.

A.4 A version-specific external consistency check

The final paragraph of §7.2, printed page 16 of the inspected 18-page 2020 preprint [35], displays the untwisted identity

$$\beta_p(n) = \beta_p(n+1) + \beta_p\left(\frac{n}{n+1}\right). \tag{A.1}$$

For its stated exterior-square symbol, this identity is false. Take $p = 7$, $n = 2$. Since $2/3 = 3$ in $\mathbb{F}_7$ and the class of 3 is inverse to that of 2 modulo sign,

$$\beta_7(3) = -\beta_7(2).$$

Equation (A.1) would therefore imply $3\beta_7(2) = 0$. But $2^2 = 4$ is neither 1 nor -1 modulo 7, so Theorem 11.9 proves $\beta_7(2) \neq 0$.

There is also a small exact matrix certificate. In the regular basis of sign classes $1, 2, 3$, the image of the defect under the coordinate model $\beta_7(a) \mapsto P_a - P_{a^{-1}}$ is

$$\begin{pmatrix} 0 & 3 & -3 \\ -3 & 0 & 3 \\ 3 & -3 & 0 \end{pmatrix} \neq 0. \tag{A.2}$$

Its entries are integers; no numerical logarithm independence test enters this counterexample.

Proposed correction and version scope. Withdraw the untwisted identity as printed, or supply a corrected coefficient action or quotient together with a proof. This article does not guess what a missing action should be. The definition of β and the boundary formula (11.8) remain valid; the latter was independently derived in Section 11.7. The statement was verified against the 18-page author preprint and its 2020 version; the separately published 2023 journal version is not covered by this particular check. The correction is therefore specifically to the inspected 2020 preprint, and must not be attributed to an unchecked version.

A.5 Two Stieltjes index and coefficient corrections

These checks concern the identified displays in the inspected source PDFs. They also illustrate why the coefficient convention and the initial summation index must be fixed before comparing harmonic spectral identities.

A.5.1 The initial term in the Hurwitz–Stieltjes limit

On printed page 1 of *On certain harmonic zeta functions*, arXiv:2403.07123v1, dated 11 March 2024 [135], the Hurwitz series begins at $n = 0$, but the following unnumbered Stieltjes-limit display begins at $n = 1$. For real $a > 0$ and integer $m \geq 0$, the correct unit-coordinate limit is

$$\gamma_m(a) = \lim_{N\to\infty}\left[\sum_{n=0}^{N} \frac{\log^m(n+a)}{n+a} - \frac{\log^{m+1}(N+a)}{m+1}\right]. \tag{A.3}$$

Equivalently, retain the lower index one and add $\log^m(a)/a$.

Proof. The standard sum–integral subtraction for the Laurent expansion of $\zeta(s,a)$ gives (A.3); changing the upper endpoint by one contributes $O((1+\log^m N)/N)$ and does not change the limit. The recurrence $\zeta(1+u,a+1)=\zeta(1+u,a)-a^{-1-u}$ gives, coefficient by coefficient,

$$\gamma_m(a+1)=\gamma_m(a)-\frac{\log^m a}{a}.$$

The sum starting at one is the sum for $a+1$, hence its printed limit is $\gamma_m(a+1)$, not $\gamma_m(a)$. □

At $m=0, a=1$, the exact counterexample is

$$\lim_{N\to\infty}\{H_{N+1}-1-\log(N+1)\}=\gamma-1.$$

At $a=1, m>0$ the omitted term vanishes; testing only those cases would miss the index defect. This correction is specifically to the inspected version-one introductory display.

A.5.2 The sign of the height-one linear coefficient

Young's published *Series of height one multiple zeta functions*, *Integers* 24 (2024), A43, printed page 12 [134], describes the symbol in Corollary 3 as the linear coefficient. Its displayed integral expression instead gives the negative of the literal coefficient $L_d=[u]Z(1+u,1^{d-1})$ in our convention. The ordinary Stieltjes value in Example 5 is consistent with the conventional $\gamma_1=-L_1$.

The sign can be checked at every depth without a decimal fit. For $0\le r\le 1$ and $s>1$, define the classical Barnes series

$$\mathcal{B}_r(s)=\sum_{n\ge 0}\frac{(r)_n}{n!}(n+1)^{-s}.$$

At $r=0$ only the initial term remains. Put

$$U_d=\int_0^1(1-r)\mathcal{B}_r(d+2)\,\mathrm{d}r,\qquad V_d=\int_0^1\mathcal{B}_r(d+1)\,\mathrm{d}r.$$

The right side of the inspected Corollary 3 is $U_d-V_d-\alpha_{d+1}$. Using $(r)_{n+1}=(r+n)(r)_n$ and termwise integration gives

$$U_d-V_d=-\sum_{n\ge 1}\frac{\rho_n}{n^{d+2}},$$

where ρ_n is defined in Proposition 8.117. All relevant series converge absolutely because $(r)_n/n!\le 1$ on $0\le r\le 1$ and $d\ge 1$. The proved Cauchy coefficient formula therefore shows exactly that

$$U_d-V_d-\alpha_{d+1}=-L_d. \tag{A.4}$$

Thus one may keep the displayed expression and describe the negative linear coefficient, or negate it when naming the literal $[u]$ coefficient. At depth one it gives γ_1, whereas the literal linear coefficient of $\zeta(1+u)$ is $-\gamma_1$. The depth-one discrepancy is nonzero rigorously: the exact rational certificate in `verification/digamma-coordinate-source-audit.json` gives

$$-\frac{3}{40}<\gamma_1<-\frac{7}{100}<0.$$

Here is its analytic error budget. For $f(x)=\log(x)/x$ and $N=10$, one Euler–Maclaurin correction gives

$$\gamma_1=\sum_{n=1}^{N}\frac{\log n}{n}-\frac{\log^2 N}{2}-\frac{\log N}{2N}+\frac{\log N-1}{12N^2}+E_N,\qquad |E_N|\le\frac{\log N-1}{12N^2}.$$

The bound follows from $|\widetilde{B}_2|/2! \le 1/12$ and $f''(x) = (2\log x - 3)/x^3 \ge 0$ for $x \ge 10$; thus $\int_N^\infty |f''| = -f'(N)$. Each logarithm is enclosed rationally by 128 terms of its atanh series, with the geometric positive-tail bound. The resulting endpoint comparisons are exact fractions, not floating sign tests. The separately replayed 100-digit Cauchy/Barnes diagnostic also confirms the sign; (A.4) proves the all-depth coefficient identity. The correction concerns that coefficient description, with Example 5's ordinary Stieltjes formula retained.

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[122] NIST Digital Library of Mathematical Functions, §5.7, *Series Expansions of the Gamma Function*, especially formula 5.7.3. https://dlmf.nist.gov/5.7.E3. Inspected 10 October 2026.

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[124] NIST Digital Library of Mathematical Functions, §16.4, *Rogers–Dougall Very Well-Poised Sum*, formula 16.4.9. https://dlmf.nist.gov/16.4.E9. Inspected 10 October 2026.

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[129] ProveIt research continuation, *Exact Resonant Identities Beyond the Diagonal*, 11 October 2026. Original package preserved under `reports/stieltjes-harmonic-resonance/continuations/exact-resonant-identities/`. The complete-Cayley formal obstruction is audited in Chapter 12; the numerical S_6 conjecture remains open.

[130] ProveIt research continuation, *Harmonic Parity and Resolvent Identities*, 11 October 2026. Original package preserved under `reports/stieltjes-harmonic-resonance/continuations/harmonic-parity-resolvents/`. The centered trigamma-square moments and their polylogarithmic generator are audited in Chapter 8; the broader classification and resolvent claims retain separate scopes.

[131] ProveIt research continuations, *Higher Digamma Powers and Ordered Gamma Sums*, *Canonical Digamma Polynomials and the Fifth Harmonic Bridge*, and *Mixed Resolvent Products, Higher Digamma Powers, and Stieltjes Collisions*, 11 October 2026. Original packages are preserved as `reports/stieltjes-harmonic-resonance/continuations/higher-digamma-powers/`, `reports/stieltjes-harmonic-resonance/continuations/canonical-digamma-fifth/`, and `reports/stieltjes-correlation-calculus/continuations/mixed-resolvents-harmonic-powers/`. Their all-power and fifth-moment cores are reconciled in Chapter 8; other Gamma, resolvent, collision and ordered branches retain separate audit scopes.

[132] NIST Digital Library of Mathematical Functions, Sections 5.5 (digamma recurrence and reflection), 5.7 (Taylor and reciprocal-Gamma series), and 5.11 (sectorial asymptotics). In particular formulas 5.5.2, 5.5.4, 5.7.4 and 5.11.2. https://dlmf.nist.gov/5. Inspected 11 October 2026 UTC.

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[135] M. Can, L. Kargın, M. Cenkci and A. Dil, *On certain harmonic zeta functions*, arXiv:2403.07123v1, 11 March 2024, printed page 1. https://arxiv.org/pdf/2403.07123v1. The initial-index correction in Appendix A is scoped to that version's introductory Stieltjes-limit display. Inspected 11 October 2026 UTC.

[136] ProveIt research continuations, *Bernoulli Determinants, Cubic Tornheim Jets, and Canonical Digamma Integrals*, *Differential Polygamma and Lerch Identities*, and *Exact Polylogarithm Parity, Harmonic-Tail Reconstruction, and Higher Polygamma Moments*, 11 October 2026. Original packages preserved under `reports/stieltjes-harmonic-resonance/continuations/bernoulli-tornheim-digamma/`, `reports/stieltjes-harmonic-resonance/continuations/differential-polygamma-lerch-jets/`, and `reports/stieltjes-harmonic-resonance/continuations/polygamma-tail-identities/`. Their common differential-polynomial analytic core is reconciled in Chapter 8; their remaining coordinate, rank and Lerch claims retain separate audit scopes.

[137] ProveIt research continuation, *Beta Mixtures Across Integer Twists: Finite spectral corrections, harmonic logarithmic moments, and Stieltjes–polylogarithm primitives*, 10 October 2026. Original article, PDF and verification evidence preserved under `reports/stieltjes-correlation-calculus/continuations/beta-crossing/`. The affine crossing, integer-total mixture and balanced-jet cores are reconciled in Chapter 8; the later logarithmic Mellin and primitive families retain their separate integration scope.

[138] ProveIt research continuations, *Alternating Tails and Spectral Defects*, *Quartic Spectral Reconstruction*, *Bernoulli Tails and Spectral Faces*, *Bilinear Tails and Harmonic Determinants*, and *Tail Kernels and Boundary Products*, 11 October 2026. Original sources and exact evidence preserved on the harmonic-resonance spine and under `reports/tail-boundaries/`. The common centered coefficient model, retained determinant and effective eventual proof are reconciled in Chapter 8. Complete orders ten and twelve retain report attribution; the collective order-fourteen extension has its full exact certificate under `verification/prefix-order-fourteen/`.

[139] ProveIt research continuations, *Universal Gamma Normal Forms: Rational Kernels, Harmonic Sums, and Stieltjes Primitives*, and *A Universal Beta Formula for Gamma Normal Forms*, 11 October 2026. Original articles, PDFs and exact evidence preserved under `reports/stieltjes-harmonic-resonance/continuations/universal-gamma-coefficients/` and `reports/stieltjes-harmonic-resonance/continuations/universal-beta-normal-forms/`. Their common all-index coefficient formula and ordinary finite projections are reconciled in Chapter 8. The collective analytic-jet extension gives an actual locally convergent infinite normal generator.

[140] ProveIt research continuations, *Higher Identities*, *Cubic Gamma Normal Form*, and *Universal Gamma Coefficients*, 11–12 October 2026. Delivered articles and exact evidence preserved on the harmonic-resonance continuation spine. The compensator construction uses the classical Euler homotopy operator: D. Poole and W. Hereman, *The homotopy operator method for symbolic integration by parts and inversion of divergences with applications*, https://arxiv.org/abs/0908.0399; W. Hereman, B. Deconinck and L. D. Poole, *Continuous and Discrete Homotopy Operators: A Theoretical Approach made Concrete*, https://arxiv.org/abs/nlin/0510064. The hierarchy specialization, analytic primitive domain and independent normal-generator proof are reconciled in Chapter 8. The collective continuation proves every primitive coefficient and full support/sign, with convergence on the full input disk; fixed-gap analytic completion remains a separate pending proof scope.

[141] ProveIt research continuation, *A Finite Polylogarithm Alphabet for Rational Hankel Sums*, 11 October 2026. Original article, compiler and recorded evidence preserved under `reports/stieltjes-harmonic-resonance/continuations/rational-hankel-polylogarithms/`. The confluent weight budget, anchored two-letter compiler and quadratic denominator evaluation are reconciled in Chapter 8. The classical tools retain their attribution: M. E. Hoffman, *Quasi-shuffle products*, J. Algebraic Combin. **11** (2000), 49–68, https://arxiv.org/abs/math/9907173; E. Panzer, *Algorithms for the symbolic integration of hyperlogarithms with applications to Feynman integrals*, Comput. Phys. Commun. **188** (2015), 148–166, https://arxiv.org/abs/1403.3385.

[142] ProveIt research continuations, *Resonant Polylogarithmic Jets and Convergent Inverse Transseries* and *Polylogarithm Order Jets and Resonant Inversion*, 11–12 October 2026. Original sources and evidence are preserved under `reports/uniform-transition-continuation/continuations/`, in `normalized-resonant-jets/` and `order-jets-resonant-inversion/`. The

fixed-sector theorem extends R1 of the preserved `resonance-correlations-nonseparable-jets/` report. Reciprocal-Gamma normalization is credited to both overlapping continuations. Section 1.6 develops the exact order-lowering and arbitrary repeated-primitive identities from that normalization. Inverse geometry and Borel reconstruction retain separate review scopes.

[143] ProveIt research continuations, *Resonant Polylogarithm Transseries* and *Boundary Completion for Quadratic Polylogarithm Feedback*, 11–12 October 2026. Original articles, PDFs and evidence are preserved under `Analysis/Transseries/docs/series-and-transseries/`, in `Resonant_Polylogarithm_Transseries/` and `Quadratic_Polylogarithm_Feedback_Boundary_Completion/`. The first report establishes the interior fold and poses the boundary question; the second proves its completion. Section 1.7 reconciles that classification and develops finite Bessel-polynomial coefficient and order-spectrum identities. Nonmarginal critical charts and crossover refinements retain separate integration scopes.

[144] NIST Digital Library of Mathematical Functions, §18.34, *Bessel Polynomials*, especially (18.34.1)–(18.34.5), https://dlmf.nist.gov/18.34. The convention here is $y_n(x) = y_n(x; 2)$.

[145] P. Flajolet and A. Odlyzko, *Singularity analysis of generating functions*, SIAM J. Discrete Math. **3** (1990), 216–240.

[146] NIST Digital Library of Mathematical Functions, §4.13, *Lambert W-function*, https://dlmf.nist.gov/4.13.

[147] A. G. Pakes, *Affine relation between an infinitely divisible distribution function and its Lévy measure*, J. Math. Anal. Appl. **499** (2021), article 125008, https://doi.org/10.1016/j.jmaa.2021.125008. Related lower-branch Laplace-transform literature; the normalization and identities used in Section 1.8 are proved there.

[148] ProveIt research continuation, *Mixed Hurwitz-Tail Moments, Zeta-Anchor Reductions, and Stieltjes Transport*, 11 October 2026, preserved article and verification evidence in `reports/stieltjes-harmonic-resonance/continuations/mixed-tail-anchor-reductions/`. Its third research question asks for rational polynomial combinations for $(2, 2, 3)$ and $(2, 3, 4)$. Section 8.48 develops the first case within a precisely specified polynomial certificate class.

[149] ProveIt research continuation, *Double Resonance and Nested Stokes Geometry in Polylogarithmic Lambert Inversion*, 12 October 2026. Preserved source and evidence in `Analysis/Transseries/docs/series-and-transseries/Resonant_Lambert_Inverse_Transseries/`. The finite inverse formula and the conjectured hyperbolic amplitudes motivate Section 1.9.

[150] ProveIt research continuation, *Summation and Singularities under Polylogarithmic Reversion*, 12 October 2026, in the preceding report's `continuations/summation-singularities/` directory. Its Lambert section proves the leading amplitudes and first correction; its all-order question is resolved by Theorem 1.26. Original source evidence and fresh isolated verification are preserved.

[151] ProveIt research continuation, *Sharp Late Coefficients and Borel Continuation of Resonant Lambert Inverses*, 12 October 2026, in the resonant Lambert inverse report's `continuations/late-coefficients/` directory. Original source evidence and a complete isolated replay are preserved. Its inverse-index all-order question is resolved by Theorem 1.26; its nonlinear continuation argument is developed in Theorem 1.28.

[152] D. Sauzin, *Nonlinear analysis with resurgent functions*, Annales scientifiques de l'École Normale Supérieure (4) **48** (2015), 667–702. arXiv:1212.4477. Theorems 3 and 5 establish analytic substitution and compositional inversion for an additive closed discrete singular set; Theorems 3′ and 5′ give the corresponding summability statements.

[153] ProveIt research continuation, *Order-Jet Filters for Polylogarithmic Transseries: Sharp cancellation costs, Stokes-neutral curves, and a Stieltjes–digamma inverse*, 12 October 2026, in the resonant polylogarithm report's `continuations/order-jet-filters/` directory. Original source evidence and two isolated replays are preserved. Theorem 1.41 answers its fixed-filter neutral inverse-transport question.

[154] NIST Digital Library of Mathematical Functions, Gamma functional relations and Gamma series expansions.

[155] NIST Digital Library of Mathematical Functions, Gamma and digamma integral representations and sectorial asymptotic expansions.

[156] M. Karatarakis and F. Wiedijk, *A formalization of the Gelfond–Schneider theorem*, arXiv:2603.24823v1 (2026), Theorem 3.1. Used for the classical transcendence theorem with a chosen complex logarithm; this citation does not formalize the present filters.

[157] ProveIt research report, *Nonlinear Stokes Transport and Logarithmic Inversion*, in the Transseries collection at `Analysis/Transseries/docs/series-and-transseries/Nonlinear_Stokes_Transport_Logarithmic_Inversion/`. The order-zero averaging defect specializes its nonlinear inverse mechanism.

[158] ProveIt research continuation, *Uniform Coefficient Transseries across the Polylogarithmic Order-Three Resonance*, 12 October 2026, in the quadratic feedback boundary report's `continuations/joint-coefficient-transseries/` directory. The original article, PDF and evidence, together with an isolated replay, are preserved. The matched-derivative comparison in Theorem 1.63 develops its third research question.

[159] ProveIt research collection. *Resonant Order Jets and Convergent Inverse Transseries: for Polylogarithms and Logarithmically Weighted Tails*. 12 October 2026. Pinned article source. In particular, the noninteger inverse charts and research question 9, "Global geometry beyond integer orders."

[160] NIST Digital Library of Mathematical Functions. *Chapter 12, Parabolic Cylinder Functions*: §12.2, definitions and initial values; §12.5, integral representations, especially formula 12.5.4; §12.9, large-variable asymptotic expansions. Accessed 12 October 2026.

[161] H. Lam, J. Blanchet, D. Burch, and M. Z. Bazant. Corrections to the Central Limit Theorem for Heavy-tailed Probability Densities. *Journal of Theoretical Probability* **24**(4) (2011), 895–927. doi:10.1007/s10959-011-0379-y. Author-hosted final text.

[162] Q. Berger, M. Birkner, and L. Yuan. Collective vs. individual behaviour for sums of i.i.d. random variables: appearance of the one-big-jump phenomenon. *Annales de la Faculté des Sciences de Toulouse*, Series 6, **33**(5) (2024), 1413–1485; published online 24 April 2025. doi:10.5802/afst.1802.

[163] ProveIt research continuation. *Balanced Stieltjes Moments: Bernoulli Closure, Harmonic Filters, and Polylogarithmic Resonances*. 11 October 2026. Preserved article, Balanced scalar

kernels and their Mellin transform; Section 7 gives mixed logarithmic moments of filtered Stieltjes functions.

[164] ProveIt research continuation. *Gamma Determinants and Colored Stieltjes Moments: Closed Normalization Constants, Finite Gaussian Moment Laws, and Clustered Polylogarithm Identities*. 12 October 2026. Preserved colored-moment section. The colored scalar transform, Stieltjes dictionary, and first exceptional moment are established there; the further-research section asks about simultaneous tail zeros.

[165] NIST Digital Library of Mathematical Functions. §2.5, Mellin inversion and Parseval, Eq. 15.6.1, Euler's hypergeometric integral, and Table 18.3.1, Laguerre orthogonality. Accessed 12 October 2026.

[166] J. Brillhart. On the Euler and Bernoulli polynomials. *Journal für die reine und angewandte Mathematik* **234** (1969), 45–64. doi:10.1515/crll.1969.234.45. The multiple-root theorem is also stated explicitly in F. T. Howard, *Roots of the Euler polynomials*, Pacific Journal of Mathematics **64**(1) (1976), 181–191, p. 181, publisher text, and NIST DLMF, §24.12(iv).

[167] NIST Digital Library of Mathematical Functions. §25.15, Dirichlet L-functions: functional equation, Gauss sums, Euler products, and trivial zeros. Accessed 12 October 2026.